CHAPTER 6

Selecting Input Probability Distributions

Recommended sections for a first reading: 6.1, 6.2, 6.4 through 6.7, 6.11

6.1 INTRODUCTION

To carry out a simulation using random inputs such as interarrival times or demand sizes, we have to specify their probability distributions. For example, in the simulation of the single-server queueing system in Sec. 1.4.3, the interarrival times were taken to be IID exponential random variables with a mean of 1 minute; the demand sizes in the inventory simulation of Sec. 1.5 were specified to be 1, 2, 3, or 4 items with respective probabilities $\frac{1}{6}$, $\frac{1}{3}$, $\frac{1}{3}$, and $\frac{1}{6}$. Then, given that the input random variables to a simulation model follow particular distributions, the simulation proceeds through time by generating random values from these distributions. Chapters 7 and 8 discuss methods for generating random values from various distributions and processes. Our concern in this chapter is with how the analyst might go about specifying these input probability distributions.

Almost all real-world systems contain one or more sources of randomness, as illustrated in Table 6.1. Furthermore, in Figs. 6.1 through 6.4 we show histograms for four data sets taken from actual simulation projects. Figure 6.1 corresponds to 890 machine processing times (in minutes) for an automotive manufacturer. It can be seen that the histogram has a longer right tail (positive skewness) and that the minimum value is approximately 25 minutes. In Fig. 6.2 we show a histogram for 219 interarrival times (in minutes) to a drive-up bank (see Example 6.4). Figure 6.3 displays a histogram for 856 ship-loading times (in days) (see Example 6.17). Finally, in Fig. 6.4 we give a histogram for the number of yards of paper (scaled for confidentiality reasons) on 1000 large rolls of paper used to make facial or bathroom tissue. In this case the histogram has a longer left tail (negative skewness).

TABLE 6.1
Sources of randomness for common simulation applications

Type of system	Sources of randomness
Manufacturing	Processing times, machine times to failure, machine repair times
Defense-related	Arrival times and payloads of missiles or airplanes, outcome of an engagement, miss distances for munitions
Communications	Interarrival times of messages, message types, message lengths
Transportation	Ship-loading times, interarrival times of customers to a subway

Note that none of the four histograms has a symmetric shape like that of a normal distribution, despite the fact that many simulation practitioners and simulation books widely use normal input distributions.

We saw in Sec. 4.7 that it is generally necessary to represent each source of system randomness by a probability distribution (rather than just its mean) in the simulation model. The following example shows that failure to choose the "correct" distribution can also affect the accuracy of a model's results, sometimes drastically.

EXAMPLE 6.1. A single-server queueing system (e.g., a single machine in a factory) has exponential interarrival times with a mean of 1 minute. Suppose that 200 service times are available from the system, but their underlying probability distribution is

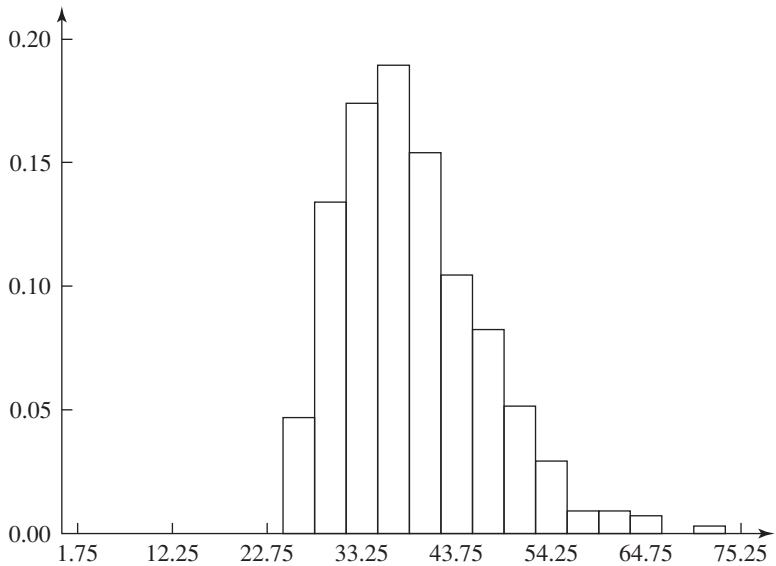


FIGURE 6.1
Histogram of 890 machine processing times for an automotive manufacturer.

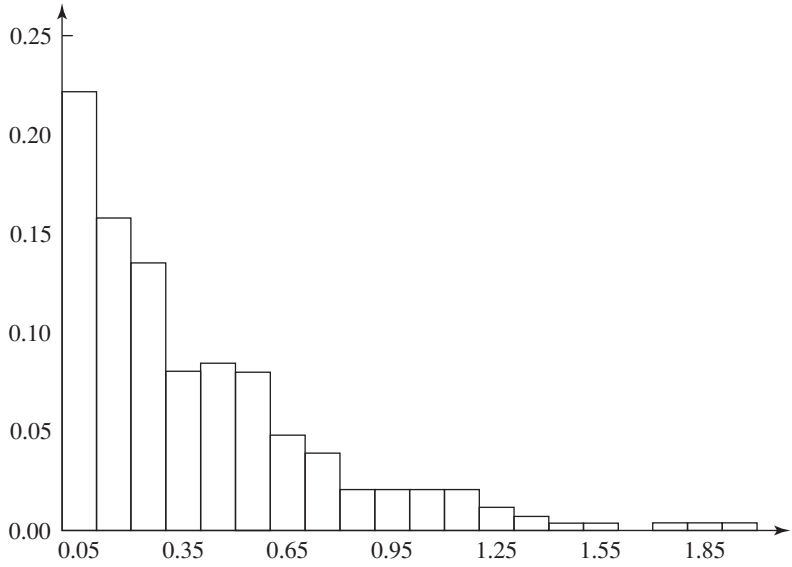


FIGURE 6.2
Histogram of 219 interarrival times to a drive-up bank.

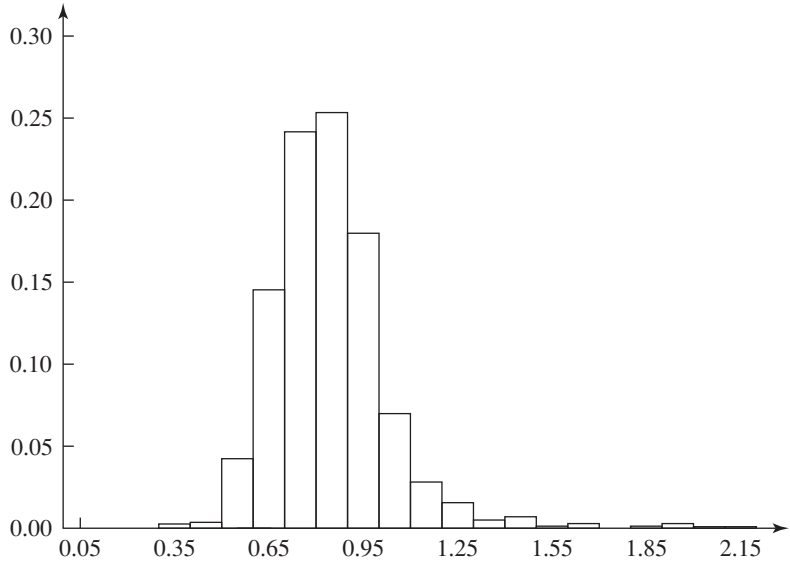


FIGURE 6.3
Histogram of 856 ship-loading times.

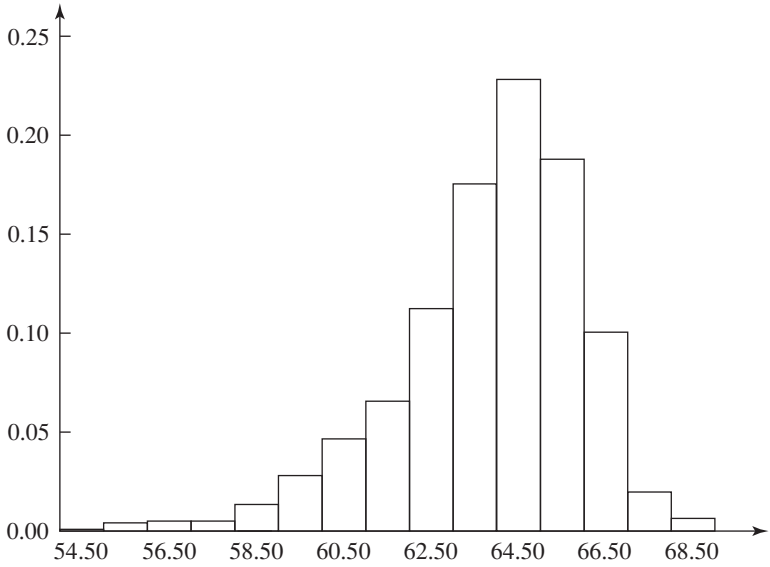


FIGURE 6.4
Histogram of the yardages for 1000 large rolls of paper for a household-products manufacturer.

unknown. Using an approach to be discussed in Sec. 6.5, we "fit" the "best" exponential, gamma, Weibull, lognormal, and normal distributions (see Sec. 6.2.2 for a discussion of these distributions) to the observed service-time data. (In the case of the exponential distribution, we chose the mean β so that the resulting distribution most closely "resembled" the available data.) We then made 100 independent simulation runs (i.e., different random numbers were used for each run, as discussed in Sec. 7.2) of the queueing system, using *each* of the five fitted distributions. (For the normal distribution, if a service time was negative, then it was generated again.) Each of the 500 simulation runs was continued until 1000 delays in queue were collected. A summary of the results from these simulation runs is given in Table 6.2. Note in the second column of the table that the average of the 100,000 delays is given for each of the service-time distributions (see Prob. 6.27). As we will see in Sec. 6.7, the Weibull distribution actually provides the best model for the service-time data. Thus, the average delay for the real system should be close to 4.36 minutes. On the other hand, the average delays for the normal and lognormal

TABLE 6.2
Simulation results for the five service-time distributions (in minutes where appropriate)

Service-time distribution	Average delay in queue	Average number in queue	Proportion of delays ≥ 20
Exponential	6.71	6.78	0.064
Gamma	4.54	4.60	0.019
Weibull	4.36	4.41	0.013
Lognormal	7.19	7.30	0.078
Normal	6.04	6.13	0.045

distributions are 6.04 and 7.19 minutes, respectively, corresponding to model output errors of 39 percent and 65 percent. This is particularly surprising for the lognormal distribution, since it has the same general shape (i.e., skewed to the right) as the Weibull distribution. However, it turns out that the lognormal distribution has a "thicker" right tail, which allows larger service times and delays to occur. The relative differences between the "tail probabilities" in column 4 of the table are even more significant. The choice of probability distributions can evidently have a large impact on the simulation output and, potentially, on the quality of the decisions made with the simulation results.

If it is possible to collect data on an input random variable of interest, these data can be used in one of the following approaches to specify a distribution (in increasing order of desirability):

1. The data values themselves are used directly in the simulation. For example, if the data represent service times, then one of the data values is used whenever a service time is needed in the simulation. This is sometimes called a *trace-driven simulation*.
2. The data values themselves are used to define an *empirical* distribution function (see Sec. 6.2.4) in some way. If these data represent service times, we would sample from this distribution when a service time is needed in the simulation.
3. Standard techniques of statistical inference are used to "fit" a *theoretical* distribution form (see Example 6.1), e.g., exponential or Poisson, to the data and to perform hypothesis tests to determine the goodness of fit. If a particular theoretical distribution with certain values for its parameters is a good model for the service-time data, then we would sample from this distribution when a service time is needed in the simulation.

Two drawbacks of approach 1 are that the simulation can only reproduce what has happened historically and that there is seldom enough data to make all the desired simulation runs. Approach 2 avoids these shortcomings since, at least for continuous data, any value between the minimum and maximum observed data points can be generated (see Sec. 8.3.16). Thus, approach 2 is generally preferable to approach 1. However, approach 1 does have its uses. For example, suppose that it is desired to compare a proposed material-handling system with the existing system for a distribution center. For each incoming order there is an arrival time, a list of the desired products, and a quantity for each product. Modeling a stream of orders for a certain period of time (e.g., for 1 month) will be difficult, if not impossible, using approach 2 or 3. Thus, in this case the existing and proposed systems will often be simulated using the historical order stream. Approach 1 is also recommended for *model validation* when model output for an existing system is compared with the corresponding output for the system itself. (See the discussion of the correlated inspection approach in Sec. 5.6.1.)

If a theoretical distribution can be found that fits the observed data reasonably well (approach 3), then this will generally be preferable to using an empirical distribution (approach 2) for the following reasons:

- An empirical distribution function may have certain "irregularities," particularly if only a small number of data values are available. A theoretical distribution, on the other hand, "smooths out" the data and may provide information on the overall underlying distribution.

- If empirical distributions are used in the *usual* way (see Sec. 6.2.4), it is not possible to generate values outside the range of the observed data in the simulation (see Sec. 8.3.16). This is unfortunate, since many measures of performance for simulated systems depend heavily on the probability of an "extreme" event's occurring, e.g., generation of a very large service time. With a fitted theoretical distribution, however, values outside the range of the observed data can be generated.
- There may be a compelling physical reason in some situations for using a certain theoretical distribution form as a model for a particular input random variable (see, for example, Sec. 6.12.1). Even when we are fortunate enough to have this kind of information, it is a good idea to use observed data to provide empirical support for the use of this particular distribution.
- A theoretical distribution is a compact way of representing a set of data values. Conversely, if n data values are available from a continuous distribution, then $2n$ values (e.g., data and corresponding cumulative probabilities) must be entered and stored in the computer to represent an empirical distribution in simulation packages. Thus, use of an empirical distribution will be cumbersome if the data set is large.
- A theoretical distribution is easier to change. For example, suppose that a set of interarrival times is found to be modeled well by an exponential distribution with a mean of 1 minute. If we want to determine the effect on the simulated system of increasing the arrival rate by 10 percent, then all we have to do is to change the mean of the exponential distribution to 0.909.

There are definitely situations for which no theoretical distribution will provide an adequate fit for the observed data, including the following:

- The data are a mixture of two or more heterogeneous populations (see the discussion of machine repair times in Sec. 6.4.2).
- The times to perform some task have been significantly rounded (effectively making the data discrete), and there are not enough distinct values in the sample to allow any continuous theoretical distribution to provide a good representation.

In situations where no theoretical distribution is appropriate, we recommend using an empirical distribution. Another possible drawback of theoretical distributions (e.g., lognormal) is that arbitrarily large values can be generated, albeit with a very small probability. Thus, if it is known that a random variable can never take on values larger than b, then it might be desirable to truncate the fitted theoretical distribution at b (see Sec. 6.8). For example, it might be known that a service time in a bank is extremely unlikely to exceed 15 minutes.

The remainder of this chapter discusses various topics related to the selection of input distributions. Section 6.2 discusses how theoretical distributions are parameterized, provides a compendium of relevant facts on most of the commonly used continuous and discrete distributions, and discusses how empirical distributions can be specified. In Sec. 6.3 we present techniques for determining whether the data are independent observations from some underlying distribution, which is a requirement of many of the statistical procedures in this chapter. In Secs. 6.4 through 6.6 we discuss the three basic activities in specifying a theoretical distribution on the basis

of observed data. The ExpertFit distribution-fitting software and a comprehensive example are discussed in Sec. 6.7. In Sec. 6.8 we indicate how certain of the theoretical distributions, e.g., gamma, Weibull, and lognormal, can be "shifted" away from 0 to make them better fit our observed data in some cases; we also discuss truncated distributions. We treat Bézier distributions, which are a fourth way to specify a distribution based on observed data, in Sec. 6.9. In Sec. 6.10 we describe how multivariate distributions are specified and estimated when observed data are available. In Sec. 6.11 we describe several possible methods for specifying input distributions when no data are available. Several useful probabilistic models for describing the manner in which "customers" arrive to a system are given in Sec. 6.12, while in Sec. 6.13 we present techniques for determining whether observations from different sources are homogeneous and can be merged.

The graphical plots and goodness-of-fit tests presented in this chapter were developed using the ExpertFit distribution-fitting software (see Sec. 6.7).

6.2 USEFUL PROBABILITY DISTRIBUTIONS

The purpose of this section is to discuss a variety of distributions that have been found to be useful in simulation modeling and to provide a unified listing of relevant properties of these distributions [see also Forbes et al. (2011); Johnson, Kotz, and Balakrishnan (1994, 1995); and Johnson, Kotz, and Kemp (1992)]. Section 6.2.1 provides a short discussion of common methods by which continuous distributions are defined, or parameterized. Then Secs. 6.2.2 and 6.2.3 contain compilations of several continuous and discrete distributions. Finally, Sec. 6.2.4 suggests how the data themselves can be used directly to define an empirical distribution.

6.2.1 Parameterization of Continuous Distributions

For a given family of continuous distributions, e.g., normal or gamma, there are usually several alternative ways to define, or *parameterize*, the probability density function. However, if the parameters are defined correctly, they can be classified, on the basis of their physical or geometric interpretation, as being one of three basic types: location, scale, or shape parameters.

A *location parameter* γ specifies an abscissa (x axis) location point of a distribution's range of values; usually γ is the midpoint (e.g., the mean μ for a normal distribution) or lower endpoint (see Sec. 6.8) of the distribution's range. (In the latter case, location parameters are sometimes called *shift parameters*.) As γ changes, the associated distribution merely shifts left or right without otherwise changing. Also, if the distribution of a random variable X has a location parameter of 0, then the distribution of the random variable $Y = X + \gamma$ has a location parameter of γ.

A *scale parameter* β determines the scale (or unit) of measurement of the values in the range of the distribution. (The standard deviation σ is a scale parameter for the normal distribution.) A change in β compresses or expands the associated

distribution without altering its basic form. Also, if the distribution of the random variable X has a scale parameter of 1, then the distribution of the random variable $Y = \beta X$ has a scale parameter of β.

A *shape parameter* α determines, distinct from location and scale, the basic form or shape of a distribution within the general family of distributions of interest. A change in α generally alters a distribution's properties (e.g., skewness) more fundamentally than a change in location or scale. Some distributions (e.g., exponential and normal) do not have a shape parameter, while others (e.g., beta) may have two.

6.2.2 Continuous Distributions

Table 6.3 gives information relevant to simulation modeling applications for 13 continuous distributions. Possible applications are given first to indicate some (certainly not all) uses of the distribution [see Hahn and Shapiro (1994) and Lawless (2003) for other applications]. Then the density function and distribution function (if it exists in simple closed form) are listed. Next is a short description of the parameters, including their possible values. The range indicates the interval where the associated random variable can take on values. Also listed are the mean (expected value), variance, and mode, i.e., the value at which the density function is maximized. MLE refers to the maximum-likelihood estimator(s) of the parameter(s), treated later in Sec. 6.5. General comments include relationships of the distribution under study to other distributions. Graphs are given of the density functions for each distribution. The notation following the name of each distribution is our abbreviation for that distribution, which includes the parameters. The symbol $\sim$ is read "is distributed as."

Note that we have included the less familiar Johnson S_B, Johnson S_U, log-logistic, Pearson type V, and Pearson type VI distributions, because we have found that these distributions often provide a better fit to data sets than standard distributions such as gamma, lognormal, and Weibull.

TABLE 6.3
Continuous distributions

Uniform	**U(a, b)**
Possible applications	Used as a "first" model for a quantity that is felt to be randomly varying between a and b but about which little else is known. The U(0, 1) distribution is essential in generating random values from all other distributions (see Chaps. 7 and 8).
Density (See Fig. 6.5)	$f(x) = \begin{cases} \dfrac{1}{b-a} & \text{if } a \le x \le b \\ 0 & \text{otherwise} \end{cases}$
Distribution	$F(x) = \begin{cases} 0 & \text{if } x < a \\ \dfrac{x-a}{b-a} & \text{if } a \le x \le b \\ 1 & \text{if } b < x \end{cases}$
Parameters	a and b real numbers with $a < b$; a is a location parameter, $b - a$ is a scale parameter

TABLE 6.3 (*continued*)

Uniform	**U(a, b)**
Range	$[a, b]$
Mean	$\dfrac{a + b}{2}$
Variance	$\dfrac{(b - a)^2}{12}$
Mode	Does not uniquely exist
MLE	$\hat{a} = \min_{1 \le i \le n} X_i, \hat{b} = \max_{1 \le i \le n} X_i$
Comments	1. The U(0, 1) distribution is a special case of the beta distribution (when $\alpha_1 = \alpha_2 = 1$). 2. If $X \sim$ U(0, 1) and $[x, x + \Delta x]$ is a subinterval of [0, 1] with $\Delta x \ge 0$, $$P(X \in [x, x + \Delta x]) = \int_x^{x+\Delta x} 1 dy = (x + \Delta x) - x = \Delta x$$ which justifies the name "uniform."

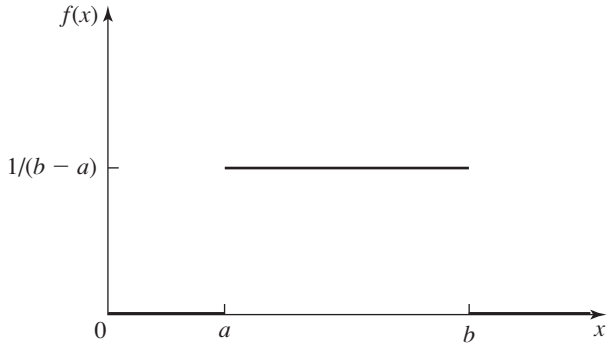


FIGURE 6.5
U(a, b) density function.

Exponential	**expo(β)**
Possible applications	Interarrival times of "customers" to a system that occur at a constant rate, time to failure of a piece of equipment.
Density (see Fig. 6.6)	$f(x) = \begin{cases} \dfrac{1}{\beta} e^{-x/\beta} & \text{if } x \ge 0 \\ 0 & \text{otherwise} \end{cases}$
Distribution	$F(x) = \begin{cases} 1 - e^{-x/\beta} & \text{if } x \ge 0 \\ 0 & \text{otherwise} \end{cases}$
Parameter	Scale parameter $\beta > 0$
Range	$[0, \infty)$
Mean	β
Variance	β^2
Mode	0

(*continued*)

TABLE 6.3 (*continued*)

Exponential	**expo(β)**
MLE	$\hat{\beta} = \overline{X}(n)$
Comments	1. The expo(β) distribution is a special case of both the gamma and Weibull distributions (for shape parameter $\alpha = 1$ and scale parameter β in both cases). 2. If $X_1, X_2, \ldots, X_m$ are independent expo(β) random variables, then $X_1 + X_2 + \cdots + X_m \sim$ gamma(m, β), also called the *m-Erlang*(β) *distribution.* 3. The exponential distribution is the only continuous distribution with the memoryless property (see Prob. 4.30).

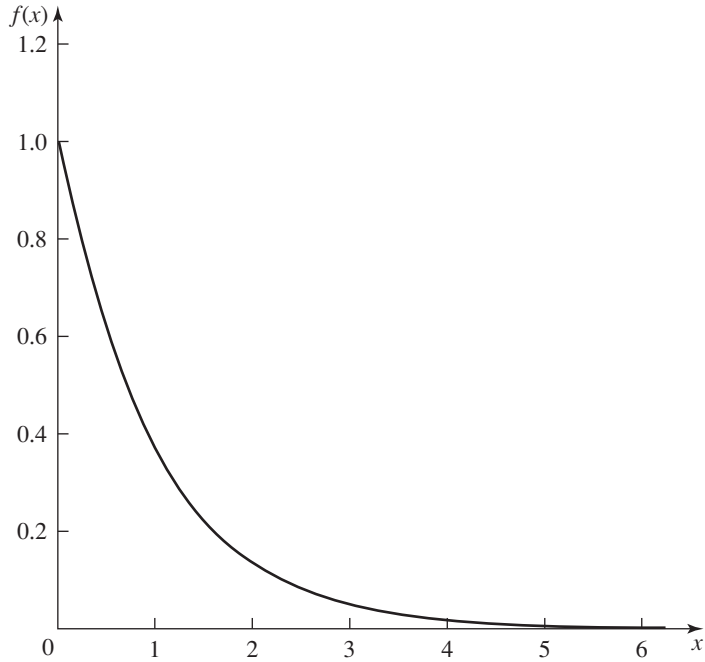


FIGURE 6.6
expo(1) density function.

Gamma	**gamma(α, β)**
Possible applications	Time to complete some task, e.g., customer service or machine repair
Density (see Fig. 6.7)	$f(x) = \begin{cases} \dfrac{\beta^{-\alpha} x^{\alpha-1} e^{-x/\beta}}{\Gamma(\alpha)} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$ where $\Gamma(\alpha)$ is the *gamma function*, defined by $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$ for any real number $z > 0$. Some properties of the gamma function: $\Gamma(z + 1) = z\Gamma(z)$ for any $z > 0$, $\Gamma(k + 1) = k!$ for any nonnegative integer k, $\Gamma(k + \frac{1}{2}) = \sqrt{\pi} \cdot 1 \cdot 3 \cdot 5 \cdots (2k - 1)/2^k$ for any positive integer k, $\Gamma(1/2) = \sqrt{\pi}$

TABLE 6.3 (*continued*)

Gamma	**gamma(α, β)**
Distribution	If α is not an integer, there is no closed form. If α is a positive integer, then $$F(x) = \begin{cases} 1 - e^{-x/\beta} \sum_{j=0}^{\alpha-1} \frac{(x/\beta)^j}{j!} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$
Parameters	Shape parameter $\alpha > 0$, scale parameter $\beta > 0$
Range	$[0, \infty)$
Mean	$\alpha\beta$
Variance	$\alpha\beta^2$
Mode	$\beta(\alpha - 1)$ if $\alpha \geq 1$, 0 if $\alpha < 1$
MLE	The following two equations must be satisfied: $$\ln \hat{\beta} + \Psi(\hat{\alpha}) = \frac{\sum_{i=1}^{n} \ln X_i}{n}, \qquad \hat{\alpha}\hat{\beta} = \bar{X}(n)$$ which could be solved numerically. [$\Psi(\hat{\alpha}) = \Gamma'(\hat{\alpha})/\Gamma(\hat{\alpha})$ and is called the *digamma function*; Γ' denotes the derivative of Γ.] Alternatively, approximations to $\hat{\alpha}$ and $\hat{\beta}$ can be obtained by letting $T = [\ln \bar{X}(n) - \sum_{i=1}^{n} \ln X_i/n]^{-1}$, using Table 6.21 (see App. 6A) to obtain $\hat{\alpha}$ as a function of T, and letting $\hat{\beta} = \bar{X}(n)/\hat{\alpha}$. [See Choi and Wette (1969) for the derivation of this procedure and of Table 6.21.]

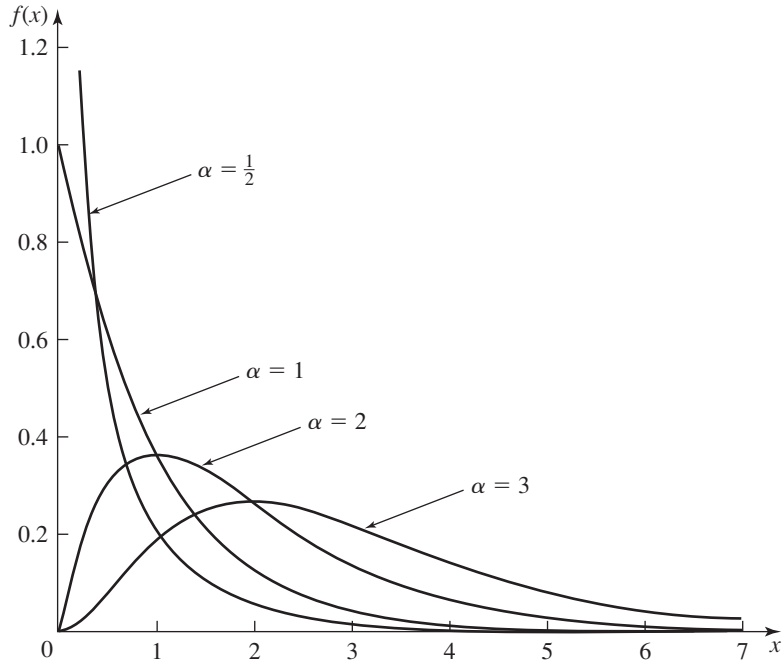


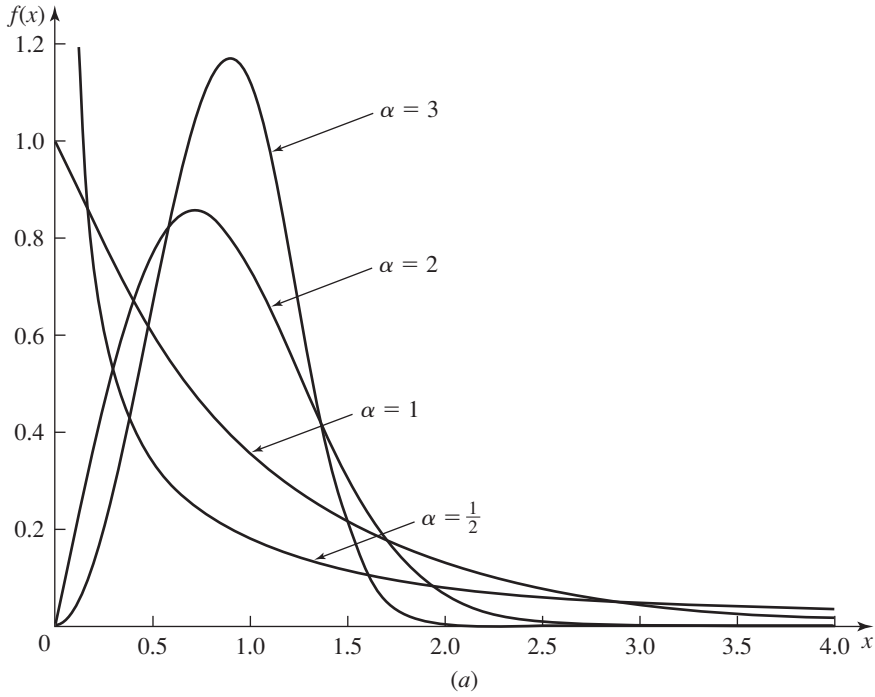
FIGURE 6.7
gamma(α, 1) density functions. (*continued*)

TABLE 6.3 (*continued*)

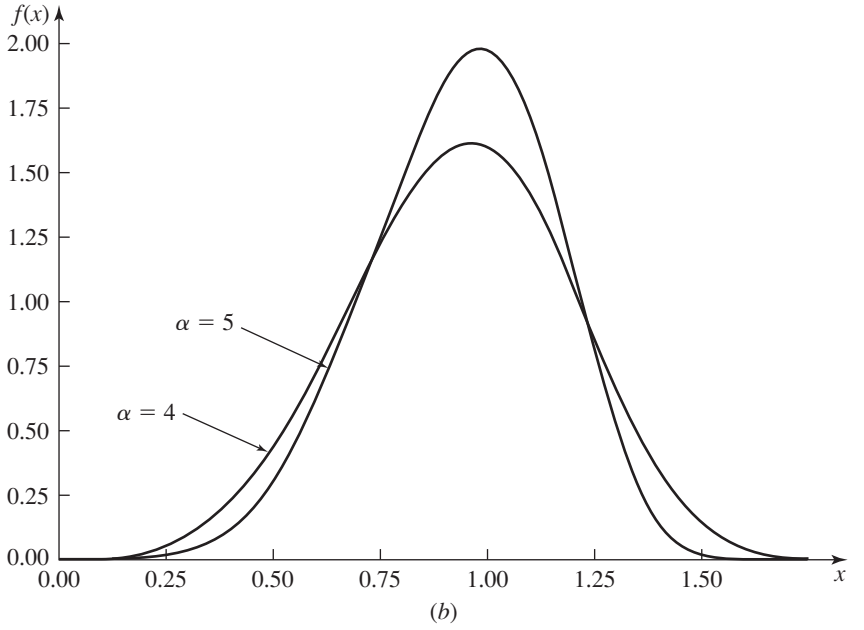
Gamma	**gamma(α, β)**
Comments	1. The expo(β) and gamma(1, β) distributions are the same. 2. For a positive integer m, the gamma(m, β) distribution is called the m-Erlang(β) distribution. 3. The chi-square distribution with k df is the same as the gamma($k/2$, 2) distribution. 4. If $X_1, X_2, \ldots, X_m$ are independent random variables with $X_i \sim$ gamma(α_i, β), then $X_1 + X_2 + \cdots + X_m \sim$ gamma($\alpha_1 + \alpha_2 + \cdots + \alpha_m$, β). 5. If X_1 and X_2 are independent random variables with $X_i \sim$ gamma(α_i, β), then $X_1/(X_1 + X_2) \sim$ beta(α_1, α_2). 6. $X \sim$ gamma(α, β) if and only if $Y = 1/X$ has a Pearson type V distribution with shape and scale parameters α and $1/\beta$, denoted PT5(α, $1/\beta$). 7. $\lim_{x\to 0} f(x) = \begin{cases} \infty & \text{if } \alpha < 1 \\ \dfrac{1}{\beta} & \text{if } \alpha = 1 \\ 0 & \text{if } \alpha > 1 \end{cases}$

Weibull	**Weibull(α, β)**
Possible applications	Time to complete some task, time to failure of a piece of equipment; used as a rough model in the absence of data (see Sec. 6.11)
Density (see Fig. 6.8)	$f(x) = \begin{cases} \alpha\beta^{-\alpha}x^{\alpha-1}e^{-(x/\beta)^{\alpha}} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$
Distribution	$F(x) = \begin{cases} 1 - e^{-(x/\beta)^{\alpha}} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$
Parameters	Shape parameter $\alpha > 0$, scale parameter $\beta > 0$
Range	$[0, \infty)$
Mean	$\dfrac{\beta}{\alpha}\Gamma\left(\dfrac{1}{\alpha}\right)$
Variance	$\dfrac{\beta^2}{\alpha}\left\{2\Gamma\left(\dfrac{2}{\alpha}\right) - \dfrac{1}{\alpha}\left[\Gamma\left(\dfrac{1}{\alpha}\right)\right]^2\right\}$
Mode	$\begin{cases} \beta\left(\dfrac{\alpha-1}{\alpha}\right)^{1/\alpha} & \text{if } \alpha \geq 1 \\ 0 & \text{if } \alpha < 1 \end{cases}$
MLE	The following two equations must be satisfied: $\dfrac{\sum_{i=1}^{n} X_i^{\hat{\alpha}} \ln X_i}{\sum_{i=1}^{n} X_i^{\hat{\alpha}}} - \dfrac{1}{\hat{\alpha}} = \dfrac{\sum_{i=1}^{n} \ln X_i}{n}, \qquad \hat{\beta} = \left(\dfrac{\sum_{i=1}^{n} X_i^{\hat{\alpha}}}{n}\right)^{1/\hat{\alpha}}$ The first can be solved for $\hat{\alpha}$ numerically by Newton's method, and the second equation then gives $\hat{\beta}$ directly. The general recursive step for the Newton iterations is $\hat{\alpha}_{k+1} = \hat{\alpha}_k + \dfrac{A + 1/\hat{\alpha}_k - C_k/B_k}{1/\hat{\alpha}_k^2 + (B_k H_k - C_k^2)/B_k^2}$

TABLE 6.3 (*continued*)



(*a*)



(*b*)

FIGURE 6.8
Weibull(α, 1) density functions. (*continued*)

TABLE 6.3 (*continued*)

Weibull	**Weibull(α, β)**
	where $$A = \frac{\sum_{i=1}^{n} \ln X_i}{n}, \qquad B_k = \sum_{i=1}^{n} X_i^{\hat{\alpha}} k, \qquad C_k = \sum_{i=1}^{n} X_i^{\hat{\alpha}} k \ln X_i$$ and $$H_k = \sum_{i=1}^{n} X_i^{\hat{\alpha}} k (\ln X_i)^2$$ [See Thoman, Bain, and Antle (1969) for these formulas, as well as for confidence intervals on the true α and β.] As a starting point for the iterations, the estimate $$\hat{\alpha}_0 = \left\{ \frac{(6/\pi^2)\left[\sum_{i=1}^{n} (\ln X_i)^2 - \left(\sum_{i=1}^{n} \ln X_i\right)^2 \Big/ n\right]}{n-1} \right\}^{-1/2}$$ [due to Menon (1963) and suggested in Thoman, Bain, and Antle (1969)] may be used. With this choice of $\hat{\alpha}_0$, it was reported in Thoman, Bain, and Antle (1969) that an average of only 3.5 Newton iterations were needed to achieve four-place accuracy.
Comments	1. The expo(β) and Weibull(1, β) distributions are the same. 2. $X \sim$ Weibull(α, β) if and only if $X^\alpha \sim$ expo(β^α) (see Prob. 6.2). 3. The (natural) logarithm of a Weibull random variable has a distribution known as the *extreme-value* or *Gumbel distribution* [see Averill M. Law & Associates (2013), Lawless (2003), and Prob. 8.1(*b*)]. 4. The Weibull(2, β) distribution is also called a *Rayleigh distribution* with parameter β, denoted Rayleigh(β). If Y and Z are independent normal random variables with mean 0 and variance β^2 (see the normal distribution), then $X = (Y^2 + Z^2)^{1/2} \sim$ Rayleigh($2^{1/2}\beta$). 5. As $\alpha \to \infty$, the Weibull distribution becomes degenerate at β. Thus, Weibull densities for large α have a sharp peak at the mode. 6. The Weibull distribution has a negative skewness when $\alpha > 3.6$ [see Fig. 6.8(*b*)]. 7. $$\lim_{x \to 0} f(x) = \begin{cases} \infty & \text{if } \alpha < 1 \\ \dfrac{1}{\beta} & \text{if } \alpha = 1 \\ 0 & \text{if } \alpha > 1 \end{cases}$$

Normal	**N(μ, σ^2)**
Possible applications	Errors of various types, e.g., in the impact point of a bomb; quantities that are the sum of a large number of other quantities (by virtue of central limit theorems)
Density (see Fig. 6.9)	$f(x) = \dfrac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/(2\sigma^2)}$ for all real numbers x
Distribution	No closed form
Parameters	Location parameter $\mu \in (-\infty, \infty)$, scale parameter $\sigma > 0$
Range	$(-\infty, \infty)$

TABLE 6.3 (*continued*)

Normal	**N(μ, σ^2)**
Mean	μ
Variance	σ^2
Mode	μ
MLE	$\hat{\mu} = \overline{X}(n), \qquad \hat{\sigma} = \left[\frac{n-1}{n} S^2(n)\right]^{1/2}$
Comments	1. If two jointly distributed normal random variables are uncorrelated, they are also independent. For distributions other than normal, this implication is not true in general. 2. Suppose that the joint distribution of $X_1, X_2, \ldots, X_m$ is multivariate normal and let $\mu_i = E(X_i)$ and $C_{ij} = \text{Cov}(X_i, X_j)$. Then for any real numbers a, b_1, $b_2, \ldots, b_m$, the random variable $a + b_1X_1 + b_2X_2 + \cdots + b_mX_m$ has a normal distribution with mean $\mu = a + \sum_{i=1}^{m} b_i\mu_i$ and variance $$\sigma^2 = \sum_{i=1}^{m}\sum_{j=1}^{m} b_i b_j C_{ij}$$ Note that we need *not* assume independence of the X_i's. If the X_i's *are* independent, then $$\sigma^2 = \sum_{i=1}^{m} b_i^2 \,\text{Var}(X_i)$$ 3. The N(0, 1) distribution is often called the *standard* or *unit normal distribution.* 4. If $X_1, X_2, \ldots, X_k$ are independent standard normal random variables, then $X_1^2 + X_2^2 + \cdots + X_k^2$ has a chi-square distribution with k df, which is also the gamma($k/2$, 2) distribution.

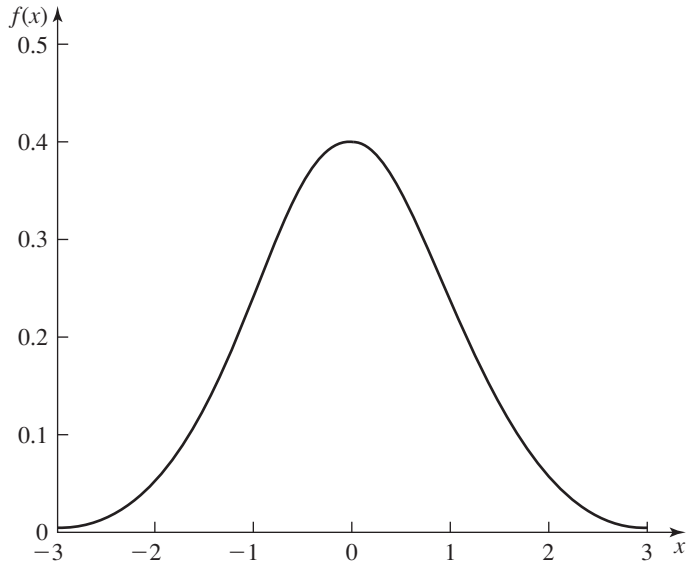


FIGURE 6.9
N(0, 1) density function. (*continued*)

TABLE 6.3 (*continued*)

Normal	**N(μ, σ^2)**
	5. If $X \sim$ N(μ, σ^2), then e^X has the *lognormal distribution* with scale parameter e^μ and shape parameter σ, denoted LN(μ, σ^2). 6. If $X \sim$ N(0, 1), if Y has a chi-square distribution with k df, and if X and Y are independent, then $X/\sqrt{Y/k}$ has a t distribution with k df (sometimes called *Student's t distribution*). 7. If the normal distribution is used to represent a nonnegative quantity (e.g., time), then its density should be truncated at $x = 0$ (see Sec. 6.8). 8. As $\sigma \to 0$, the normal distribution becomes degenerate at μ.

Lognormal	**LN(μ, σ^2)**
Possible applications	Time to perform some task [density takes on shapes similar to gamma(α, β) and Weibull(α, β) densities for $\alpha > 1$, but can have a large "spike" close to $x = 0$ that is often useful]; quantities that are the product of a large number of other quantities (by virtue of central limit theorem); used as a rough model in the absence of data (see Sec. 6.11)
Density (see Fig. 6.10)	$f(x) = \begin{cases} \dfrac{1}{x\sqrt{2\pi\sigma^2}} \exp \dfrac{-(\ln x - \mu)^2}{2\sigma^2} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$
Distribution	No closed form
Parameters	Shape parameter $\sigma > 0$, scale parameter $e^\mu > 0$
Range	$[0, \infty)$
Mean	$e^{\mu+\sigma^2/2}$
Variance	$e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)$

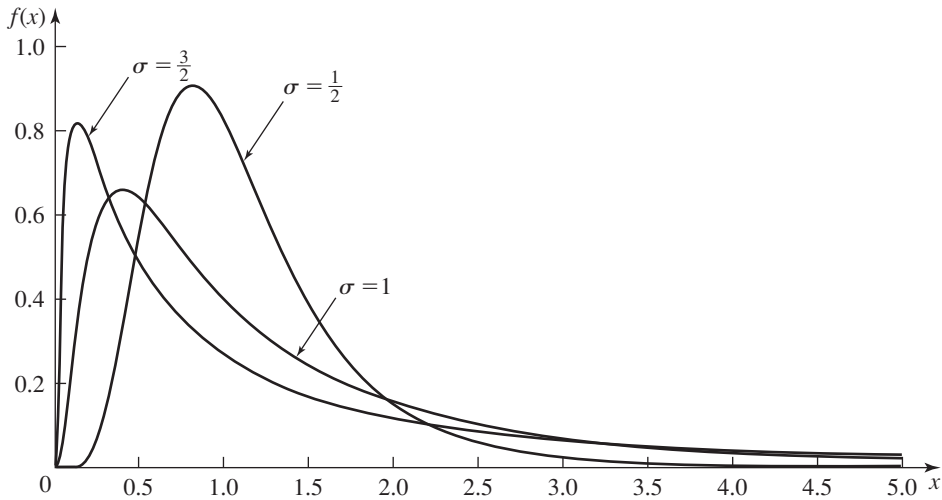


FIGURE 6.10
LN(0, σ^2) density functions.

TABLE 6.3 (*continued*)

Lognormal	**LN(μ, σ^2)**
Mode	$e^{\mu-\sigma^2}$
MLE	$\hat{\mu} = \dfrac{\sum_{i=1}^{n} \ln X_i}{n}$, $\hat{\sigma} = \left[\dfrac{\sum_{i=1}^{n} (\ln X_i - \hat{\mu})^2}{n}\right]^{1/2}$, MLE for scale parameter $= e^{\hat{\mu}}$
Comments	1. $X \sim \text{LN}(\mu, \sigma^2)$ if and only if $\ln X \sim \text{N}(\mu, \sigma^2)$. Thus, if one has data X_1, $X_2, \ldots, X_n$ that are thought to be lognormal, the logarithms of the data points $\ln X_1$, $\ln X_2, \ldots, \ln X_n$ can be treated as normally distributed data for purposes of hypothesizing a distribution, parameter estimation, and goodness-of-fit testing. 2. As $\sigma \to 0$, the lognormal distribution becomes degenerate at e^{μ}. Thus, lognormal densities for small σ have a sharp peak at the mode. 3. $\lim_{x \to 0} f(x) = 0$, regardless of the parameter values.

Beta	**beta(α_1, α_2)**
Possible applications	Used as a rough model in the absence of data (see Sec. 6.11); distribution of a random proportion, such as the proportion of defective items in a shipment; time to complete a task, e.g., in a PERT network
Density (see Fig. 6.11)	$f(x) = \begin{cases} \dfrac{x^{\alpha_1-1}(1-x)^{\alpha_2-1}}{B(\alpha_1,\alpha_2)} & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$ where $B(\alpha_1, \alpha_2)$ is the *beta function,* defined by $B(z_1, z_2) = \int_0^1 t^{z_1-1}(1-t)^{z_2-1}\,dt$ for any real numbers $z_1 > 0$ and $z_2 > 0$. Some properties of the beta function: $B(z_1, z_2) = B(z_2, z_1), \qquad B(z_1, z_2) = \dfrac{\Gamma(z_1)\Gamma(z_2)}{\Gamma(z_1 + z_2)}$
Distribution	No closed form, in general. If either α_1 or α_2 is a positive integer, a binomial expansion can be used to obtain $F(x)$, which will be a polynomial in x, and the powers of x will be, in general, positive real numbers ranging from 0 through $\alpha_1 + \alpha_2 - 1$.
Parameters	Shape parameters $\alpha_1 > 0$ and $\alpha_2 > 0$
Range	[0, 1]
Mean	$\dfrac{\alpha_1}{\alpha_1 + \alpha_2}$
Variance	$\dfrac{\alpha_1\alpha_2}{(\alpha_1 + \alpha_2)^2(\alpha_1 + \alpha_2 + 1)}$
Mode	$\begin{cases} \dfrac{\alpha_1 - 1}{\alpha_1 + \alpha_2 - 2} & \text{if } \alpha_1 > 1, \alpha_2 > 1 \\ 0 \text{ and } 1 & \text{if } \alpha_1 < 1, \alpha_2 < 1 \\ 0 & \text{if } (\alpha_1 < 1, \alpha_2 \geq 1) \quad \text{or} \quad \text{if } (\alpha_1 = 1, \alpha_2 > 1) \\ 1 & \text{if } (\alpha_1 \geq 1, \alpha_2 < 1) \quad \text{or} \quad \text{if } (\alpha_1 > 1, \alpha_2 = 1) \\ \text{does not uniquely exist} & \text{if } \alpha_1 = \alpha_2 = 1 \end{cases}$

(*continued*)

TABLE 6.3 (*continued*)

Beta	**beta(α_1, α_2)**
MLE	The following two equations must be satisfied: $$\Psi(\hat{\alpha}_1) - \Psi(\hat{\alpha}_1 + \hat{\alpha}_2) = \ln G_1, \qquad \Psi(\hat{\alpha}_2) - \Psi(\hat{\alpha}_1 + \hat{\alpha}_2) = \ln G_2$$ where Ψ is the digamma function, $G_1 = (\prod_{i=1}^{n} X_i)^{1/n}$, and $G_2 = [\prod_{i=1}^{n}(1 - X_i)]^{1/n}$ [see Gnanadesikan, Pinkham, and Hughes (1967)]; note that $G_1 + G_2 \leq 1$. These equations could be solved numerically [see Beckman and Tietjen (1978)], or approximations to $\hat{\alpha}_1$ and $\hat{\alpha}_2$ can be obtained from Table 6.22 (see App. 6A), which was computed for particular (G_1, G_2) pairs by modifications of the methods in Beckman and Tietjen (1978).
Comments	1. The U(0, 1) and beta(1, 1) distributions are the same. 2. If X_1 and X_2 are independent random variables with $X_i \sim$ gamma(α_i, β), then $X_1/(X_1 + X_2) \sim$ beta(α_1, α_2). 3. A beta random variable X on [0, 1] can be rescaled and relocated to obtain a beta random variable on [a, b] of the same shape by the transformation $a + (b - a)X$. 4. $X \sim$ beta(α_1, α_2) if and only if $1 - X \sim$ beta(α_2, α_1). 5. $X \sim$ beta(α_1, α_2) if and only if $Y = X/(1 - X)$ has a Pearson type VI distribution with shape parameters α_1, α_2 and scale parameter 1, denoted PT6(α_1, α_2, 1).

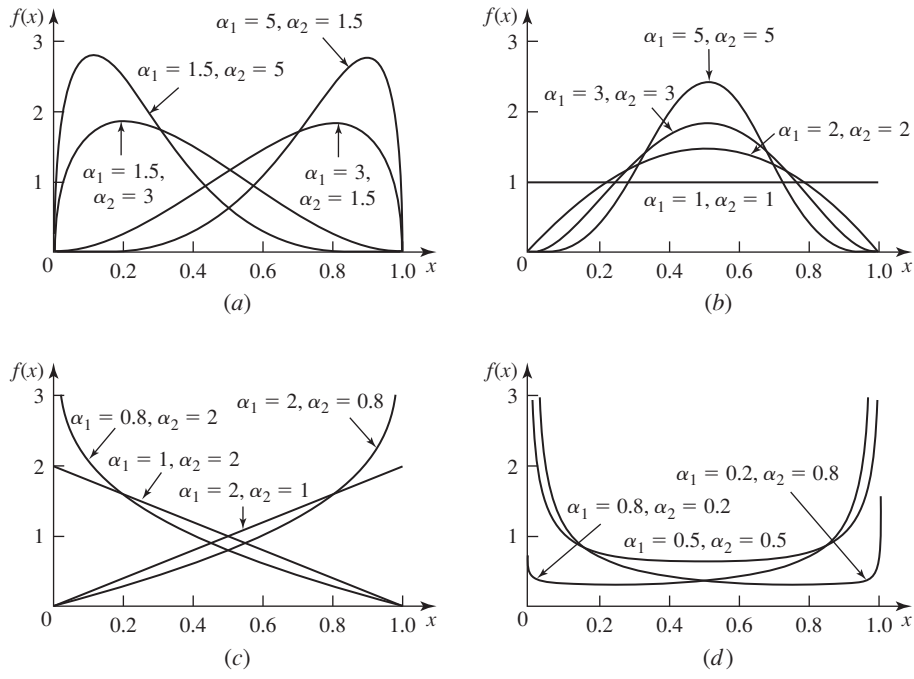


FIGURE 6.11
beta(α_1, α_2) density functions.

TABLE 6.3 (*continued*)

Beta	**beta(α_1, α_2)**
	6. The beta(1, 2) density is a left triangle, and the beta(2, 1) density is a right triangle.
	7. $\lim_{x\to 0} f(x) = \begin{cases} \infty & \text{if } \alpha_1 < 1 \\ \alpha_2 & \text{if } \alpha_1 = 1, \\ 0 & \text{if } \alpha_1 > 1 \end{cases}$ $\lim_{x\to 1} f(x) = \begin{cases} \infty & \text{if } \alpha_2 < 1 \\ \alpha_1 & \text{if } \alpha_2 = 1 \\ 0 & \text{if } \alpha_2 > 1 \end{cases}$
	8. The density is symmetric about $x = \frac{1}{2}$ if and only if $\alpha_1 = \alpha_2$. Also, the mean and the mode are equal if and only if $\alpha_1 = \alpha_2 > 1$.
Pearson type V	**PT5(α, β)**
Possible applications	Time to perform some task (density takes on shapes similar to lognormal, but can have a larger "spike" close to $x = 0$)
Density (see Fig. 6.12)	$f(x) = \begin{cases} \dfrac{x^{-(\alpha+1)}e^{-\beta/x}}{\beta^{-\alpha}\Gamma(\alpha)} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$
Distribution	$F(x) = \begin{cases} 1 - F_G\left(\dfrac{1}{x}\right) & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$ where $F_G(x)$ is the distribution function of a gamma(α, $1/\beta$) random variable
Parameters	Shape parameter $\alpha > 0$, scale parameter $\beta > 0$

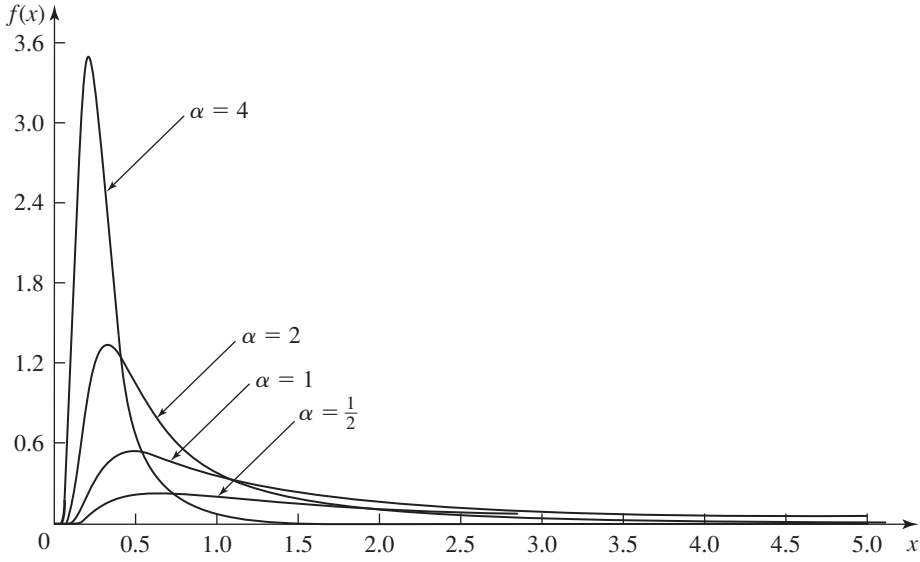


FIGURE 6.12
PT5(α, 1) density functions.

(*continued*)

TABLE 6.3 (*continued*)

Pearson type V	**PT5(α, β)**
Range	$[0, \infty)$
Mean	$\dfrac{\beta}{\alpha - 1}$ for $\alpha > 1$
Variance	$\dfrac{\beta^2}{(\alpha - 1)^2(\alpha - 2)}$ for $\alpha > 2$
Mode	$\dfrac{\beta}{\alpha + 1}$
MLE	If one has data $X_1, X_2, \ldots, X_n$, then fit a gamma(α_G, β_G) distribution to $1/X_1$, $1/X_2, \ldots, 1/X_n$, resulting in the maximum-likelihood estimators $\hat{\alpha}_G$ and $\hat{\beta}_G$. Then the maximum-likelihood estimators for the PT5(α, β) are $\hat{\alpha} = \hat{\alpha}_G$ and $\hat{\beta} = 1/\hat{\beta}_G$ (see comment 1 below).
Comments	1. $X \sim$ PT5(α, β) if and only if $Y = 1/X \sim$ gamma(α, $1/\beta$). Thus, the Pearson type V distribution is sometimes called the *inverted gamma distribution*. 2. Note that the mean and variance exist only for certain values of the shape parameter.

Pearson type VI	**PT6(α_1, α_2, β)**
Possible applications	Time to perform some task
Density (see Fig. 6.13)	$f(x) = \begin{cases} \dfrac{(x/\beta)^{\alpha_1 - 1}}{\beta B(\alpha_1, \alpha_2)[1 + (x/\beta)]^{\alpha_1 + \alpha_2}} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$
Distribution	$F(x) = \begin{cases} F_B\left(\dfrac{x}{x + \beta}\right) & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$ where $F_B(x)$ is the distribution function of a beta(α_1, α_2) random variable
Parameters	Shape parameters $\alpha_1 > 0$ and $\alpha_2 > 0$, scale parameter $\beta > 0$
Range	$[0, \infty)$
Mean	$\dfrac{\beta\alpha_1}{\alpha_2 - 1}$ for $\alpha_2 > 1$
Variance	$\dfrac{\beta^2\alpha_1(\alpha_1 + \alpha_2 - 1)}{(\alpha_2 - 1)^2(\alpha_2 - 2)}$ for $\alpha_2 > 2$
Mode	$\begin{cases} \dfrac{\beta(\alpha_1 - 1)}{\alpha_2 + 1} & \text{if } \alpha_1 \geq 1 \\ 0 & \text{otherwise} \end{cases}$
MLE	If one has data $X_1, X_2, \ldots, X_n$ that are thought to be PT6(α_1, α_2, 1), then fit a beta(α_1, α_2) distribution to $X_i/(1 + X_i)$ for $i = 1, 2, \ldots, n$, resulting in the maximum-likelihood estimators $\hat{\alpha}_1$ and $\hat{\alpha}_2$. Then the maximum-likelihood estimators for the PT6(α_1, α_2, 1) (note that $\beta = 1$) distribution are also $\hat{\alpha}_1$ and $\hat{\alpha}_2$ (see comment 1 below).

TABLE 6.3 (*continued*)

Pearson type VI	**PT6(α_1, α_2, β)**
Comments	1. $X \sim$ PT6(α_1, α_2, 1) if and only if $Y = X/(1 + X) \sim$ beta(α_1, α_2). 2. If X_1 and X_2 are independent random variables with $X_1 \sim$ gamma(α_1, β) and $X_2 \sim$ gamma(α_2, 1), then $Y = X_1/X_2 \sim$ PT6(α_1, α_2, β) (see Prob. 6.3). 3. Note that the mean and variance exist only for certain values of the shape parameter α_2.

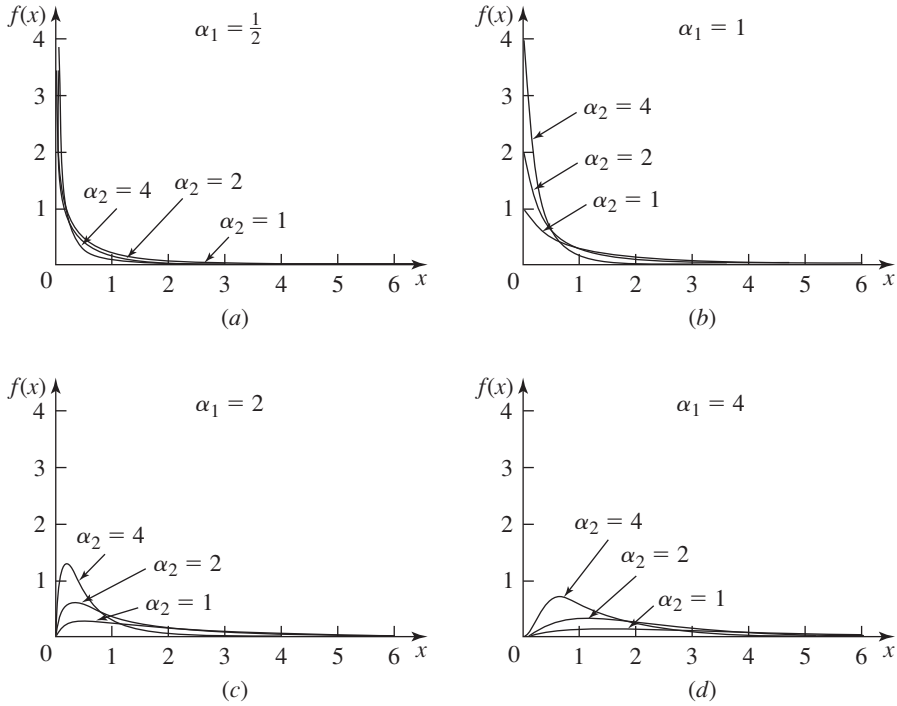


FIGURE 6.13
PT6(α_1, α_2, 1) density functions.

Log-logistic	**LL(α, β)**
Possible applications	Time to perform some task
Density (see Fig. 6.14)	$f(x) = \begin{cases} \dfrac{\alpha(x/\beta)^{\alpha-1}}{\beta[1 + (x/\beta)^{\alpha}]^2} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$
Distribution	$F(x) = \begin{cases} \dfrac{1}{1 + (x/\beta)^{-\alpha}} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$
Parameters	Shape parameter $\alpha > 0$, scale parameter $\beta > 0$

(*continued*)

TABLE 6.3 (*continued*)

Log-logistic	**LL(α, β)**
Range	$[0, \infty)$
Mean	$\beta\theta$ cosecant(θ) for $\alpha > 1$, where $\theta = \pi/\alpha$
Variance	$\beta^2\theta\{2 \text{ cosecant}(2\theta) - \theta[\text{cosecant}(\theta)]^2\}$ for $\alpha > 2$
Mode	$\begin{cases} \beta\left(\dfrac{\alpha - 1}{\alpha + 1}\right)^{1/\alpha} & \text{if } \alpha > 1 \\ 0 & \text{otherwise} \end{cases}$
MLE	Let $Y_i = \ln X_i$. Solve the following two equations for $\hat{a}$ and $\hat{b}$: $$\sum_{i=1}^{n}\left[1 + e^{(Y_i - \hat{a})/\hat{b}}\right]^{-1} = \frac{n}{2} \quad (6.1)$$ and $$\sum_{i=1}^{n}\left(\frac{Y_i - \hat{a}}{\hat{b}}\right)\frac{1 - e^{(Y_i - \hat{a})/\hat{b}}}{1 + e^{(Y_i - \hat{a})/\hat{b}}} = n \quad (6.2)$$ Then the MLEs for the log-logistic distribution are $\hat{\alpha} = 1/\hat{a}$ and $\hat{\beta} = e^{\hat{b}}$. Johnson, Kotz, and Balakrishnan (1995, chap. 23) suggest solving Eqs. (6.1) and (6.2) by using Newton's method.

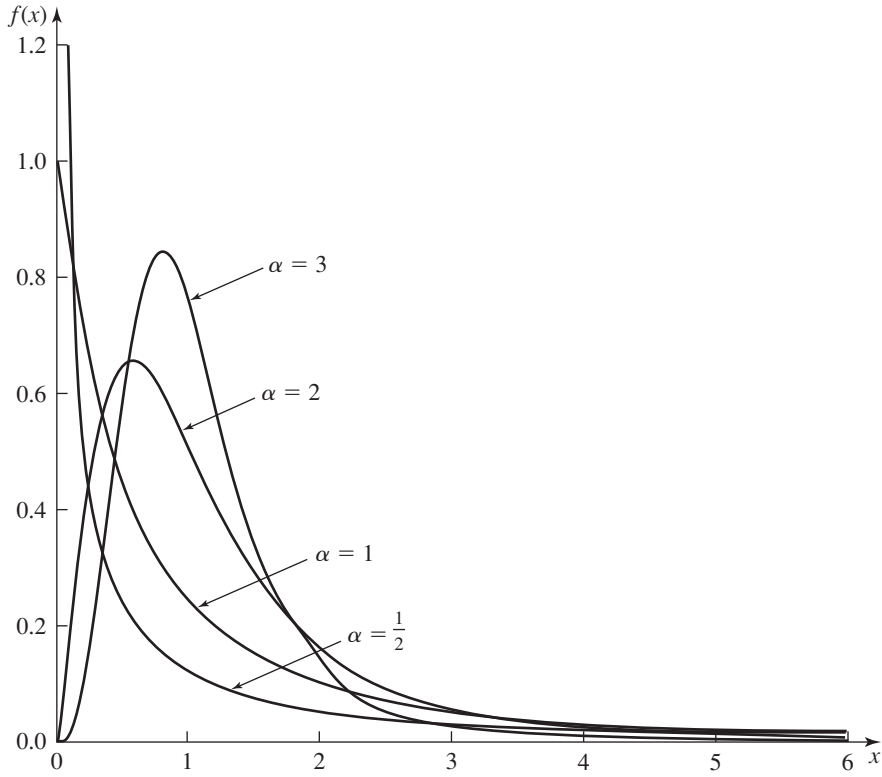


FIGURE 6.14
LL(α, 1) density functions.

TABLE 6.3 (*continued*)

Log-logistic	**LL(α, β)**
Comment	$X \sim$ LL(α, β) if and only if ln X is distributed as a logistic distribution (see Prob. 8.1) with location parameter ln β and scale parameter $1/\alpha$. Thus, if one has data $X_1, X_2, \ldots, X_n$ that are thought to be log-logistic, the logarithms of the data points ln X_1, ln X_2, . . . , ln X_n can be treated as having a logistic distribution for purposes of hypothesizing a distribution, parameter estimation, and goodness-of-fit testing.

Johnson S_B	**JSB(α_1, α_2, a, b)**
Density (see Fig. 6.15)	$f(x) = \begin{cases} \dfrac{\alpha_2(b-a)}{(x-a)(b-x)\sqrt{2\pi}}\, e^{-\frac{1}{2}\left[\alpha_1+\alpha_2 \ln\left(\frac{x-a}{b-x}\right)\right]^2} & \text{if } a < x < b \\ 0 & \text{otherwise} \end{cases}$
Distribution	$F(x) = \begin{cases} \Phi\left[\alpha_1 + \alpha_2 \ln\left(\dfrac{x-a}{b-x}\right)\right] & \text{if } a < x < b \\ 0 & \text{otherwise} \end{cases}$
	where $\Phi(x)$ is the distribution function of a normal random variable with $\mu = 0$ and $\sigma^2 = 1$
Parameters	Location parameter $a \in (-\infty, \infty)$, scale parameter $b - a$ ($b > a$), shape parameters $\alpha_1 \in (-\infty, \infty)$ and $\alpha_2 > 0$
Range	$[a, b]$
Mean	All moments exist but are extremely complicated [see Johnson, Kotz, and Balakrishnan (1994, p. 35)]
Mode	The density is bimodal when $\alpha_2 < \dfrac{1}{\sqrt{2}}$ and $$\lvert\alpha_1\rvert < \frac{\sqrt{1-2\alpha_2^2}}{\alpha_2} - 2\alpha_2 \tanh^{-1}\left(\sqrt{1-2\alpha_2^2}\right)$$ [$\tanh^{-1}$ is the inverse hyperbolic tangent]; otherwise the density is unimodal. The equation satisfied by any mode x, other than at the endpoints of the range, is $$\frac{2(x-a)}{b-a} = 1 + \alpha_1\alpha_2 + \alpha_2^2 \ln\left(\frac{x-a}{b-x}\right)$$
Comments	1. $X \sim$ JSB(α_1, α_2, a, b) if and only if $$\alpha_1 + \alpha_2 \ln\left(\frac{X-a}{b-X}\right) \sim \mathrm{N}(0, 1)$$ 2. The density function is skewed to the left, symmetric, or skewed to the right if $\alpha_1 > 0$, $\alpha_1 = 0$, or $\alpha_1 < 0$, respectively. 3. $\lim_{x\to a} f(x) = \lim_{x\to b} f(x) = 0$ for all values of α_1 and α_2. 4. The four parameters may be estimated using a number of methods [see, for example, Swain, Venkatraman, and Wilson (1988) and Slifker and Shapiro (1980)].

(*continued*)

TABLE 6.3 (*continued*)

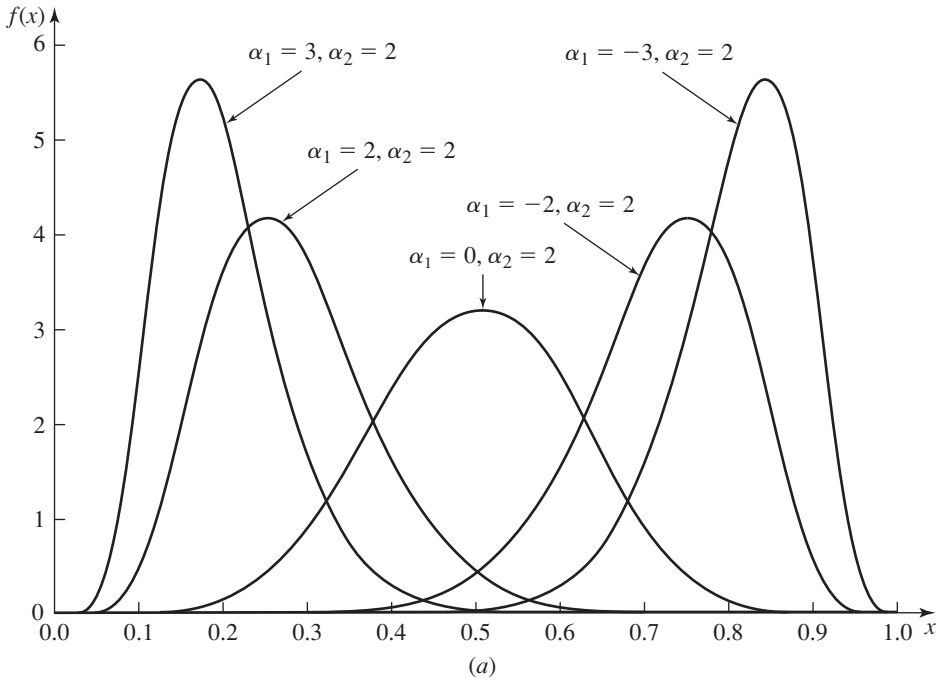


FIGURE 6.15
JSB(α_1, α_2, 0, 1) density functions.

TABLE 6.3 (*continued*)

Johnson S_U	**JSU(α_1, α_2, γ, β)**
Density (see Fig. 6.16)	$f(x) = \dfrac{\alpha_2}{\sqrt{2\pi}\sqrt{(x-\gamma)^2+\beta^2}}\, e^{-\frac{1}{2}\left\{\alpha_1+\alpha_2 \ln\left[\frac{x-\gamma}{\beta}+\sqrt{\left(\frac{x-\gamma}{\beta}\right)^2+1}\right]\right\}^2}$ for $-\infty < x < \infty$
Distribution	$F(x) = \Phi\left\{\alpha_1 + \alpha_2 \ln\left[\dfrac{x-\gamma}{\beta} + \sqrt{\left(\dfrac{x-\gamma}{\beta}\right)^2 + 1}\right]\right\}$ for $-\infty < x < \infty$
Parameters	Location parameter $\gamma \in (-\infty, \infty)$, scale parameter $\beta > 0$, shape parameters $\alpha_1 \in (-\infty, \infty)$ and $\alpha_2 > 0$
Range	$(-\infty, \infty)$
Mean	$\gamma - \beta e^{1/(2\alpha_2^2)} \sinh\left(\dfrac{\alpha_1}{\alpha_2}\right)$, where sinh is the hyperbolic sine
Mode	The equation satisfied by the mode, other than at the endpoints of the range, is $\gamma + \beta y$, where y satisfies $y + \alpha_1\alpha_2\sqrt{y^2+1} + \alpha_2^2\sqrt{y^2+1}\,\ln(y + \sqrt{y^2+1}) = 0$
Comments	1. $X \sim$ JSU(α_1, α_2, γ, β) if and only if $\alpha_1 + \alpha_2 \ln\left[\dfrac{X-\gamma}{\beta} + \sqrt{\left(\dfrac{X-\gamma}{\beta}\right)^2 + 1}\right] \sim \mathrm{N}(0, 1)$ 2. The density function is skewed to the left, symmetric, or skewed to the right if $\alpha_1 > 0$, $\alpha_1 = 0$, or $\alpha_1 < 0$, respectively. 3. The four parameters may be estimated by a number of methods [see, for example, Swain, Venkatraman, and Wilson (1988) and Slifker and Shapiro (1980)].

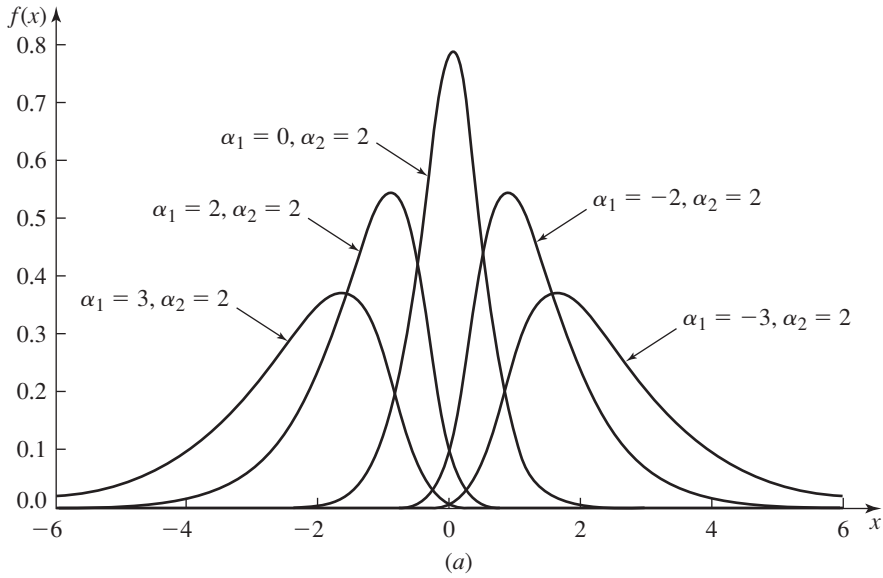


FIGURE 6.16
JSU(α_1, α_2, 0, 1) density functions.

(*continued*)

TABLE 6.3 (*continued*)

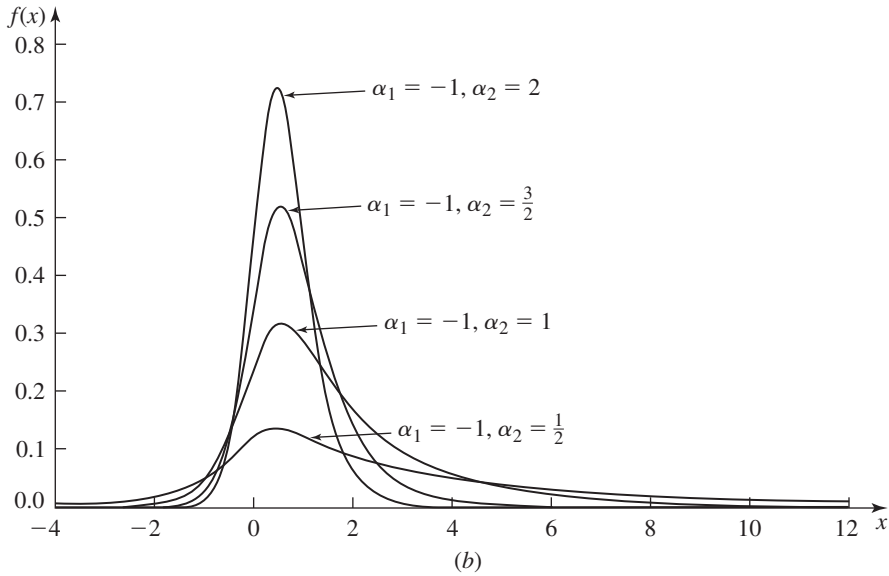


FIGURE 6.16
(*continued*)

Triangular	**triang(*a*, *b*, *m*)**
Possible applications	Used as a rough model in the absence of data (see Sec. 6.11)
Density (see Fig. 6.17)	$f(x) = \begin{cases} \dfrac{2(x-a)}{(b-a)(m-a)} & \text{if } a \le x \le m \\ \dfrac{2(b-x)}{(b-a)(b-m)} & \text{if } m < x \le b \\ 0 & \text{otherwise} \end{cases}$
Distribution	$F(x) = \begin{cases} 0 & \text{if } x < a \\ \dfrac{(x-a)^2}{(b-a)(m-a)} & \text{if } a \le x \le m \\ 1 - \dfrac{(b-x)^2}{(b-a)(b-m)} & \text{if } m < x \le b \\ 1 & \text{if } b < x \end{cases}$
Parameters	a, b, and m real numbers with $a < m < b$. a is a location parameter, $b - a$ is a scale parameter, m is a shape parameter
Range	$[a, b]$
Mean	$\dfrac{a + b + m}{3}$

TABLE 6.3 (*continued*)

Triangular	**triang(a, b, m)**
Variance	$\dfrac{a^2 + b^2 + m^2 - ab - am - bm}{18}$
Mode	m
MLE	Our use of the triangular distribution, as described in Sec. 6.11, is as a rough model when there are no data. Thus, MLEs are not relevant.
Comment	The limiting cases as $m \to b$ and $m \to a$ are called the *right triangular* and *left triangular distributions*, respectively, and are discussed in Prob. 8.7. For $a = 0$ and $b = 1$, both the left and right triangular distributions are special cases of the beta distribution.

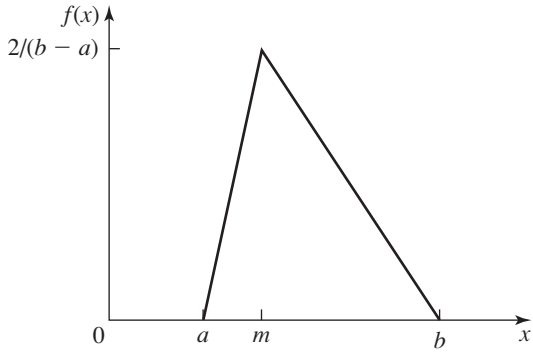


FIGURE 6.17
triang(a, b, m) density function.

6.2.3 Discrete Distributions

The descriptions of the six discrete distributions in Table 6.4 follow the same pattern as for the continuous distributions in Table 6.3.

6.2.4 Empirical Distributions

In some situations we might want to use the observed data themselves to specify directly (in some sense) a distribution, called an *empirical distribution,* from which random values are generated during the simulation, rather than fitting a theoretical distribution to the data. For example, it could happen that we simply cannot find a theoretical distribution that fits the data adequately (see Secs. 6.4 through 6.6). This section explores ways of specifying empirical distributions.

For continuous random variables, the type of empirical distribution that can be defined depends on whether we have the actual values of the individual original observations $X_1, X_2, \ldots, X_n$ rather than only the *number* of X_i's that fall into each of several specified intervals. (The latter case is called *grouped data* or *data in the form of a histogram*.) If the original data are available, we can define a continuous, piecewise-linear distribution function F by first sorting the X_i's into increasing

TABLE 6.4
Discrete distributions

Bernoulli	**Bernoulli(p)**
Possible applications	Random occurrence with two possible outcomes; used to generate other discrete random variates, e.g., binomial, geometric, and negative binomial
Mass (see Fig. 6.18)	$p(x) = \begin{cases} 1-p & \text{if } x = 0 \\ p & \text{if } x = 1 \\ 0 & \text{otherwise} \end{cases}$
Distribution	$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1-p & \text{if } 0 \le x < 1 \\ 1 & \text{if } 1 \le x \end{cases}$
Parameter	$p \in (0, 1)$
Range	$\{0, 1\}$
Mean	p
Variance	$p(1-p)$
Mode	$\begin{cases} 0 & \text{if } p < \frac{1}{2} \\ 0 \text{ and } 1 & \text{if } p = \frac{1}{2} \\ 1 & \text{if } p > \frac{1}{2} \end{cases}$
MLE	$\hat{p} = \overline{X}(n)$
Comments	1. A Bernoulli(p) random variable X can be thought of as the outcome of an experiment that either "fails" or "succeeds." If the probability of success is p, and we let $X = 0$ if the experiment fails and $X = 1$ if it succeeds, then $X \sim$ Bernoulli(p). Such an experiment, often called a *Bernoulli trial,* provides a convenient way of relating several other discrete distributions to the Bernoulli distribution. 2. If t is a positive integer and $X_1, X_2, \ldots, X_t$ are independent Bernoulli(p) random variables, then $X_1 + X_2 + \cdots + X_t$ has the binomial distribution with parameters t and p. Thus, a binomial random variable can be thought of as the number of successes in a fixed number of independent Bernoulli trials. 3. Suppose we begin making independent replications of a Bernoulli trial with probability p of success on each trial. Then the number of failures *before* observing the first success has a geometric distribution with parameter p. For a positive integer s, the number of failures before

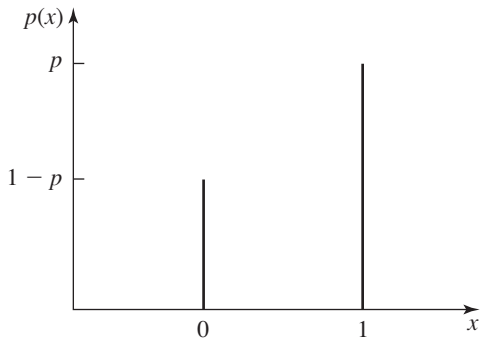


FIGURE 6.18
Bernoulli(p) mass function ($p > 0.5$ here).

TABLE 6.4 (*continued*)

Bernoulli	**Bernoulli(p)**
	observing the sth success has a negative binomial distribution with parameters s and p. 4. The Bernoulli(p) distribution is a special case of the binomial distribution (with $t = 1$ and the same value for p).

Discrete uniform	**DU(i, j)**
Possible applications	Random occurrence with several possible outcomes, each of which is equally likely; used as a "first" model for a quantity that is varying among the integers i through j but about which little else is known
Mass (see Fig. 6.19)	$p(x) = \begin{cases} \dfrac{1}{j - i + 1} & \text{if } x \in \{i, i+1, \ldots, j\} \\ 0 & \text{otherwise} \end{cases}$
Distribution	$F(x) = \begin{cases} 0 & \text{if } x < i \\ \dfrac{\lfloor x \rfloor - i + 1}{j - i + 1} & \text{if } i \le x \le j \\ 1 & \text{if } j < x \end{cases}$ where $\lfloor x \rfloor$ denotes the largest integer $\le x$
Parameters	i and j integers with $i \le j$; i is a location parameter, $j - i$ is a scale parameter
Range	$\{i, i+1, \ldots, j\}$
Mean	$\dfrac{i + j}{2}$
Variance	$\dfrac{(j - i + 1)^2 - 1}{12}$
Mode	Does not uniquely exist
MLE	$\hat{i} = \min_{1 \le k \le n} X_k, \qquad \hat{j} = \max_{1 \le k \le n} X_k$
Comment	The DU(0, 1) and Bernoulli($\frac{1}{2}$) distributions are the same.

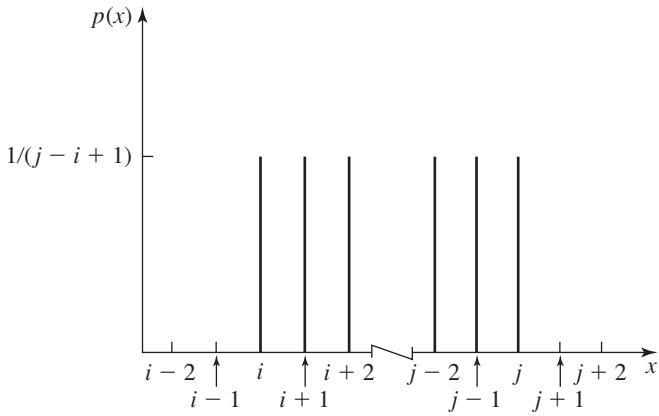


FIGURE 6.19
DU(i, j) mass function.
(*continued*)

TABLE 6.4 (*continued*)

Binomial	**bin(t, p)**
Possible applications	Number of successes in t independent Bernoulli trials with probability p of success on each trial; number of "defective" items in a batch of size t; number of items in a batch (e.g., a group of people) of random size; number of items demanded from an inventory
Mass (see Fig. 6.20)	$p(x) = \begin{cases} \binom{t}{x} p^x (1-p)^{t-x} & \text{if } x \in \{0, 1, \ldots, t\} \\ 0 & \text{otherwise} \end{cases}$ where $\binom{t}{x}$ is the *binomial coefficient*, defined by $\binom{t}{x} = \frac{t!}{x!(t-x)!}$
Distribution	$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ \sum_{i=0}^{\lfloor x \rfloor} \binom{t}{i} p^i (1-p)^{t-i} & \text{if } 0 \le x \le t \\ 1 & \text{if } t < x \end{cases}$
Parameters	t a positive integer, $p \in (0, 1)$
Range	$\{0, 1, \ldots, t\}$

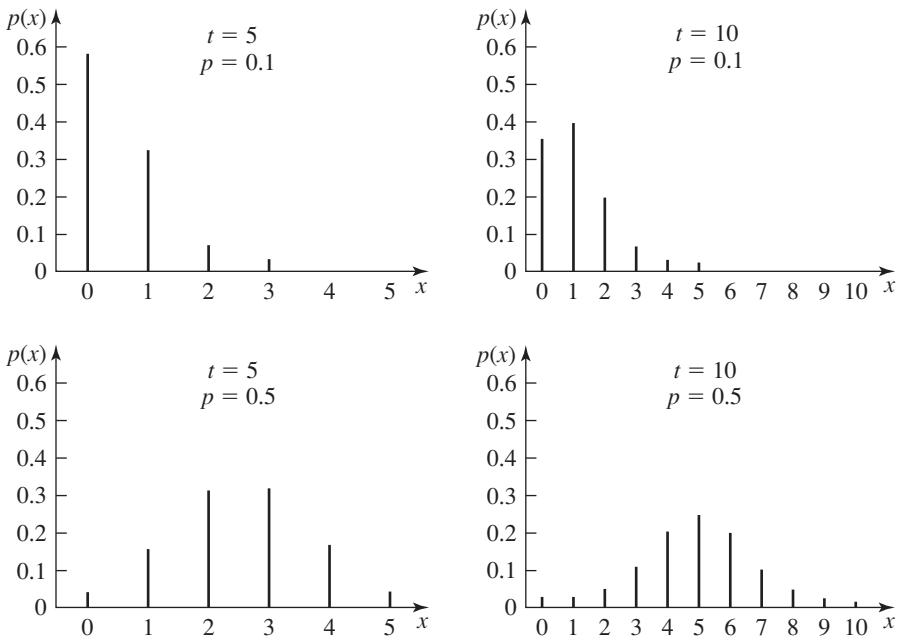


FIGURE 6.20
bin(t, p) mass functions.

TABLE 6.4 (*continued*)

Binomial	**bin(t, p)**
Mean	tp
Variance	$tp(1-p)$
Mode	$\begin{cases} p(t+1)-1 \text{ and } p(t+1) & \text{if } p(t+1) \text{ is an integer} \\ \lfloor p(t+1) \rfloor & \text{otherwise} \end{cases}$
MLE	If t is known, then $\hat{p} = \overline{X}(n)/t$. If both t and p are unknown, then $\hat{t}$ and $\hat{p}$ exist if and only if $\overline{X}(n) > (n-1)S^2(n)/n = V(n)$. Then the following approach could be taken. Let $M = \max_{1 \le i \le n} X_i$, and for $k = 0, 1, \ldots, M$, let f_k be the number of X_i's $\ge k$. Then it can be shown that $\hat{t}$ and $\hat{p}$ are the values for t and p that maximize the function $$g(t,p) = \sum_{k=1}^{M} f_k \ln(t-k+1) + nt\ln(1-p) + n\overline{X}(n)\ln\frac{p}{1-p}$$ subject to the constraints that $t \in \{M, M+1, \ldots\}$ and $0 < p < 1$. It is easy to see that for a fixed value of t, say t_0, the value of p that maximizes $g(t_0, p)$ is $\overline{X}(n)/t_0$, so $\hat{t}$ and $\hat{p}$ are the values of t and $\overline{X}(n)/t$ that lead to the largest value of $g[t, \overline{X}(n)/t]$ for $t \in \{M, M+1, \ldots, M'\}$, where M' is given by [see DeRiggi (1983)] $$M' = \left\lfloor \frac{\overline{X}(n)(M-1)}{1-[V(n)/\overline{X}(n)]} \right\rfloor$$ Note also that $g[t, \overline{X}(n)/t]$ is a unimodal function of t.
Comments	1. If $Y_1, Y_2, \ldots, Y_t$ are independent Bernoulli(p) random variables, then $Y_1 + Y_2 + \cdots + Y_t \sim \text{bin}(t, p)$. 2. If $X_1, X_2, \ldots, X_m$ are independent random variables and $X_i \sim \text{bin}(t_i, p)$, then $X_1 + X_2 + \cdots + X_m \sim \text{bin}(t_1 + t_2 + \cdots + t_m, p)$. 3. The bin($t, p$) distribution is symmetric if and only if $p = \frac{1}{2}$. 4. $X \sim \text{bin}(t, p)$ if and only if $t - X \sim \text{bin}(t, 1-p)$. 5. The bin(1, p) and Bernoulli(p) distributions are the same.

Geometric	**geom(p)**
Possible applications	Number of failures before the first success in a sequence of independent Bernoulli trials with probability p of success on each trial; number of items inspected before encountering the first defective item; number of items in a batch of random size; number of items demanded from an inventory
Mass (see Fig. 6.21)	$p(x) = \begin{cases} p(1-p)^x & \text{if } x \in \{0, 1, \ldots\} \\ 0 & \text{otherwise} \end{cases}$
Distribution	$F(x) = \begin{cases} 1-(1-p)^{\lfloor x \rfloor + 1} & \text{if } x \ge 0 \\ 0 & \text{otherwise} \end{cases}$
Parameter	$p \in (0, 1)$
Range	$\{0, 1, \ldots\}$
Mean	$\dfrac{1-p}{p}$

(*continued*)

TABLE 6.4 (*continued*)

Geometric	**geom(p)**
Variance	$\dfrac{1-p}{p^2}$
Mode	0
MLE	$\hat{p} = \dfrac{1}{\overline{X}(n) + 1}$
Comments	1. If $Y_1, Y_2, \ldots$ is a sequence of independent Bernoulli(p) random variables and $X = \min\{i: Y_i = 1\} - 1$, then $X \sim$ geom(p). 2. If $X_1, X_2, \ldots, X_s$ are independent geom(p) random variables, then $X_1 + X_2 + \cdots + X_s$ has a negative binomial distribution with parameters s and p. 3. The geometric distribution is the discrete analog of the exponential distribution, in the sense that it is the only discrete distribution with the memoryless property (see Prob. 4.31). 4. The geom(p) distribution is a special case of the negative binomial distribution (with $s = 1$ and the same value for p).

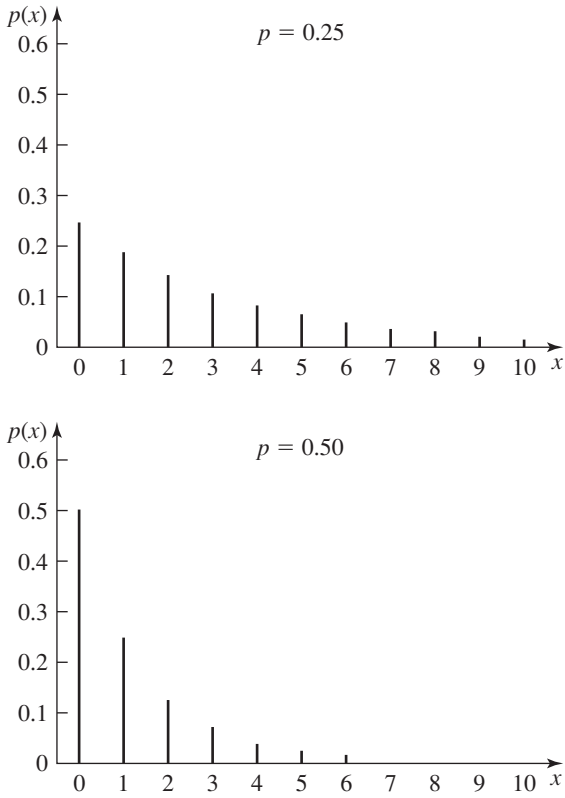


FIGURE 6.21
geom(p) mass functions.

TABLE 6.4 (*continued*)

Negative binomial	**negbin(s, p)**
Possible applications	Number of failures before the sth success in a sequence of independent Bernoulli trials with probability p of success on each trial; number of good items inspected before encountering the sth defective item; number of items in a batch of random size; number of items demanded from an inventory
Mass (see Fig. 6.22)	$p(x) = \begin{cases} \binom{s+x-1}{x} p^s(1-p)^x & \text{if } x \in \{0, 1, \ldots\} \\ 0 & \text{otherwise} \end{cases}$
Distribution	$F(x) = \begin{cases} \sum_{i=0}^{\lfloor x \rfloor} \binom{s+i-1}{i} p^s(1-p)^i & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$
Parameters	s a positive integer, $p \in (0, 1)$
Range	$\{0, 1, \ldots\}$
Mean	$\dfrac{s(1-p)}{p}$
Variance	$\dfrac{s(1-p)}{p^2}$
Mode	Let $y = [s(1-p) - 1]/p$; then $\text{Mode} = \begin{cases} y \text{ and } y+1 & \text{if } y \text{ is an integer} \\ \lfloor y \rfloor + 1 & \text{otherwise} \end{cases}$
MLE	If s is known, then $\hat{p} = s/[\overline{X}(n) + s]$. If both s and p are unknown, then $\hat{s}$ and $\hat{p}$ exist if and only if $V(n) = (n-1)S^2(n)/n > \overline{X}(n)$. Let $M = \max_{1 \leq i \leq n} X_i$, and for $k = 0, 1, \ldots, M$, let f_k be the number of X_i's $\geq k$. Then we can show that $\hat{s}$ and $\hat{p}$ are the values for s and p that maximize the function $$h(s, p) = \sum_{k=1}^{M} f_k \ln(s + k - 1) + ns \ln p + n\overline{X}(n) \ln(1-p)$$ subject to the constraints that $s \in \{1, 2, \ldots\}$ and $0 < p < 1$. For a fixed value of s, say s_0, the value of p that maximizes $h(s_0, p)$ is $s_0/[\overline{X}(n) + s_0]$, so that we could examine $h(1, 1/[\overline{X}(n) + 1])$, $h(2, 2/[\overline{X}(n) + 2])$, ... Then $\hat{s}$ and $\hat{p}$ are chosen to be the values of s and $s/[\overline{X}(n) + s]$ that lead to the biggest observed value of $h(s, s/[\overline{X}(n) + s])$. However, since $h(s, s/[\overline{X}(n) + s])$ is a unimodal function of s [see Levin and Reeds (1977)], it is clear when to terminate the search.
Comments	1. If $Y_1, Y_2, \ldots, Y_s$ are independent geom(p) random variables, then $Y_1 + Y_2 + \cdots + Y_s \sim$ negbin(s, p). 2. If $Y_1, Y_2, \ldots$ is a sequence of independent Bernoulli(p) random variables and $X = \min\{i: \sum_{j=1}^{i} Y_j = s\} - s$, then $X \sim$ negbin(s, p). 3. If $X_1, X_2, \ldots, X_m$ are independent random variables and $X_i \sim$ negbin(s_i, p), then $X_1 + X_2 + \cdots + X_m \sim$ negbin($s_1 + s_2 + \cdots + s_m, p$). 4. The negbin(1, p) and geom(p) distributions are the same.

(*continued*)

TABLE 6.4 (*continued*)

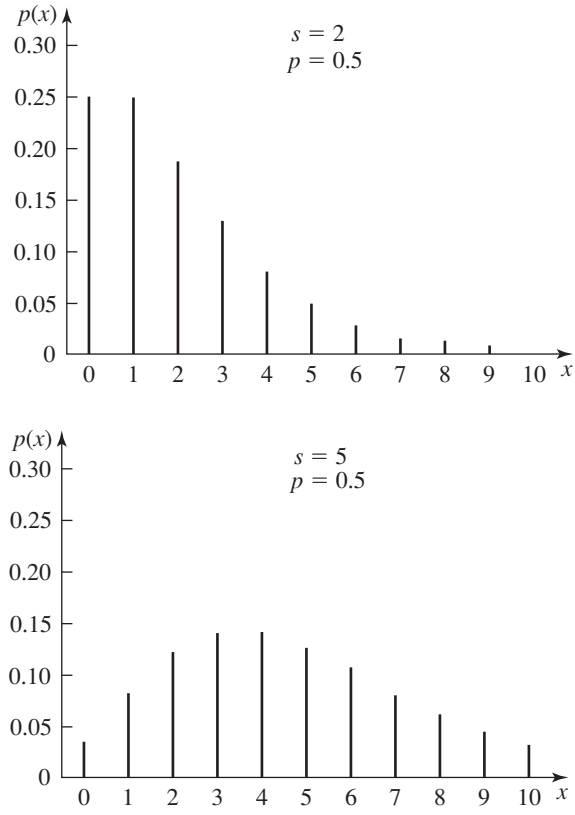


FIGURE 6.22
negbin(s, p) mass functions.

Poisson	**Poisson(λ)**
Possible applications	Number of events that occur in an interval of time when the events are occurring at a constant rate (see Sec. 6.12); number of items in a batch of random size; number of items demanded from an inventory
Mass (see Fig. 6.23)	$p(x) = \begin{cases} \dfrac{e^{-\lambda}\lambda^x}{x!} & \text{if } x \in \{0, 1, \ldots\} \\ 0 & \text{otherwise} \end{cases}$
Distribution	$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ e^{-\lambda} \displaystyle\sum_{i=0}^{\lfloor x \rfloor} \frac{\lambda^i}{i!} & \text{if } x \geq 0 \end{cases}$
Parameter	$\lambda > 0$
Range	$\{0, 1, \ldots\}$
Mean	λ
Variance	λ

TABLE 6.4 (*continued*)

Poisson	**Poisson(λ)**
Mode	$\begin{cases} \lambda - 1 \text{ and } \lambda & \text{if } \lambda \text{ is an integer} \\ \lfloor \lambda \rfloor & \text{otherwise} \end{cases}$
MLE	$\hat{\lambda} = \overline{X}(n)$.
Comments	1. Let $Y_1, Y_2, \ldots$ be a sequence of nonnegative IID random variables, and let $X = \max\{i\colon \sum_{j=1}^{i} Y_j \le 1\}$. Then the distribution of the Y_i's is expo$(1/\lambda)$ if and only if $X \sim$ Poisson(λ). Also, if $X' = \max\{i\colon \sum_{j=1}^{i} Y_j \le \lambda\}$, then the Y_i's are expo(1) if and only if $X' \sim$ Poisson(λ) (see also Sec. 6.12). 2. If $X_1, X_2, \ldots, X_m$ are independent random variables and $X_i \sim$ Poisson(λ_i), then $X_1 + X_2 + \cdots + X_m \sim$ Poisson$(\lambda_1 + \lambda_2 + \cdots + \lambda_m)$.

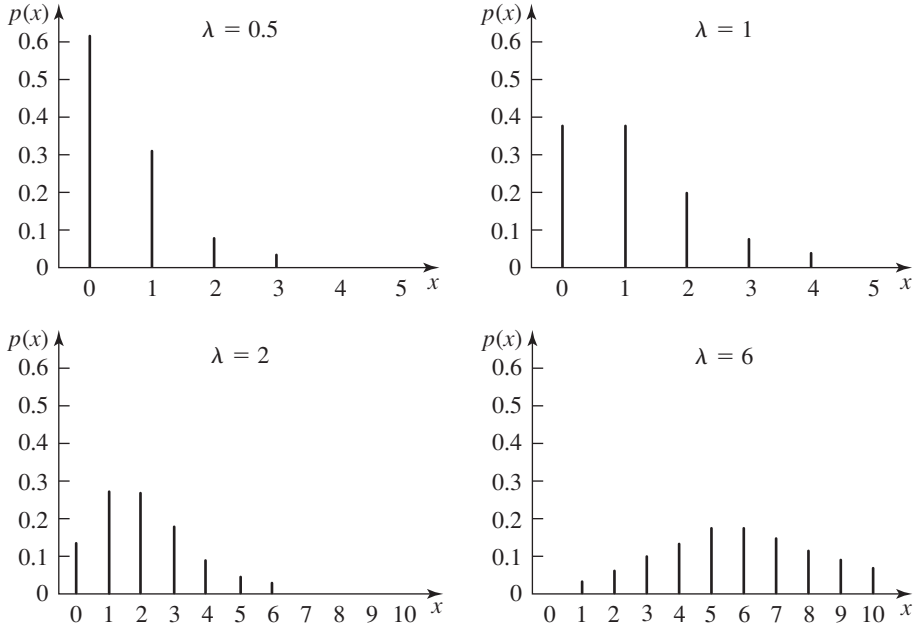


FIGURE 6.23
Poisson(λ) mass functions.

order. Let $X_{(i)}$ denote the ith smallest of the X_j's, so that $X_{(1)} \le X_{(2)} \le \cdots \le X_{(n)}$. Then F is given by

$$F(x) = \begin{cases} 0 & \text{if } x < X_{(1)} \\ \dfrac{i-1}{n-1} + \dfrac{x - X_{(i)}}{(n-1)(X_{(i+1)} - X_{(i)})} & \text{if } X_{(i)} \le x < X_{(i+1)} \\ & \quad \text{for } i = 1, 2, \ldots, n-1 \\ 1 & \text{if } X_{(n)} \le x \end{cases}$$

Figure 6.24 gives an illustration for $n = 6$. Note that $F(x)$ rises most rapidly over those ranges of x in which the X_i's are most densely distributed, as desired.

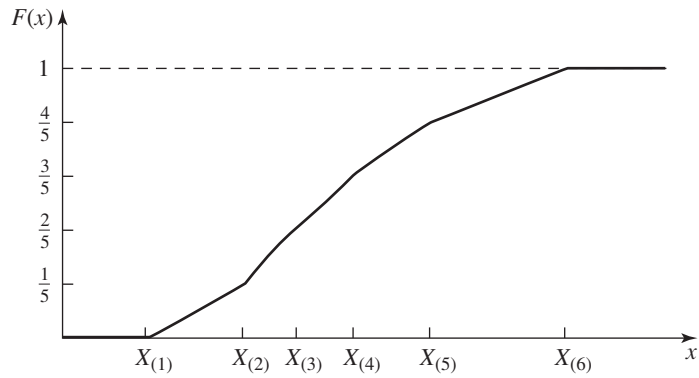


FIGURE 6.24
Continuous, piecewise-linear empirical distribution function from original data.

Also, for each i, $F(X_{(i)}) = (i - 1)/(n - 1)$, which is approximately (for large n) the proportion of the X_j's that are *less than* $X_{(i)}$; this is also the way we would like a *continuous* distribution function to behave. (See Prob. 6.5 for a discussion of another way to define F.) However, one clear disadvantage of specifying this particular empirical distribution is that random values generated from it during a simulation run can never be less than $X_{(1)}$ or greater than $X_{(n)}$ (see Sec. 8.3.16). Also, the mean of $F(x)$ is not equal to the sample mean $\overline{X}(n)$ of the X_i's (see Prob. 6.6).

If, however, the data are grouped, then a different approach must be taken since we do not know the values of the individual X_i's. Suppose that the n X_i's are grouped into k adjacent intervals $[a_0, a_1), [a_1, a_2), \ldots, [a_{k-1}, a_k)$, so that the jth interval contains n_j observations, where $n_1 + n_2 + \cdots + n_k = n$. (Often the a_j's will be equally spaced, but we need not make this assumption.) A reasonable piecewise-linear empirical distribution function G could be specified by first letting $G(a_0) = 0$ and $G(a_j) = (n_1 + n_2 + \cdots + n_j)/n$ for $j = 1, 2, \ldots, k$. Then, interpolating linearly between the a_j's, we define

$$G(x) = \begin{cases} 0 & \text{if } x < a_0 \\ G(a_{j-1}) + \dfrac{x - a_{j-1}}{a_j - a_{j-1}}\,[G(a_j) - G(a_{j-1})] & \text{if } a_{j-1} \le x < a_j \\ & \quad \text{for } j = 1, 2, \ldots, k \\ 1 & \text{if } a_k \le x \end{cases}$$

Figure 6.25 illustrates this specification of an empirical distribution for $k = 4$. In this case, $G(a_j)$ is the proportion of the X_i's that are less than a_j, and $G(x)$ rises most rapidly over ranges of x where the observations are most dense. The random values generated from this distribution, however, will still be bounded both below (by a_0) and above (by a_k); see Sec. 8.3.16.

In practice, many continuous distributions are skewed to the right and have a density with a shape similar to that in Fig. 6.26. Thus, if the sample size n is not very large, we are likely to have few, if any, observations from the right tail of the true underlying distribution (since these tail probabilities are usually small). Moreover, the above empirical distributions do not allow random values to be generated

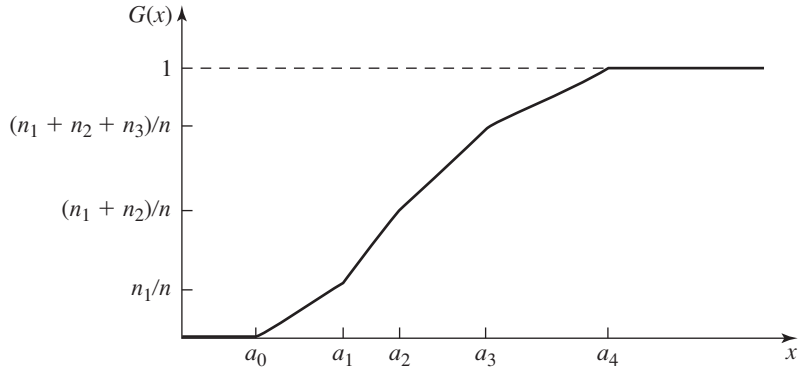


FIGURE 6.25
Continuous, piecewise-linear empirical distribution function from grouped data.

beyond the largest observation. On the other hand, very large generated values can have a significant impact on the disposition of a simulation run. For example, a large service time can cause considerable congestion in a queueing-type system. As a result, Bratley, Fox, and Schrage (1987, pp. 131–133, 150–151) suggest appending an exponential distribution to the right side of the empirical distribution, which allows values larger than $X_{(n)}$ to be generated.

For discrete data, it is quite simple to define an empirical distribution, provided that the original data values $X_1, X_2, \ldots, X_n$ are available. For each possible value x, an empirical mass function $p(x)$ can be defined to be the proportion of the X_i's that are equal to x. For grouped discrete data we could define a mass function such that the *sum* of the $p(x)$'s over all possible values of x in an interval is equal to the

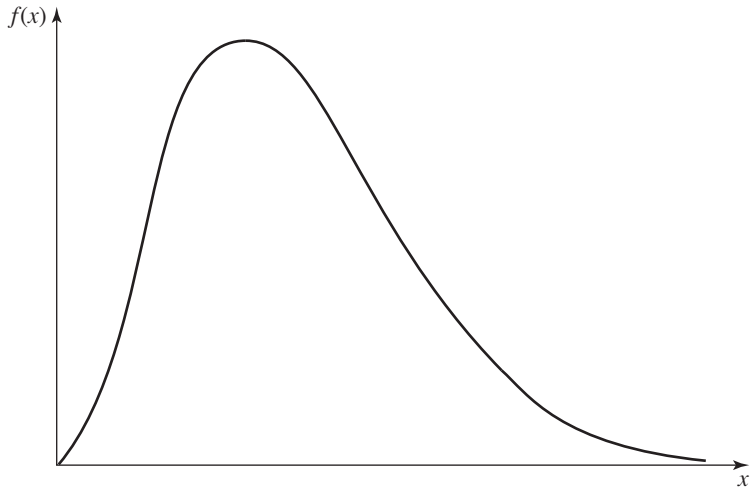


FIGURE 6.26
Typical density function experienced in practice.

proportion of the X_i's in that interval. How the individual $p(x)$'s are allocated for the possible values of x within an interval is essentially arbitrary.

6.3 TECHNIQUES FOR ASSESSING SAMPLE INDEPENDENCE

An important assumption made by many of the statistical techniques discussed in this chapter is that the observations $X_1, X_2, \ldots, X_n$ are an *independent* (or *random*) sample from some underlying distribution. For example, maximum-likelihood estimation (see Sec. 6.5) and chi-square tests (see Sec. 6.6.2) assume independence. If the assumption of independence is not satisfied, then these statistical techniques may not be valid. However, even when the data are not independent, heuristic techniques such as histograms can still be used.

Sometimes observations collected over time are dependent. For example, suppose that $X_1, X_2, \ldots$ represent hourly temperatures in a certain city starting at noon on a particular day. We would not expect these data to be independent, since hourly temperatures close together in time should be positively correlated. As a second example, consider the single-server queueing system in Sec. 1.4. Let $X_1, X_2, \ldots$ be the delays in queue of the successive customers arriving to the system. If the arrival rate of customers is close to the service rate, the system will be congested and the X_i's will be highly positively correlated (see Sec. 4.3).

We now describe two graphical techniques for informally assessing whether the data $X_1, X_2, \ldots, X_n$ (listed in time order of collection) are independent. The *correlation plot* is a graph of the sample correlation $\hat{\rho}_j$ (see Sec. 4.4) for $j = 1, 2, \ldots, l$ (l is a positive integer). The sample correlation $\hat{\rho}_j$ is an estimate of the true correlation ρ_j between two observations that are j observations apart in time. (Note that $-1 \le \rho_j \le 1$.) If the observations $X_1, X_2, \ldots, X_n$ are independent, then $\rho_j = 0$ for $j = 1, 2, \ldots, n - 1$. However, the $\hat{\rho}_j$'s will not be exactly zero even when the X_i's are independent, since $\hat{\rho}_j$ is an observation of a random variable whose mean is not equal to 0 (see Sec. 4.4). If the $\hat{\rho}_j$'s differ from 0 by a significant amount, then this is strong evidence that the X_i's are not independent.

The *scatter diagram* of the observations $X_1, X_2, \ldots, X_n$ is a plot of the pairs (X_i, X_{i+1}) for $i = 1, 2, \ldots, n - 1$. Suppose for simplicity that the X_i's are nonnegative. If the X_i's are independent, one would expect the points (X_i, X_{i+1}) to be scattered randomly throughout the first quadrant of the (X_i, X_{i+1}) plane. The nature of the scattering will, however, depend on the underlying distributions of the X_i's. If the X_i's are positively correlated, then the points will tend to lie along a line with positive slope in the first quadrant. If the X_i's are negatively correlated, then the points will tend to lie along a line with negative slope in the first quadrant.

EXAMPLE 6.2. In Figs. 6.27 and 6.28 we give the correlation plot and scatter diagram for 100 *independent* observations from an exponential distribution with a mean of 1. Note in Fig. 6.27 that the sample correlations are close to 0, but have absolute values as large as 0.16. The scattering of the points in Fig. 6.28 substantiates the independence of the exponential data.

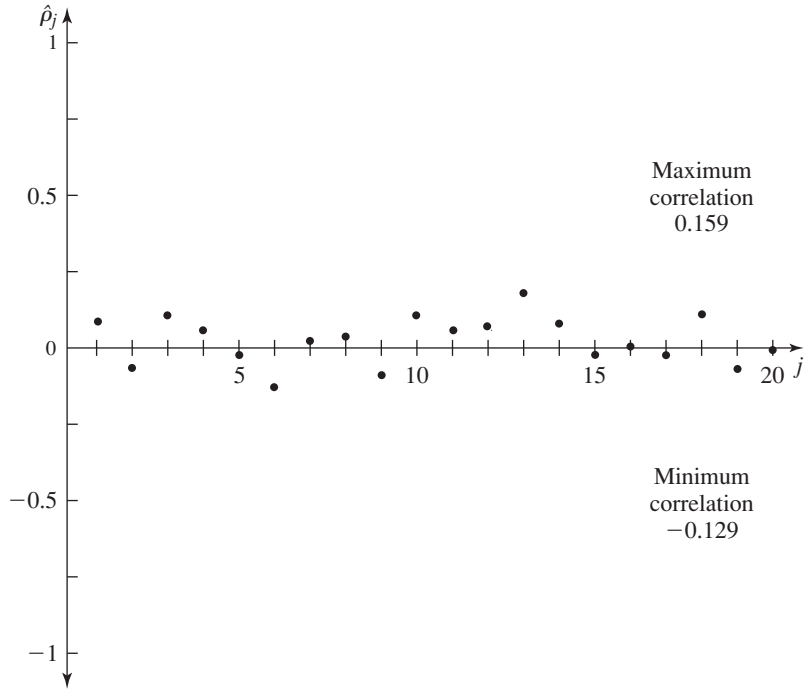


FIGURE 6.27
Correlation plot for independent exponential data.

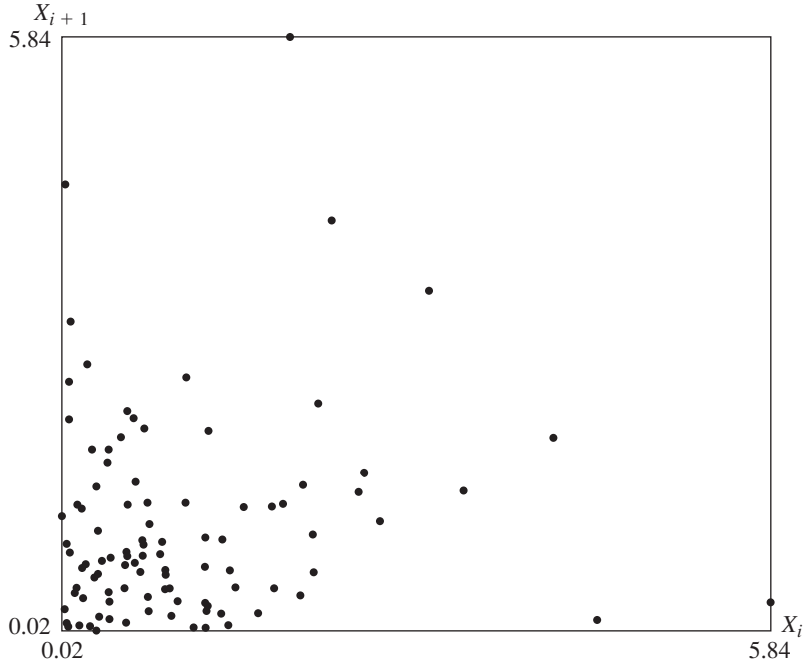


FIGURE 6.28
Scatter diagram for independent exponential data.

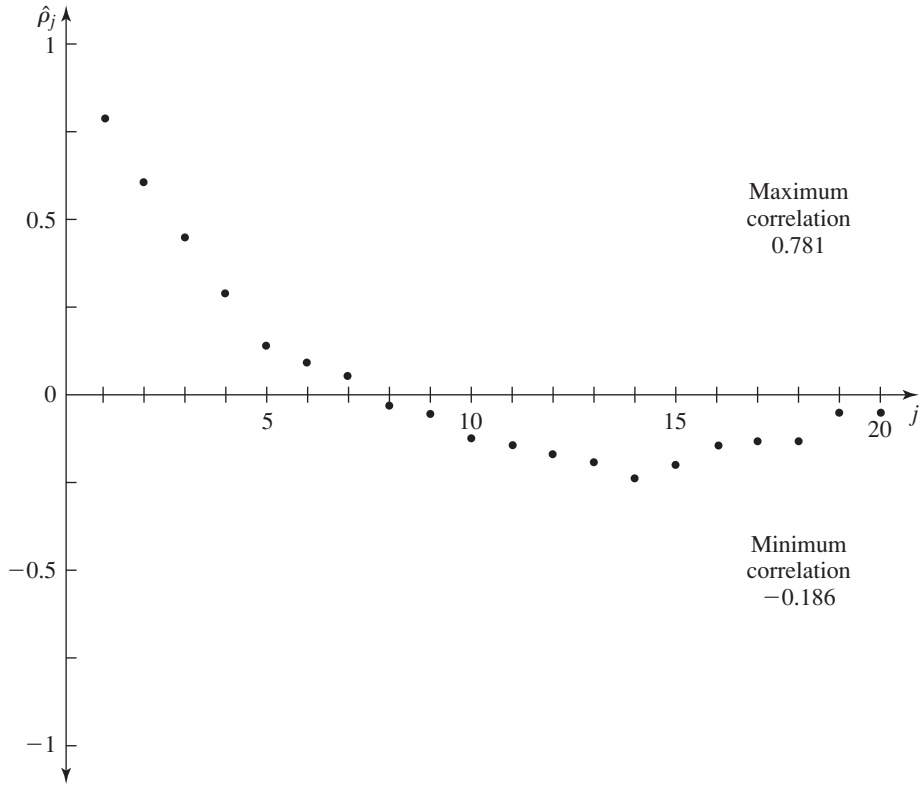


FIGURE 6.29
Correlation plot for correlated queueing data.

EXAMPLE 6.3. In Figs. 6.29 and 6.30 we present the correlation plot and scatter diagram for 100 delays in queue from an *M*/*M*/1 queueing system (see Sec. 1.4.3) with utilization factor $\rho = 0.8$. Note that the $\hat{\rho}_j$'s are large for small values of j and that the points in the scatter diagram tend to lie along a line with positive slope. These facts are consistent with our statement that delays in queue are positively correlated.

There are also several nonparametric (i.e., no assumptions are made about the distributions of the X_i's) statistical tests that can be used to test formally whether $X_1, X_2, \ldots, X_n$ are independent. Bartels (1982) proposes a rank version of von Neumann's ratio as a test statistic for independence and provides the necessary critical values to carry out the test. However, one potential drawback is that the test assumes that there are no "ties" in the data, where a tie means $X_i = X_j$ for $i \neq j$. This requirement will generally not be met for discrete data, and may not even be satisfied for continuous data if they are recorded with only a few decimal places of accuracy. (See the interarrival times in Table 6.7.) Bartels states that his critical values may still be reasonably accurate if the number of ties is small.

There are several versions of the runs test [see, for example, Sec. 7.4.1 and Gibbons (1985)], which can also be used to assess the independence of the X_i's.

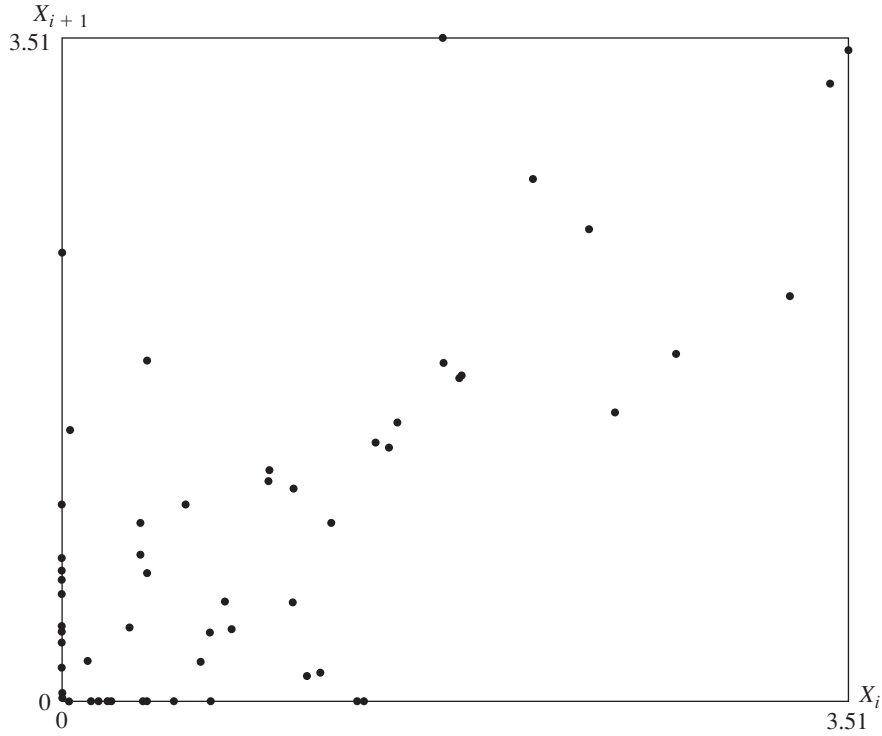


FIGURE 6.30
Scatter diagram for correlated queueing data.

They should have less difficulty with ties than the rank von Neumann test, since runs tests require only that $X_i \neq X_{i+1}$ for $i = 1, 2, \ldots, n - 1$. On the other hand, Bartels showed empirically that the rank von Neumann test is considerably more powerful than one of the runs tests against certain types of alternatives to the X_i's being independent.

6.4 ACTIVITY I: HYPOTHESIZING FAMILIES OF DISTRIBUTIONS

The first step in selecting a particular input distribution is to decide what general families—e.g., exponential, normal, or Poisson—appear to be appropriate on the basis of their shapes, without worrying (yet) about the specific parameter values for these families. This section describes some general techniques that can be used to *hypothesize* families of distributions that might be representative of a simulation input random variable.

In some situations, use can be made of *prior knowledge* about a certain random variable's role in a system to select a modeling distribution or at least rule out some

distributions; this is done on theoretical grounds and does not require any data at all. For example, if we feel that customers arrive to a service facility one at a time, at a constant rate, and so that the numbers of customers arriving in disjoint time intervals are independent, then there are theoretical reasons (see Sec. 6.12.1) for postulating that the interarrival times are IID exponential random variables. Recall also that several discrete distributions—binomial, geometric, and negative binomial—were developed from a physical model. Often the range of a distribution rules it out as a modeling distribution. Service times, for example, should not be generated directly from a normal distribution (at least in principle), since a random value from *any* normal distribution can be negative. The proportion of defective items in a large batch should not be assumed to have a gamma distribution, since proportions must be between 0 and 1, whereas gamma random variables have no upper bound. Prior information should be used whenever available, but confirming the postulated distribution with data is also strongly recommended.

In practice, we seldom have enough of this kind of theoretical prior information to select a single distribution, and the task of hypothesizing a distribution family from observed data is somewhat less structured. In the remainder of this section, we discuss various *heuristics*, or guidelines, that can be used to help one choose appropriate families of distributions.

6.4.1 Summary Statistics

Some distributions are characterized at least partially by functions of their *true* parameters. In Table 6.5 we give a number of such functions, formulas to estimate these functions from IID data $X_1, X_2, \ldots, X_n$ [these estimates are called *summary* (or *descriptive*) statistics], an indication of whether they are applicable to continuous (C) or discrete (D) data, and comments about their interpretation or use. (We have included the sample minimum and maximum because of their utility, even though they may not be a direct function of a distribution's parameters.) Further discussion of many of these functions may be found in Chap. 4.

These functions might be used in some cases to suggest an appropriate distribution family. For a symmetric continuous distribution (e.g., normal), the mean μ is equal to the median $x_{0.5}$. (For a symmetric discrete distribution, the population mean and median may be only approximately equal; see the definition of the median in Sec. 4.2.) Thus, if the estimates $\overline{X}(n)$ and $\hat{x}_{0.5}(n)$ are "almost equal," this is some indication that the underlying distribution may be symmetric. One should keep in mind that $\overline{X}(n)$ and $\hat{x}_{0.5}(n)$ are observations of random variables, and thus their relationship does not necessarily provide definitive information about the true relationship between μ and $x_{0.5}$.

The *coefficient of variation* cv can sometimes provide useful information about the form of a continuous distribution. In particular, cv = 1 for the exponential distribution, regardless of the scale parameter β. Thus, $\widehat{\text{cv}}(n)$ being close to 1 suggests that the underlying distribution is exponential. For the gamma and Weibull distributions, cv is greater than, equal to, or less than 1 when the shape parameter α is less than, equal to, or greater than 1, respectively. Furthermore, these distributions

TABLE 6.5
Useful summary statistics

Function	Sample estimate (summary statistic)	Continuous (C) or discrete (D)	Comments
Minimum, maximum	$X_{(1)}, X_{(n)}$	C, D	$[X_{(1)}, X_{(n)}]$ is a rough estimate of the range
Mean μ	$\overline{X}(n)$	C, D	Measure of central tendency
Median $x_{0.5}$	$\hat{x}_{0.5}(n) = \begin{cases} X_{((n+1)/2)} & \text{if } n \text{ is odd} \\ [X_{(n/2)} + X_{((n/2)+1)}]/2 & \text{if } n \text{ is even} \end{cases}$	C, D	Alternative measure of central tendency
Variance σ^2	$S^2(n)$	C, D	Measure of variability
Coefficient of variation, $\text{cv} = \dfrac{\sqrt{\sigma^2}}{\mu}$	$\widehat{\text{cv}}(n) = \dfrac{\sqrt{S^2(n)}}{\overline{X}(n)}$	C	Alternative measure of variability
Lexis ratio, $\tau = \dfrac{\sigma^2}{\mu}$	$\hat{\tau}(n) = \dfrac{S^2(n)}{\overline{X}(n)}$	D	Alternative measure of variability
Skewness, $\nu = \dfrac{E[(X-\mu)^3]}{(\sigma^2)^{3/2}}$	$\hat{\nu}(n) = \dfrac{n^2}{(n-1)(n-2)} \dfrac{\sum_{i=1}^{n} [X_i - \overline{X}(n)]^3/n}{[S^2(n)]^{3/2}}$	C, D	Measure of symmetry

will have a shape similar to the density function in Fig. 6.26 when $\alpha > 1$, which implies that cv < 1. On the other hand, the lognormal distribution always has a density with a shape similar to that in Fig. 6.26, but its cv can be any positive real number. Thus, if the underlying distribution (observed histogram) has this shape and $\widehat{cv}(n) > 1$, the lognormal may be a better model than either the gamma or Weibull. For the remainder of the distributions in Table 6.3, the cv is not particularly useful. [In fact, cv is not even well defined for distributions such as $U(-c, c)$ (for $c > 0$) or $N(0, \sigma^2)$, since the mean μ is zero.]

For a discrete distribution, the *lexis ratio* τ plays the same role that the coefficient of variation does for a continuous distribution. We have found the lexis ratio to be very useful in discriminating among the Poisson, binomial, and negative binomial distributions, since $\tau = 1$, $\tau < 1$, and $\tau > 1$, respectively, for these distributions. (Note that the geometric distribution is a special case of the negative binomial.)

The *skewness* ν is a measure of the symmetry of a distribution. For symmetric distributions like the normal, $\nu = 0$. If $\nu > 0$ (e.g., $\nu = 2$ for the exponential distribution), the distribution is skewed to the right; if $\nu < 0$, the distribution is skewed to the left. Thus, the estimated skewness $\hat{\nu}(n)$ [see Joanes and Gill (1998)] can be used to ascertain the shape of the underlying distribution. Our experience indicates that many distributions encountered in practice are skewed to the right.

It is possible to define another function of a distribution's parameters, called the *kurtosis*, which is a measure of the "tail weight" of a distribution [see, for example, Kendall, Stuart, and Ord (1987, pp. 107–108)]. However, we have not found the kurtosis to be very useful for discriminating among distributions.

6.4.2 Histograms

For a continuous data set, a histogram is essentially a graphical estimate (see the discussion below) of the plot of the density function corresponding to the distribution of our data $X_1, X_2, \ldots, X_n$. Density functions, as shown in Figs. 6.5 through 6.16, tend to have recognizable shapes in many cases. Therefore, a graphical estimate of a density should provide a good clue to the distributions that might be tried as a model for the data.

To make a *histogram*, we break up the range of values covered by the data into k disjoint adjacent intervals $[b_0, b_1), [b_1, b_2), \ldots, [b_{k-1}, b_k)$. All the intervals should be the same width, say, $\Delta b = b_j - b_{j-1}$, which might necessitate throwing out a few extremely large or small X_i's to avoid getting an unwieldy-looking histogram plot. For $j = 1, 2, \ldots, k$, let h_j be the *proportion* of the X_i's that are in the jth interval $[b_{j-1}, b_j)$. Finally, we define the function

$$h(x) = \begin{cases} 0 & \text{if } x < b_0 \\ h_j & \text{if } b_{j-1} \le x < b_j \\ 0 & \text{if } b_k \le x \end{cases} \qquad \text{for } j = 1, 2, \ldots, k$$

which we plot as a function of x. (See Example 6.4 below for an illustration of a histogram.) The plot of h, which is piecewise-constant, is then compared with plots of densities of various distributions on the basis of shape alone (location and scale

differences are ignored) to see what distributions have densities that resemble the histogram h.

To see why the shape of h should resemble the true density f of the data, let X be a random variable with density f, so that X is distributed as the X_i's. Then, for any fixed j ($j = 1, 2, \ldots, k$),

$$P(b_{j-1} \leq X < b_j) = \int_{b_{j-1}}^{b_j} f(x)\, dx = \Delta b\, f(y)$$

for some number $y \in (b_{j-1}, b_j)$. (The first equation is by the definition of a continuous random variable, and the second follows from the mean-value theorem of calculus.) On the other hand, the probability that X falls in the jth interval is approximated by h_j, which is the value of $h(y)$. Therefore,

$$h(y) = h_j \approx \Delta b\, f(y)$$

so that $h(y)$ is roughly proportional to $f(y)$; that is, h and f have roughly the same shape. (Actually, an *estimate* of f is obtained by dividing the function h by the constant Δb.)

Histograms are applicable to any continuous distribution and provide a readily interpreted visual synopsis of the data. Furthermore, it is easy to "eyeball" a histogram in reference to certain density functions. There are, however, certain difficulties. Most vexing is the absence of a definitive guide for choosing the number of intervals k (or, equivalently, their width Δb).

Several rules of thumb have been suggested for choosing the number of intervals k {e.g., Sturges's rule [see Hoaglin, Mosteller, and Tukey (1983, pp. 23–24)] and a normal approximation due to Scott (1979)}. The best known of these guidelines is probably Sturges's rule, which says that k should be chosen according to the following formula:

$$k = \lfloor 1 + \log_2 n \rfloor = \lfloor 1 + 3.322 \log_{10} n \rfloor$$

However, in general, we do not believe that such rules are very useful (see Example 6.4). We recommend trying several different values of Δb and choosing the smallest one that gives a "smooth" histogram. This is clearly a matter of some subjectivity and represents the major problem in using histograms. If Δb is chosen too small, the histogram will have a "ragged" shape since the variances of the h_j's will be large. If Δb is chosen too large, then the histogram will have a "block-like" shape, and the true shape of the underlying density will be masked since the data have been overaggregated. In particular, a large spike in the density function near $x = 0$ or elsewhere (see Fig. 6.12) could be missed if Δb is too large.

As we have noted, a histogram is an estimate (except for rescaling) of the density function. There are many other ways in which the density function can be estimated from data, some of which are quite sophisticated. We refer the interested reader to the survey paper of Wegman (1982, pp. 309–315) and the book by Silverman (1986).

The probability mass function corresponding to a discrete data set can also be estimated by using a histogram. For each possible value x_j that can be assumed by the data, let h_j be the proportion of the X_i's that are equal to x_j. Vertical bars of height h_j are plotted versus x_j, and this is compared with the mass functions of the discrete distributions in Sec. 6.2.3 on the basis of shape.

For a discrete data set, h_j (which is a random variable) is an unbiased estimator of $p(x_j)$, where $p(x)$ is the true (unknown) mass function of the data (see Prob. 6.7).

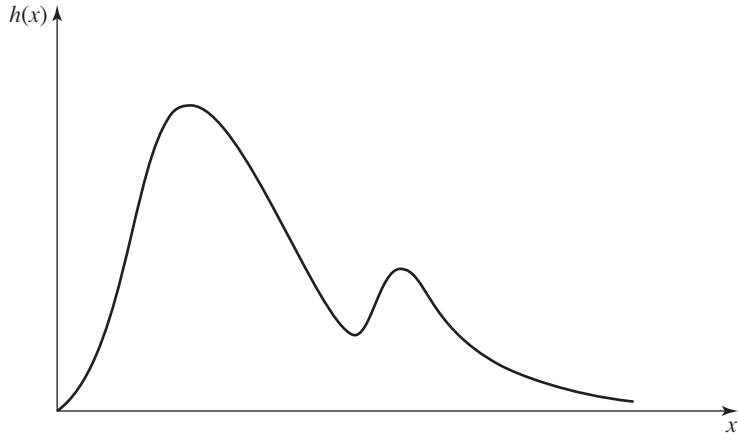


FIGURE 6.31
Histogram corresponding to a density function with two local modes.

However, in this case we need not make any arbitrary subjective decisions about interval width and placement.

There are certain situations in which the histogram will have several local modes (or "humps"), in which case none of the standard distributions discussed in Sec. 6.2 may provide an adequate representation. In Fig. 6.31 we give such an example, which might represent (continuous) times to repair some machine collected over a 1-year period. There are two types of breakdowns for the machine. Most of the breakdowns are minor, and the corresponding repair time is relatively small; this case corresponds to the left hump in Fig. 6.31. A small proportion of the breakdowns are major and have large repair times, since spare parts need to be ordered. This results in the right hump in Fig. 6.31. If the observed repair times can be separated into these two cases (corresponding to minor and major repairs), with p_j being the proportion of observations for case j ($j = 1, 2$), then a density $f_j(x)$ is fit to the class j observations ($j = 1, 2$) using the methods discussed in Secs. 6.4 through 6.6. Thus, the overall repair-time density $f(x)$ is given by

$$f(x) = p_1 f_1(x) + p_2 f_2(x)$$

and random repair times can be generated from $f(x)$ during a simulation by using the composition technique (see Sec. 8.2.2). See Sec. 6.9 and Prob. 6.8 for alternative methods for modeling a histogram with several modes.

6.4.3 Quantile Summaries and Box Plots

The *quantile summary* [see, for example, Tukey (1970)] is a synopsis of the sample that is useful for determining whether the underlying probability density function or probability mass function is symmetric or skewed to the right or to the left. It is applicable to either continuous or discrete data sets; however, for expository convenience we will explain it only for the continuous case.

TABLE 6.6
Structure of the quantile summary for the sample $X_1, X_2, \ldots, X_n$

Quantile	Depth	Sample value(s)		Midpoint
Median	$i = (n + 1)/2$	$X_{(i)}$		$X_{(i)}$
Quartiles	$j = (\lfloor i \rfloor + 1)/2$	$X_{(j)}$	$X_{(n-j+1)}$	$[X_{(j)} + X_{(n-j+1)}]/2$
Octiles	$k = (\lfloor j \rfloor + 1)/2$	$X_{(k)}$	$X_{(n-k+1)}$	$[X_{(k)} + X_{(n-k+1)}]/2$
Extremes	1	$X_{(1)}$	$X_{(n)}$	$[X_{(1)} + X_{(n)}]/2$

Suppose that $F(x)$ is the distribution function for a continuous random variable. Suppose further that $F(x)$ is continuous and strictly increasing when $0 < F(x) < 1$. [This means that if $x_1 < x_2$ and $0 < F(x_1) \le F(x_2) < 1$, then in fact $F(x_1) < F(x_2)$.] For $0 < q < 1$, the *q-quantile* of $F(x)$ is that number x_q such that $F(x_q) = q$. If F^{-1} denotes the inverse of $F(x)$, then $x_q = F^{-1}(q)$. $\{F^{-1}$ is that function such that $F[F^{-1}(x)] = F^{-1}[F(x)] = x.\}$ Here we are particularly interested in the median $x_{0.5}$, the lower and upper quartiles $x_{0.25}$ and $x_{0.75}$, and the lower and upper octiles $x_{0.125}$ and $x_{0.875}$. A quantile summary for the sample $X_1, X_2, \ldots, X_n$ has a form that is given in Table 6.6.

In Table 6.6, if the "depth" subscript l (l is equal to i, j, or k) is halfway between the integers m and $m' = m + 1$, then $X_{(l)}$ is defined to be the average of $X_{(m)}$ and $X_{(m')}$. The value $X_{(i)}$ is an estimate of the median, $X_{(j)}$ and $X_{(n-j+1)}$ are estimates of the quartiles, and $X_{(k)}$ and $X_{(n-k+1)}$ are estimates of the octiles. If the underlying distribution of the X_i's is symmetric, then the four midpoints should be approximately equal. On the other hand, if the underlying distribution of the X_i's is skewed to the right (left), then the four midpoints (from the top to the bottom of the table) should be increasing (decreasing).

A *box plot* is a graphical representation of the quantile summary (see Fig. 6.33). Fifty percent of the observations fall within the horizontal boundaries of the box $[x_{0.25}, x_{0.75}]$. (In some books, box plots do not contain octiles.)

The following two examples illustrate the use of the techniques discussed in Sec. 6.4.

EXAMPLE 6.4. A simulation model was developed for a drive-up banking facility, and data were collected on the arrival pattern for cars. Over a fixed 90-minute interval, 220 cars arrived, and we noted the (continuous) interarrival time X_i (in minutes) between cars i and $i + 1$, for $i = 1, 2, \ldots, 219$. Table 6.7 lists these $n = 219$ interarrival times after they have been sorted into increasing order. The numbers of cars arriving in the six consecutive 15-minute intervals were counted and found to be approximately equal, suggesting that the arrival rate is somewhat constant over *this* 90-minute interval. Furthermore, cars arrive one at a time, and there is no reason to believe that the numbers of arrivals in disjoint intervals are not independent. Thus, on theoretical grounds (see Sec. 6.12.1) we postulate that the interarrival times are exponential. To substantiate this hypothesis, we first look at the summary statistics given in Table 6.8. Since $\bar{X}(219) = 0.399 > 0.270 = \hat{x}_{0.5}(219)$ and $\hat{\nu}(219) = 1.478$, this suggests that the underlying distribution is skewed to the right, rather than symmetric. Furthermore, $\widehat{\text{cv}}(219) = 0.953$, which is close to the theoretical value of 1 for the exponential distribution. Next we made three different histograms of the data, using $b_0 = 0$ in each case and $\Delta b = 0.050$, 0.075, and 0.100, as shown in Fig. 6.32 (see also Fig. 6.2). The smoothest-looking

TABLE 6.7
$n = 219$ interarrival times (minutes) sorted into increasing order

0.01	0.06	0.12	0.23	0.38	0.53	0.88
0.01	0.07	0.12	0.23	0.38	0.53	0.88
0.01	0.07	0.12	0.24	0.38	0.54	0.90
0.01	0.07	0.13	0.25	0.39	0.54	0.93
0.01	0.07	0.13	0.25	0.40	0.55	0.93
0.01	0.07	0.14	0.25	0.40	0.55	0.95
0.01	0.07	0.14	0.25	0.41	0.56	0.97
0.01	0.07	0.14	0.25	0.41	0.57	1.03
0.02	0.07	0.14	0.26	0.43	0.57	1.05
0.02	0.07	0.15	0.26	0.43	0.60	1.05
0.03	0.07	0.15	0.26	0.43	0.61	1.06
0.03	0.08	0.15	0.26	0.44	0.61	1.09
0.03	0.08	0.15	0.26	0.45	0.63	1.10
0.04	0.08	0.15	0.27	0.45	0.63	1.11
0.04	0.08	0.15	0.28	0.46	0.64	1.12
0.04	0.09	0.17	0.28	0.47	0.65	1.17
0.04	0.09	0.18	0.29	0.47	0.65	1.18
0.04	0.10	0.19	0.29	0.47	0.65	1.24
0.04	0.10	0.19	0.30	0.48	0.69	1.24
0.05	0.10	0.19	0.31	0.49	0.69	1.28
0.05	0.10	0.20	0.31	0.49	0.70	1.33
0.05	0.10	0.21	0.32	0.49	0.72	1.38
0.05	0.10	0.21	0.35	0.49	0.72	1.44
0.05	0.10	0.21	0.35	0.50	0.72	1.51
0.05	0.10	0.21	0.35	0.50	0.74	1.72
0.05	0.10	0.21	0.36	0.50	0.75	1.83
0.05	0.11	0.22	0.36	0.51	0.76	1.96
0.05	0.11	0.22	0.36	0.51	0.77	
0.05	0.11	0.22	0.37	0.51	0.79	
0.06	0.11	0.23	0.37	0.52	0.84	
0.06	0.11	0.23	0.38	0.52	0.86	
0.06	0.12	0.23	0.38	0.53	0.87	

TABLE 6.8
Summary statistics for the interarrival-time data

Summary statistic	Value
Minimum	0.010
Maximum	1.960
Mean	0.399
Median	0.270
Variance	0.144
Coefficient of variation	0.953
Skewness	1.478

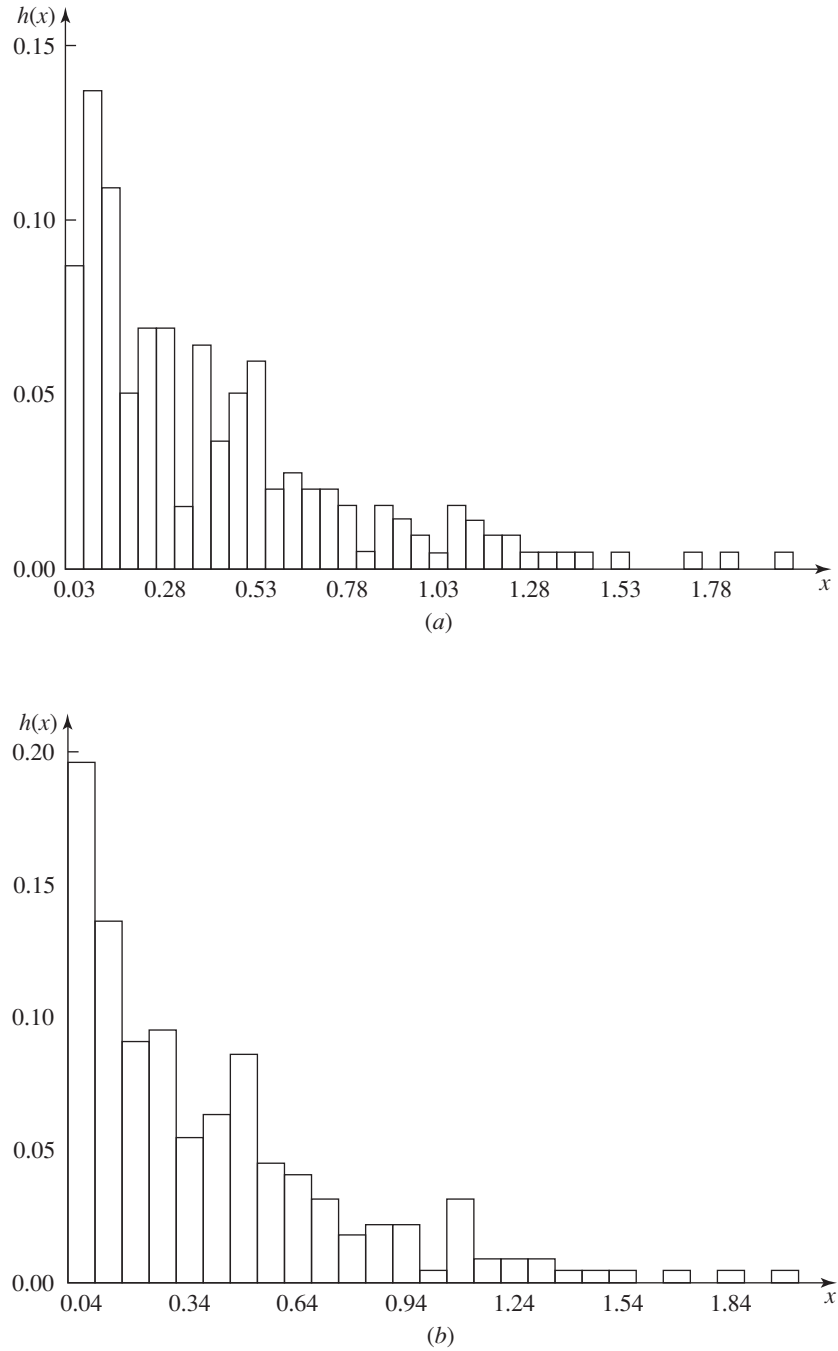


FIGURE 6.32
Histograms of the interarrival-time data in Table 6.7: (*a*) $\Delta b = 0.050$; (*b*) $\Delta b = 0.075$; (*c*) $\Delta b = 0.100$.

(*continued*)

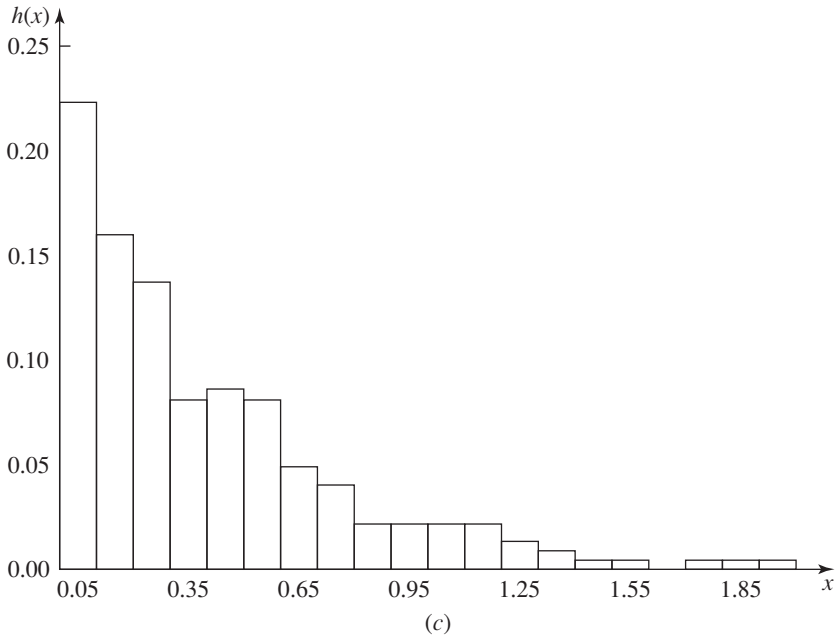


FIGURE 6.32
(*continued*)

histogram appears to be for $\Delta b = 0.100$ and its shape resembles that of an exponential density. (Note that Sturges's rule gives $k = 8$ and $\Delta b = 0.250$, resulting in overaggregation of the data.) Finally, in Fig. 6.33 we give the quantile summary and box plot for the interarrival times. The increasing midpoints and the elongated nature of the right side of the box plot reaffirm that the underlying distribution is exponential. In summary, for both theoretical and empirical reasons, we hypothesize that the interarrival times are exponential.

Quantile	Depth	Sample value(s)		Midpoint
Median	110	0.270		0.270
Quartiles	55.5	0.100	0.545	0.323
Octiles	28	0.050	0.870	0.460
Extremes	1	0.010	1.960	0.985

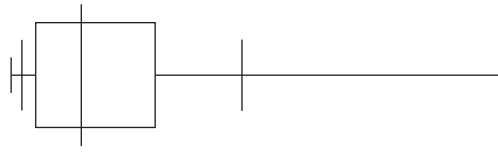


FIGURE 6.33
Quantile summary and box plot for the interarrival-time data.

TABLE 6.9
Values and counts for $n = 156$ demand sizes arranged into increasing order

0(59),	1(26),	2(24),	3(18),	4(12),
5(5),	6(4),	7(3),	9(3),	11(2)

TABLE 6.10
Summary statistics for the demand-size data

Summary statistic	Value
Minimum	0.000
Maximum	11.000
Mean	1.891
Median	1.000
Variance	5.285
Lexis ratio	2.795
Skewness	1.687

EXAMPLE 6.5. Table 6.9 gives the values and counts for $n = 156$ observations on the (discrete) number of items demanded in a week from an inventory over a 3-year period, arranged into increasing order. Rather than giving all the individual values, we give the frequency counts; 59 X_i's were equal to 0, 26 X_i's were equal to 1, etc. Summary statistics and a histogram for these data are given in Table 6.10 and Fig. 6.34, respectively.

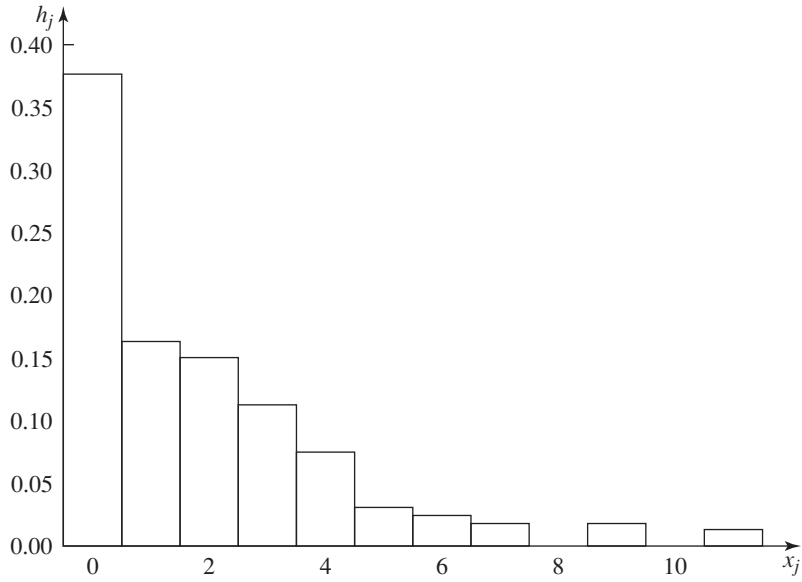


FIGURE 6.34
Histogram of the demand-size data in Table 6.9.

Since the lexis ratio $\hat{\tau}(156) = 2.795$, the binomial and Poisson distributions do not seem likely models. Furthermore, the large positive value of the skewness $\hat{\nu}(156) = 1.687$ would appear to rule out the discrete uniform distribution that is symmetric. Therefore, the possible discrete models (of those considered in this book) are the geometric and negative binomial distributions, with the former being a special case of the latter when $s = 1$. However, based on the monotonically decreasing histogram in Fig. 6.34 (and the mass functions in Fig. 6.21), we hypothesize that the demand data are from a geometric distribution.

6.5 ACTIVITY II: ESTIMATION OF PARAMETERS

After one or more candidate families of distributions have been hypothesized in Activity I, we must somehow specify the values of their parameters in order to have completely specified distributions for possible use in the simulation. Our IID data $X_1, X_2, \ldots, X_n$ were used to help us hypothesize distributions, and these same data can also be used to estimate their parameters. When data are used directly in this way to specify a numerical value for an unknown parameter, we say that we are *estimating* that parameter from the data.

An *estimator* is a numerical function of the data. There are many ways to specify the form of an estimator for a particular parameter of a given distribution, and many alternative ways to evaluate the quality of an estimator. We shall consider explicitly only one type, *maximum-likelihood estimators* (MLEs), for three reasons: (1) MLEs have several desirable properties often not enjoyed by alternative methods of estimation, e.g., least-squares estimators, unbiased estimators, and the method of moments; (2) the use of MLEs turns out to be important in justifying the chi-square goodness-of-fit test (see Sec. 6.6.2); and (3) the central idea of maximum-likelihood estimation has a strong intuitive appeal.

The basis for MLEs is most easily understood in the discrete case. Suppose that we have hypothesized a discrete distribution for our data that has one unknown parameter θ. Let $p_\theta(x)$ denote the probability mass function for this distribution, so that the parameter θ is part of the notation. *Given that we have already observed* the IID data $X_1, X_2, \ldots, X_n$, we define the *likelihood function* $L(\theta)$ as follows:

$$L(\theta) = p_\theta(X_1)p_\theta(X_2) \cdots p_\theta(X_n)$$

Now $L(\theta)$, which is just the joint probability mass function since the data are independent (see Sec. 4.2), gives the probability (likelihood) of obtaining our observed data if θ is the value of the unknown parameter. Then the MLE of the unknown value of θ, which we denote by $\hat{\theta}$, is defined to be the value of θ that maximizes $L(\theta)$; that is, $L(\hat{\theta}) \geq L(\theta)$ for all possible values of θ. Thus, $\hat{\theta}$ "best explains" the data we have collected. In the continuous case, MLEs do not have quite as simple an intuitive explanation, since the probability that a continuous random variable *equals* any fixed number is always 0 [see Prob. 6.26 and Breiman (1973, pp. 67–68) for intuitive justification of MLEs in the continuous case]. Nevertheless, MLEs for continuous distributions are defined analogously to the discrete case. If $f_\theta(x)$ denotes

the hypothesized density function (again we assume that there is only one unknown parameter θ), the likelihood function is given by

$$L(\theta) = f_\theta(X_1) f_\theta(X_2) \cdots f_\theta(X_n)$$

The MLE $\hat{\theta}$ of θ is defined to be the value of θ that maximizes $L(\theta)$ over all permissible values of θ. The following two examples show how to compute MLEs for the distributions hypothesized earlier in Examples 6.4 and 6.5.

EXAMPLE 6.6. For the exponential distribution, $\theta = \beta$ ($\beta > 0$) and $f_\beta(x) = (1/\beta)e^{-x/\beta}$ for $x \geq 0$. The likelihood function is

$$\begin{aligned} L(\beta) &= \left(\frac{1}{\beta} e^{-X_1/\beta}\right)\left(\frac{1}{\beta} e^{-X_2/\beta}\right) \cdots \left(\frac{1}{\beta} e^{-X_n/\beta}\right) \\ &= \beta^{-n} \exp\left(-\frac{1}{\beta} \sum_{i=1}^{n} X_i\right) \end{aligned}$$

and we seek the value of β that maximizes $L(\beta)$ over all $\beta > 0$. This task is more easily accomplished if, instead of working directly with $L(\beta)$, we work with its logarithm. Thus, we define the *log-likelihood function* as

$$l(\beta) = \ln L(\beta) = -n \ln \beta - \frac{1}{\beta} \sum_{i=1}^{n} X_i$$

Since the logarithm function is strictly increasing, maximizing $L(\beta)$ is equivalent to maximizing $l(\beta)$, which is much easier; that is, $\hat{\beta}$ maximizes $L(\beta)$ if and only if $\hat{\beta}$ maximizes $l(\beta)$. Standard differential calculus can be used to maximize $l(\beta)$ by setting its derivative to zero and solving for β. That is,

$$\frac{dl}{d\beta} = \frac{-n}{\beta} + \frac{1}{\beta^2} \sum_{i=1}^{n} X_i$$

which equals zero if and only if $\beta = \sum_{i=1}^{n} X_i/n = \overline{X}(n)$. To make sure that $\beta = \overline{X}(n)$ is a maximizer of $l(\beta)$ (as opposed to a minimizer or an inflection point), a sufficient (but not necessary) condition is that $d^2l/d\beta^2$, evaluated at $\beta = \overline{X}(n)$, be negative. But

$$\frac{d^2l}{d\beta^2} = \frac{n}{\beta^2} - \frac{2}{\beta^3} \sum_{i=1}^{n} X_i$$

which is easily seen to be negative when $\beta = \overline{X}(n)$ since the X_i's are positive. Thus, the MLE of β is $\hat{\beta} = \overline{X}(n)$. Notice that the MLE is quite natural here, since β is the mean of the hypothesized distribution and the MLE is the *sample* mean. For the data of Example 6.4, $\hat{\beta} = \overline{X}(219) = 0.399$.

EXAMPLE 6.7. The discrete data of Example 6.5 were hypothesized to come from a geometric distribution. Here $\theta = p$ ($0 < p < 1$) and $p_p(x) = p(1 - p)^x$ for $x = 0, 1, \ldots$. The likelihood function is

$$L(p) = p^n (1 - p)^{\sum_{i=1}^{n} X_i}$$

which is again amenable to the logarithmic transformation to obtain

$$l(p) = \ln L(p) = n \ln p + \sum_{i=1}^{n} X_i \ln (1 - p)$$

Differentiating $l(p)$, we get

$$\frac{dl}{dp} = \frac{n}{p} - \frac{\sum_{i=1}^{n} X_i}{1-p}$$

which equals zero if and only if $p = 1/[\overline{X}(n) + 1]$. To make sure that this is a maximizer, note that

$$\frac{d^2l}{dp^2} = -\frac{n}{p^2} - \frac{\sum_{i=1}^{n} X_i}{(1-p)^2} < 0$$

for any valid p. Thus, the MLE of p is $\hat{p} = 1/[\overline{X}(n) + 1]$, which is intuitively appealing (see Prob. 6.9). For the demand-size data of Example 6.5, $\hat{p} = 0.346$.

The above two examples illustrate two important practical tools for deriving MLEs, namely, the use of the log-likelihood function and setting its derivative (with respect to the parameter being estimated) equal to zero to find the MLE. While these tools are often useful in finding MLEs, the reader should be cautioned against assuming that finding a MLE is always a simple matter of setting a derivative to zero and solving easily for $\hat{\theta}$. For some distributions, neither the log-likelihood function nor differentiation is useful; probably the best-known example is the uniform distribution (see Prob. 6.10). For other distributions, both tools are useful, but solving $dl/d\theta = 0$ cannot be accomplished by simple algebra, and numerical methods must be used; the gamma, Weibull, and beta distributions are (multiparameter) examples of this general situation. We refer the reader to Breiman (1973, pp. 65–84) for examples of techniques used to find MLEs for a variety of distributions.

We have said that MLEs have several desirable statistical properties, some of which are as follows [see Breiman (1973, pp. 85–88) and Kendall and Stuart (1979, chap. 18)]:

1. For most of the common distributions, the MLE is unique; that is, $L(\hat{\theta})$ is *strictly* greater than $L(\theta)$ for any other value of θ.
2. Although MLEs need not be unbiased, in general, the asymptotic distribution (as $n \to \infty$) of $\hat{\theta}$ has mean equal to θ (see property 4 below).
3. MLEs are *invariant*; that is, if $\phi = h(\theta)$ for some function h, then the MLE of ϕ is $h(\hat{\theta})$. (Unbiasedness is not invariant.) For example, the variance of an expo(β) random variable is β^2, so the MLE of this variance is $[\overline{X}(n)]^2$.
4. MLEs are asymptotically normally distributed; that is, $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{\mathcal{D}} \mathrm{N}(0, \delta(\theta))$, where $\delta(\theta) = -n/E(d^2l/d\theta^2)$ (the expectation is with respect to X_i, assuming that X_i has the hypothesized distribution) and $\xrightarrow{\mathcal{D}}$ denotes convergence in distribution. Furthermore, if $\tilde{\theta}$ is any other estimator such that $\sqrt{n}(\tilde{\theta} - \theta) \xrightarrow{\mathcal{D}} \mathrm{N}(0, \sigma^2)$, then $\delta(\theta) \le \sigma^2$. (Thus, MLEs are called *best asymptotically normal.*)
5. MLEs are *strongly consistent*; that is, $\lim_{n\to\infty} \hat{\theta} = \theta$ (w.p. 1).

The proofs of these and other properties sometimes require additional mild "regularity" assumptions; see Kendall and Stuart (1979).

Property 4 is of special interest, since it allows us to establish an approximate confidence interval for θ. If we define $\delta(\theta)$ as in property 4 above, it can be shown

that

$$\frac{\hat{\theta} - \theta}{\sqrt{\delta(\hat{\theta})/n}} \xrightarrow{\mathcal{D}} \mathrm{N}(0, 1)$$

as $n \to \infty$. Thus, for large n an approximate $100(1 - \alpha)$ percent confidence interval for θ is

$$\hat{\theta} \pm z_{1-\alpha/2} \sqrt{\frac{\delta(\hat{\theta})}{n}} \tag{6.3}$$

EXAMPLE 6.8. Construct a 90 percent confidence interval for the parameter p of the geometric distribution, and specialize it to the data of Example 6.5. It is easy to show that

$$E\left[\frac{d^2 l}{dp^2}\right] = -\frac{n}{p^2} - \frac{n(1-p)/p}{(1-p)^2} = -\frac{n}{p^2(1-p)}$$

so that $\delta(p) = p^2(1 - p)$ and, for large n, an approximate 90 percent confidence interval for p is given by

$$\hat{p} \pm 1.645 \sqrt{\frac{\hat{p}^2(1-\hat{p})}{n}}$$

For the data of Example 6.5, we get 0.346 ± 0.037.

This suggests a way of checking how sensitive a simulation output measure of performance is to a particular input parameter. The simulation could be run for θ set at, say, the left endpoint, the center $(\hat{\theta})$, and the right endpoint of the confidence interval in (6.3). If the measure of performance appeared to be insensitive to values of θ in this range, we could feel confident that we have an adequate estimate of θ for our purposes. On the other hand, if the simulation appeared to be sensitive to θ, we might seek a better estimate of θ; this would usually entail collecting more data.

The general form of the above problem may be stated as follows. A simulation model's performance measures depend on the choice of input probability distributions and their associated parameters. When we choose the distributions to use for a simulation model, we generally don't know with absolute certainty whether these are the correct distributions to use, and this lack of complete knowledge results in what we might call *model uncertainty*. Also, given that certain input distributions have been selected, we typically do not know with complete certainty what parameters to use for these distributions, and we might call this *parameter uncertainty*. (Parameters are typically *estimated* from observed data or are specified *subjectively* based on expert opinion.) The term *input-model uncertainty* is used to refer to these two issues collectively. Ideally, we would like to have a method for constructing a confidence interval for a simulation performance measure that takes into account *both* the sampling variability of the simulation model (see Chap. 9) and input-model uncertainty. Henderson (2003) says that such a method should be understandable by the simulation practitioner, based on sound statistical procedures, relatively easy to implement, and computationally efficient.

There have been a number of methods suggested for addressing the problem of input-model uncertainty, including the following [see Barton (2012) and Henderson (2003)]:

- Bayesian model averaging [Chick (2001), Zouaoui and Wilson (2003, 2004)]
- Delta-method approaches [Cheng and Holland (1997, 1998, 2004)]
- Metamodel-assisted bootstrapping [Barton et al. (2013), Chapter 12]; see Cheng (2006) and Efron and Tibshirani (1993) for a discussion of bootstrap resampling
- Nonparametic bootstrapping [Barton and Schruben (1993, 2001)]
- Quick method based on a random-effects model [Ankenman and Nelson (2012)]

Unfortunately, most of these methods are reasonably complicated and make assumptions that may not always be satisfied in practice [see Barton (2012) and Barton et al. (2013)]. For example, Bayesian model averaging and the delta-method approaches assume that the family (or families) of distributions (but not the parameter values) that best represents a source of system randomness is known in advance, which is unlikely to be true in most real-world applications.

So far, we have explicitly treated only distributions with a single unknown parameter. If a distribution has several parameters, we can define MLEs of these parameters in a natural way. For instance, the gamma distribution has two parameters (α and β), and the likelihood function is defined to be

$$L(\alpha, \beta) = \frac{\beta^{-n\alpha}\left(\prod_{i=1}^{n} X_i\right)^{\alpha-1} \exp\left[-(1/\beta)\sum_{i=1}^{n} X_i\right]}{[\Gamma(\alpha)]^n}$$

The MLEs $\hat{\alpha}$ and $\hat{\beta}$ of the unknown values of α and β are defined to be the values of α and β that (jointly) maximize $L(\alpha, \beta)$. [Finding $\hat{\alpha}$ and $\hat{\beta}$ usually proceeds by letting $l(\alpha, \beta) = \ln L(\alpha, \beta)$ and trying to solve the equations $\partial l/\partial\alpha = 0$ and $\partial l/\partial\beta = 0$ simultaneously for α and β.] Analogs of the properties of MLEs listed above also hold in this multiparameter case. Unfortunately, the process of finding MLEs when there are several parameters is usually quite difficult. (The normal distribution is a notable exception.)

For each of the distributions in Secs. 6.2.2 (except for the Johnson S_B, Johnson S_U, and triangular distributions) and 6.2.3, we listed either formulas for the MLEs or a method for obtaining them numerically. For the gamma MLEs, Table 6.21 can be used with standard linear interpolation. For the beta MLEs, Table 6.22 can be used; one could either simply pick $(\hat{\alpha}_1, \hat{\alpha}_2)$ corresponding to the closest table values of G_1 and G_2 or devise a scheme for two-dimensional interpolation.

6.6 ACTIVITY III: DETERMINING HOW REPRESENTATIVE THE FITTED DISTRIBUTIONS ARE

After determining one or more probability distributions that might fit our observed data in Activities I and II, we must now closely examine these distributions to see how well they represent the true underlying distribution for our data. If several of

these distributions are "representative," we must also determine which distribution provides the best fit. In general, none of our fitted distributions will probably be *exactly* correct. What we are really trying to do is to determine a distribution that is accurate enough for the intended purposes of the model.

In this section we discuss both heuristic procedures and goodness-of-fit hypothesis tests for determining the "quality" of fitted distributions.

6.6.1 Heuristic Procedures

We will discuss five heuristic or graphical procedures for comparing fitted distributions with the true underlying distribution; several additional techniques can be found in Averill M. Law & Associates (2013).

Density-Histogram Plots and Frequency Comparisons

For continuous data, a *density-histogram plot* can be made by plotting $\Delta b\, \hat{f}(x)$ over the histogram $h(x)$ and looking for similarities. [Recall that the area under $h(x)$ is Δb.] A *frequency comparison* is an alternative graphical comparison of a histogram of the data with the density function $\hat{f}(x)$ of a fitted distribution. Let $[b_0, b_1), [b_1, b_2), \ldots, [b_{k-1}, b_k)$ be a set of k histogram intervals each with width $\Delta b = b_j - b_{j-1}$. Let h_j be the *observed* proportion of the X_i's in the jth interval $[b_{j-1}, b_j)$ and let r_j be the *expected* proportion of the n observations that would fall in the jth interval if the fitted distribution were in fact the true one; i.e., r_j is given by (see Prob. 6.13)

$$r_j = \int_{b_{j-1}}^{b_j} \hat{f}(x)\, dx \tag{6.4}$$

Then the frequency comparison is made by plotting both h_j and r_j in the jth histogram interval for $j = 1, 2, \ldots, k$.

For discrete data, a *frequency comparison* is a graphical comparison of a histogram of the data with the mass function $\hat{p}(x)$ of a fitted distribution. Let h_j be the observed proportion of the X_i's that are equal to x_j, and let r_j be the expected proportion of the n observations that would be equal to x_j if the fitted distribution were in fact the true one, i.e., $r_j = \hat{p}(x_j)$. Then the frequency comparison is made by plotting both h_j and r_j versus x_j for all relevant values of x_j.

For either the continuous or discrete case, if the fitted distribution is a good representation for the true underlying distribution of the data (and if the sample size n is sufficiently large), then r_j and h_j should closely agree.

EXAMPLE 6.9. For the interarrival-time data of Example 6.4, we hypothesized an exponential distribution and obtained the MLE $\hat{\beta} = 0.399$ in Example 6.6. Thus, the density of the fitted distribution is

$$\hat{f}(x) = \begin{cases} 2.506e^{-x/0.399} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

For the histogram in Fig. 6.32(c), we give a density-histogram plot in Fig. 6.35.

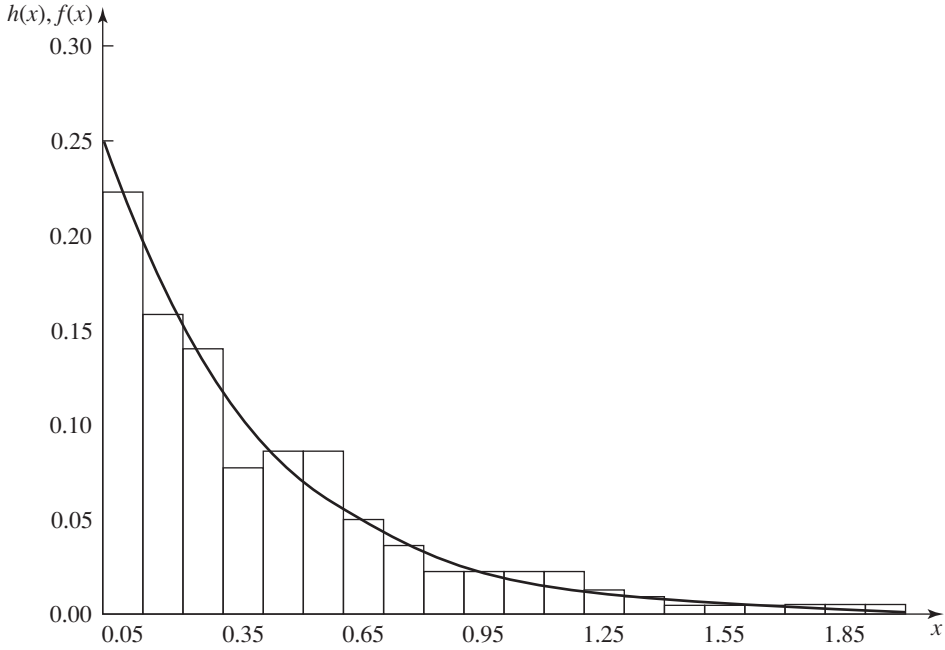


FIGURE 6.35
Density-histogram plot for the fitted exponential distribution and the interarrival-time data.

EXAMPLE 6.10. The demand-size data of Example 6.5 were hypothesized to have come from a geometric distribution, and the MLE of the parameter p was found in Example 6.7 to be $\hat{p} = 0.346$. Thus, the mass function of the fitted distribution is

$$\hat{p}(x) = \begin{cases} 0.346(0.654)^x & \text{if } x = 0, 1, 2, \ldots \\ 0 & \text{otherwise} \end{cases}$$

For the histogram in Fig. 6.34, $r_j = \hat{p}(x_j) = \hat{p}(j - 1)$ for $j = 1, 2, \ldots, 12$, and the frequency comparison is given in Fig. 6.36, where the h_j's are represented by the white vertical bars and the r_j's by the gray vertical bars. Once again, the agreement is good except possibly for $x_2 = 1$.

Distribution-Function-Differences Plots

The density-histogram plot can be thought of as a comparison of the individual probabilities of the fitted distribution and of the individual probabilities of the true underlying distribution. We can also make a graphical comparison of the cumulative probabilities (distribution functions). Define a (new) empirical distribution function $F_n(x)$ as follows:

$$F_n(x) = \frac{\text{number of } X_i\text{'s} \le x}{n} \tag{6.5}$$

which is the proportion of the observations that are less than or equal to x. Then we could plot $\hat{F}(x)$ (the distribution function of the fitted distribution) and $F_n(x)$ on the

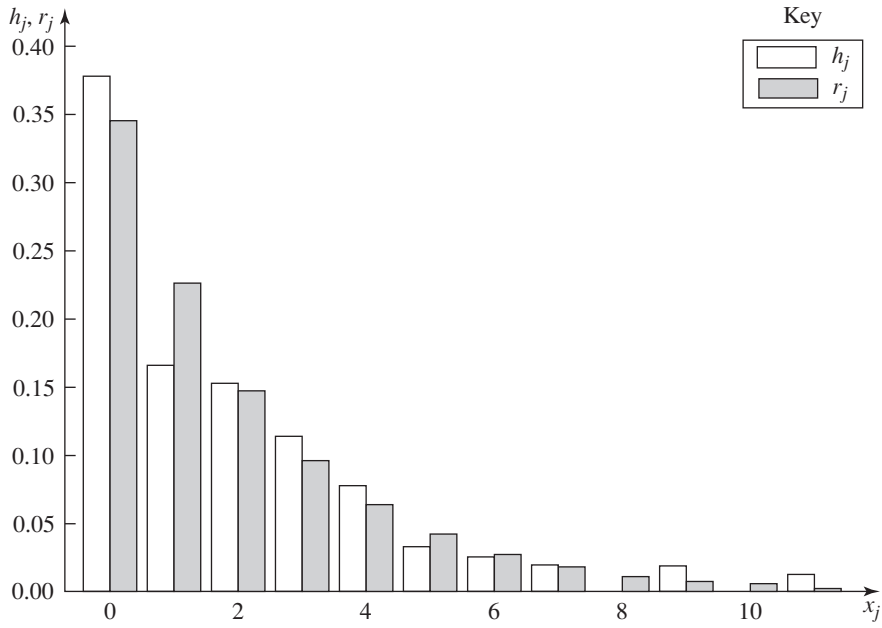


FIGURE 6.36
Frequency comparison for the fitted geometric distribution and the demand-size data.

same graph and look for similarities. However, distribution functions generally do not have as characteristic an appearance as density or mass functions do. In fact, many distribution functions have some sort of "S" shape, and eyeballing for differences or similarities in S-shaped curves is somewhat perplexing. We therefore define the *distribution-function-differences plot* to be a plot of the differences between $\hat{F}(x)$ and $F_n(x)$, over the range of the data. If the fitted distribution is a perfect fit and the sample size is infinite, then this plot will be a horizontal line at height 0. Thus, the greater the vertical deviations from this line, the worse the quality of fit. Many fitted distributions that are bad in an absolute sense have large deviations at the lower end of the range of the data.

EXAMPLE 6.11. A distribution-function-differences plot for the interarrival-time data of Example 6.4 and the fitted exponential distribution is given in Fig. 6.37. This plot indicates a good fit except possibly at the lower end of the range of the observed data. (The dotted horizontal lines are error bounds that depend on the sample size n. If a differences plot crosses these lines, then this is a strong indication of a bad fit. These error bounds were determined from the differences for each of the 35,000 data sets discussed in Sec. 6.7.)

EXAMPLE 6.12. A distribution-function-differences plot for the demand-size data of Example 6.5 and the fitted geometric distribution is given in Fig. 6.38. This plot indicates a good fit except possibly at the lower end of the range of the observed data.

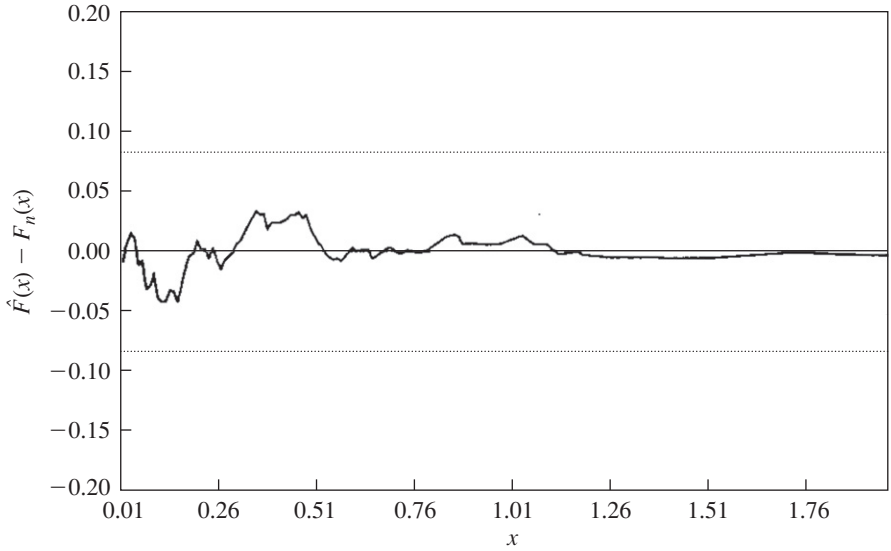


FIGURE 6.37
Distribution-function-differences plot for the fitted exponential distribution and the interarrival-time data.

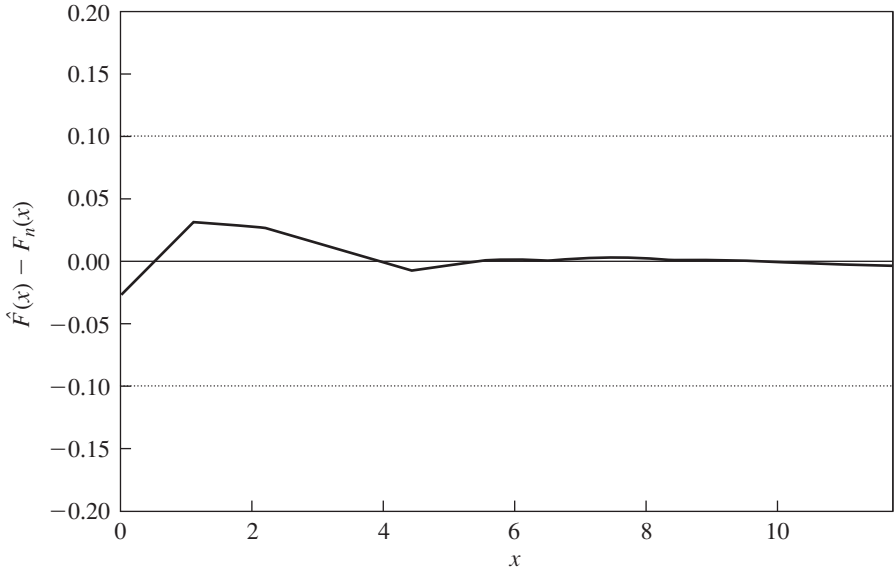


FIGURE 6.38
Distribution-function-differences plot for the fitted geometric distribution and the demand-size data.

Probability Plots

A probability plot can be thought of as another graphical comparison of an estimate of the true distribution function of our data $X_1, X_2, \ldots, X_n$ with the distribution function of a fitted distribution. There are many kinds (and uses) of probability plots, only two of which we describe here; see Barnett (1975), Hahn and Shapiro (1994), and Wilk and Gnanadesikan (1968) for additional discussions.

As in Sec. 6.2.4, let $X_{(i)}$ be the ith smallest of the X_j's, sometimes called the ith *order statistic* of the X_j's. A reasonable estimate of the distribution function $F(x)$ of a random variable X is $F_n(x)$, which was defined by Eq. (6.5). Note that $F_n(X_{(i)}) = i/n$. For purposes of probability plotting, however, it turns out to be somewhat inconvenient to have $F_n(X_{(n)}) = 1$, that is, to have an empirical distribution function that is equal to 1 for a finite value of x (see Prob. 6.14). We therefore will use the following empirical distribution function here:

$$\tilde{F}_n(X_{(i)}) = F_n(X_{(i)}) - \frac{0.5}{n} = \frac{i - 0.5}{n}$$

for $i = 1, 2, \ldots, n$. [Clearly, for moderately large n, this adjustment is quite small. Other adjustments have been suggested, such as $i/(n + 1)$.] A straightforward procedure would then be to plot the n points $(X_{(1)}, 0.5/n), (X_{(2)}, 1.5/n), \ldots, (X_{(n)}, (n - 0.5)/n)$, compare this result with a plot of the distribution function of a distribution being considered as a model for the data, and look for similarities. However, as stated above, many distribution functions have some sort of "S" shape, and eyeballing S-shaped curves for similarities or differences is difficult. Most of us, however, *can* recognize whether a set of plotted points appears to lie more or less along a *straight* line, and probability-plotting techniques reduce the problem of comparing distribution functions to one of looking for a straight line.

Let $q_i = (i - 0.5)/n$ for $i = 1, 2, \ldots, n$, so that $0 < q_i < 1$. For any *continuous* data set (see Prob. 6.15), a *quantile–quantile* (*Q–Q*) *plot* (see Sec. 6.4.3 for the definition of a quantile) is a graph of the q_i-quantile of a fitted (model) distribution function $\hat{F}(x)$, namely, $x_{q_i}^M = \hat{F}^{-1}(q_i)$, versus the q_i-quantile of the empirical distribution function $\tilde{F}_n(x)$, namely, $x_{q_i}^S = \tilde{F}_n^{-1}(q_i) = X_{(i)}$, for $i = 1, 2, \ldots, n$. The definition of a *Q–Q* plot is illustrated in Fig. 6.39, where we have represented $\tilde{F}_n(x)$ as a smooth curve for convenience. Corresponding to each ordinate value q are the two quantiles x_q^M and x_q^S.

If $\hat{F}(x)$ is the same distribution as the true underlying distribution $F(x)$, and if the sample size n is large, then $\hat{F}(x)$ and $\tilde{F}_n(x)$ will be close together and the *Q–Q* plot will be approximately linear with an intercept of 0 and a slope of 1. Even if $\hat{F}(x)$ is the correct distribution, there will be departures from linearity for small to moderate sample sizes.

A *probability–probability* (*P–P*) *plot* is a graph of the model probability $\hat{F}(X_{(i)})$ versus the sample probability $\tilde{F}_n(X_{(i)}) = q_i$, for $i = 1, 2, \ldots, n$; it is valid for both continuous and discrete data sets. This definition is also illustrated in Fig. 6.39. Corresponding to each abscissa value p are the two probabilities $\hat{F}(p)$ and $\tilde{F}_n(p)$. If

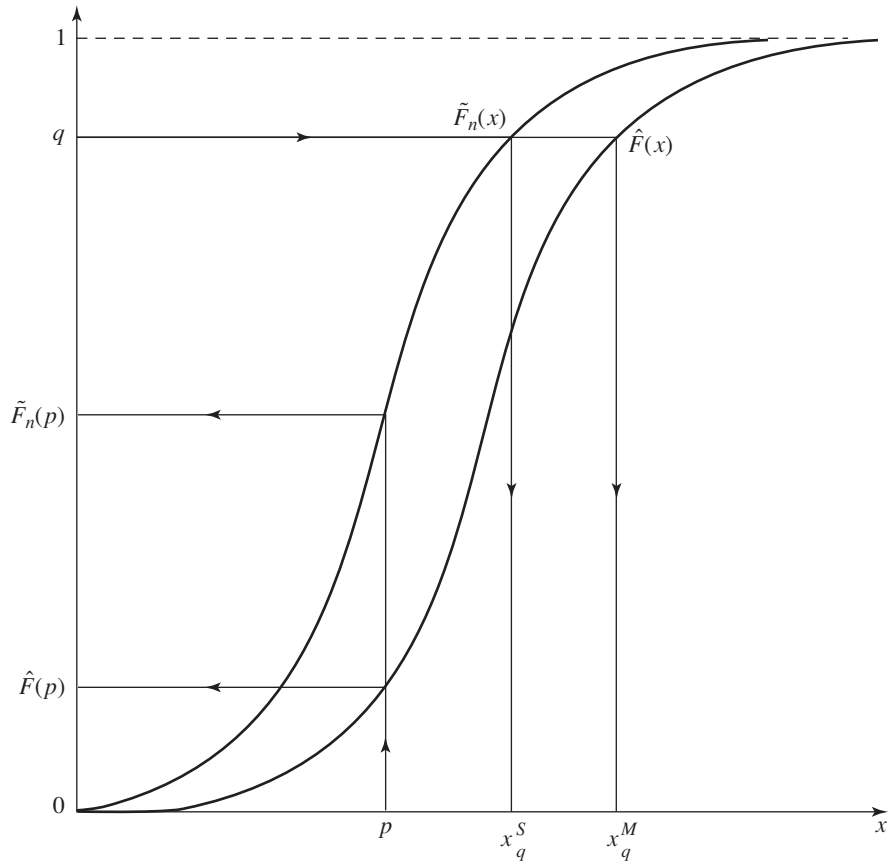


FIGURE 6.39
Definitions of *Q–Q* and *P–P* plots.

$\hat{F}(x)$ and $\tilde{F}_n(x)$ are close together, then the *P–P* plot will also be approximately linear with an intercept of 0 and a slope of 1.

The *Q–Q* plot will amplify differences that exist between the tails of the model distribution function $\hat{F}(x)$ and the tails of the empirical distribution function $\tilde{F}_n(x)$, whereas the *P–P* plot will amplify differences between the middle of $\hat{F}(x)$ and the middle of $\tilde{F}_n(x)$. The difference between the right tails of the distribution functions in Fig. 6.40 is amplified by the *Q–Q* plot but not the *P–P* plot. On the other hand, the difference between the "middles" of the two distribution functions in Fig. 6.41 is amplified by the *P–P* plot.

The above formulations of *Q–Q* and *P–P* plots implicitly assumed that the X_i's were distinct (no ties); this certainly will not always be the case. To modify the definitions when the X_i's may not be distinct, let $Y_1, Y_2, \ldots, Y_l$ be the distinct values in the sample $X_1, X_2, \ldots, X_n$ arranged in increasing order,

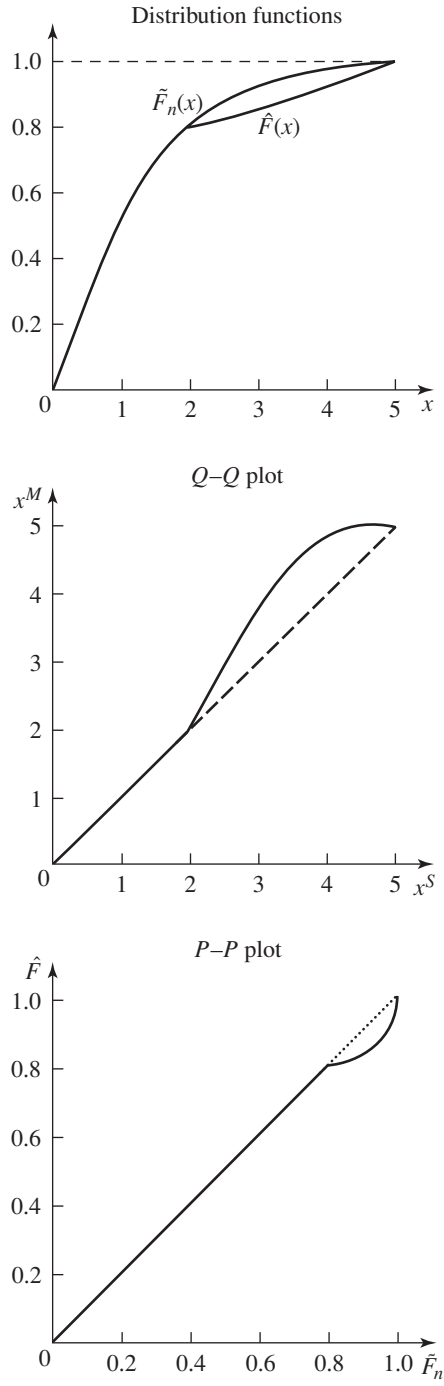


FIGURE 6.40
The difference between the right tails of $\hat{F}(x)$ and $\tilde{F}_n(x)$ amplified by the Q–Q plot.

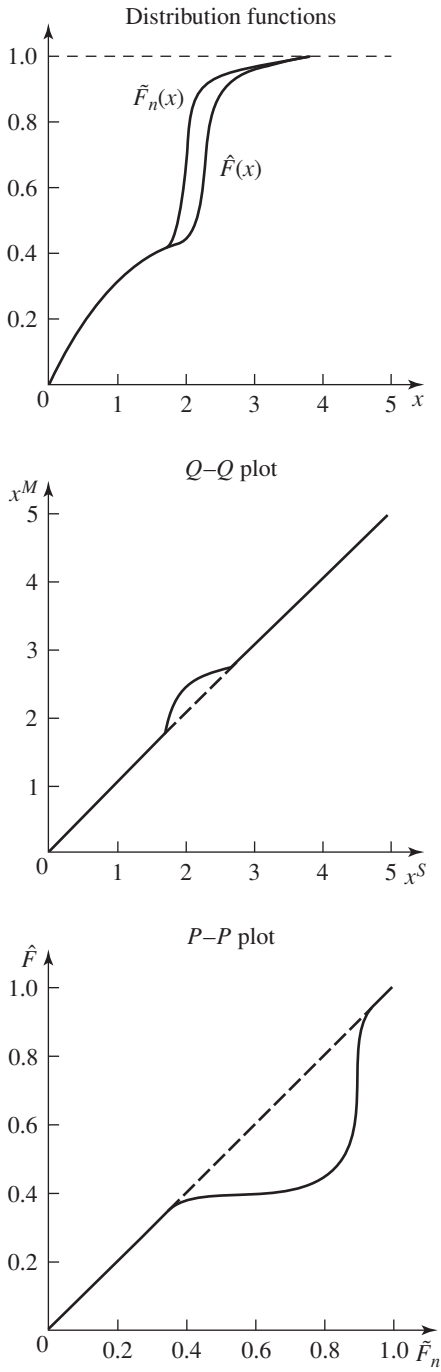


FIGURE 6.41
The difference between the "middles" of $\hat{F}(x)$ and $\tilde{F}_n(x)$ amplified by the *P–P* plot.

TABLE 6.11
Approaches for computing $\hat{F}$ or $\hat{F}^{-1}$ for certain mathematically intractable distributions

	$\hat{F}$	$\hat{F}^{-1}$
Gamma	See Bhattacharjee (1970)	See Best and Roberts (1975)
Normal	See Milton and Hotchkiss (1969)	See Moro (1995)
Lognormal	Fit a normal distribution to $Y_i = \ln X_i$ for $i = 1, 2, \ldots, n$; see Sec. 6.2.2	Same as $\hat{F}$
Beta	See Bosten and Battiste (1974)	See Cran, Martin, and Thomas (1977)
Pearson type V	Fit a gamma distribution to $Y_i = 1/X_i$ for $i = 1, 2, \ldots, n$; see Sec. 6.2.2	Same as $\hat{F}$
Pearson type VI	Fit a beta distribution to $Y_i = X_i/(1 + X_i)$ for $i = 1, 2, \ldots, n$; see Sec. 6.2.2	Same as $\hat{F}$
Johnson S_B	See the normal distribution	Same as $\hat{F}$
Johnson S_U	See the normal distribution	Same as $\hat{F}$

where $l \leq n$. (If the X_i's are distinct, then $Y_i = X_{(i)}$ for $i = 1, 2, \ldots, n$.) Let q_i' be defined by

$$q_i' = (\text{proportion of } X_j\text{'s} \leq Y_i) - \frac{0.5}{n}$$

In other words, $q_i' = \tilde{F}_n(Y_i)$. Then q_i' replaces q_i and Y_i replaces $X_{(i)}$ in the definitions of *Q–Q* and *P–P* plots.

The construction of a *Q–Q* plot requires the calculation of the model quantile $\hat{F}^{-1}(q_i)$. For the uniform, exponential, Weibull, and log-logistic distributions, there is no problem, since a closed-form expression for $\hat{F}^{-1}$ is available. For the other continuous distributions, we give in Table 6.11 either a transformation for addressing the problem or a reference to a numerical approximation for $\hat{F}^{-1}$. Also given in Table 6.11 are similar prescriptions for computing the model probability $\hat{F}(X_{(i)})$, which is required for a *P–P* plot. Functions for computing $\hat{F}$ or $\hat{F}^{-1}$ are also available in the IMSL Statistical Library [Rogue Wave (2013)], and the ExpertFit statistical package (see Sec. 6.7) performs *Q–Q* and *P–P* plots automatically.

EXAMPLE 6.13. A *Q–Q* plot for the fitted exponential distribution and the interarrival-time data is given in Fig. 6.42. The plot is fairly linear except for large values of q_i'. This is not uncommon, since the *Q–Q* plot will amplify small differences between $\hat{F}(x)$ and $\tilde{F}_n(x)$ when they are both close to 1. The corresponding *P–P* plot is given in Fig. 6.43. Its linearity indicates that the middle of the fitted exponential agrees closely with the middle of the true underlying distribution.

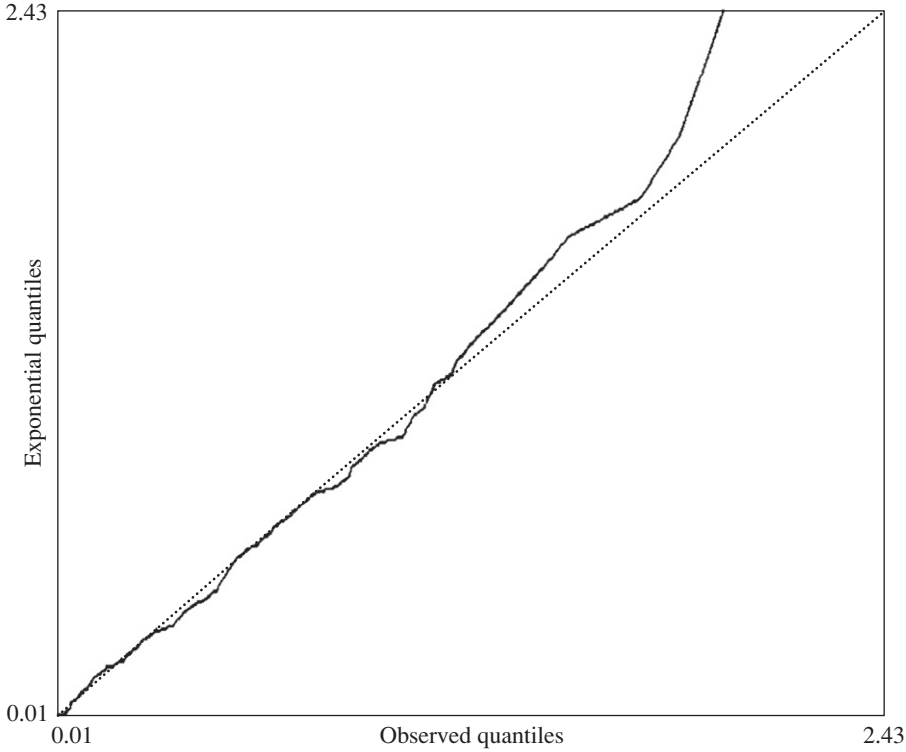


FIGURE 6.42
Q–Q plot for exponential distribution and interarrival-time data.

EXAMPLE 6.14. The *P–P* plot for the fitted geometric distribution and the demand-size data is given in Fig. 6.44. Once again we find the *P–P* plot to be reasonably linear, indicating agreement between the geometric and true distributions.

6.6.2 Goodness-of-Fit Tests

A *goodness-of-fit test* is a statistical hypothesis test (see Sec. 4.5) that is used to assess formally whether the observations $X_1, X_2, \ldots, X_n$ are an independent sample from a particular distribution with distribution function $\hat{F}$. That is, a goodness-of-fit test can be used to test the following null hypothesis:

H_0: The X_i's are IID random variables with distribution function $\hat{F}$

Before proceeding with a discussion of several specific goodness-of-fit tests, we feel compelled to comment on the formal structure and properties of these tests. First, failure to reject H_0 should *not* be interpreted as "accepting H_0 as being true."

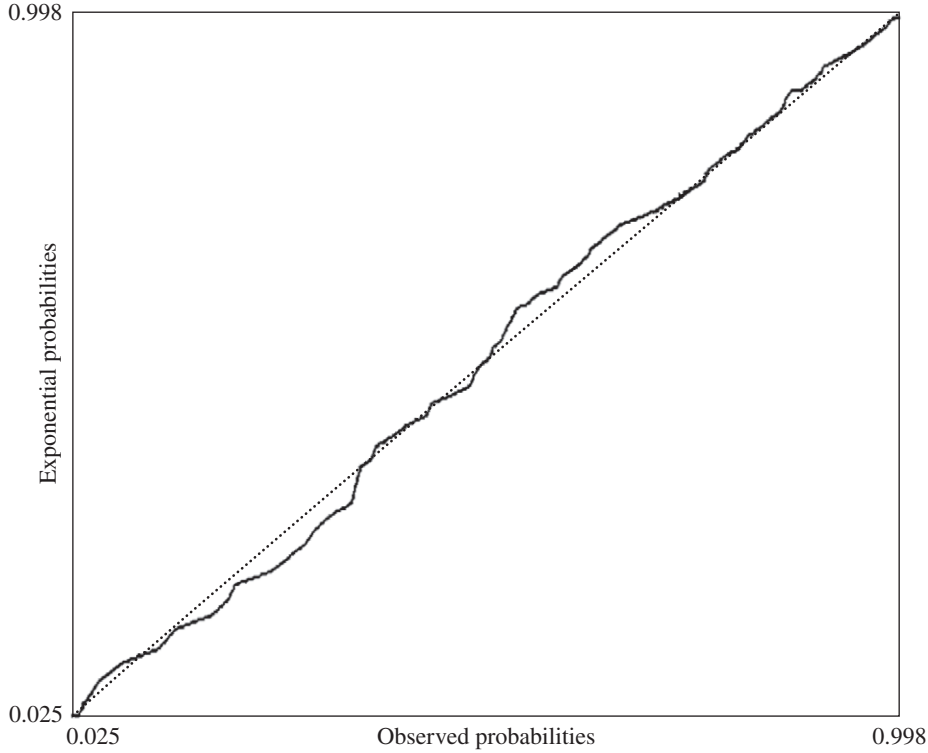


FIGURE 6.43
P–P plot for exponential distribution and interarrival-time data.

These tests are often not very powerful for small to moderate sample sizes n; that is, they are not very sensitive to subtle disagreements between the data and the fitted distribution. Instead, they should be regarded as a systematic approach for detecting fairly gross differences. On the other hand, if n is very large, then these tests will almost always reject H_0 [see Gibbons (1985, p. 76)]. Since H_0 is virtually never *exactly* true, even a minute departure from the hypothesized distribution will be detected for large n. This is an unfortunate property of these tests, since it is usually sufficient to have a distribution that is "nearly" correct.

Chi-Square Tests

The oldest goodness-of-fit hypothesis test is the *chi-square test*, which dates back at least to the paper of K. Pearson (1900). As we shall see, a chi-square test may be thought of as a more formal comparison of a histogram with the fitted density or mass function (see the frequency comparison in Sec. 6.6.1).

To compute the chi-square test statistic in either the continuous or discrete case, we must first divide the entire range of the fitted distribution into k adjacent intervals

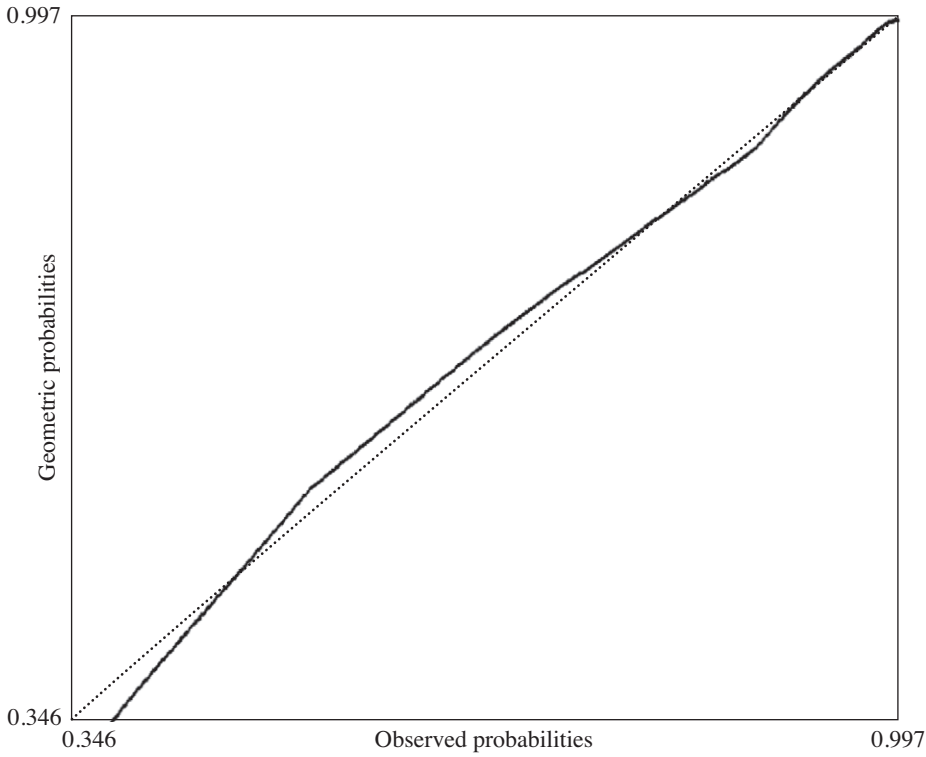


FIGURE 6.44
P–P plot for geometric distribution and demand-size data.

$[a_0, a_1)$, $[a_1, a_2)$, . . . , $[a_{k-1}, a_k)$, where it could be that $a_0 = -\infty$, in which case the first interval is $(-\infty, a_1)$, or $a_k = +\infty$, or both. Then we tally

$$N_j = \text{number of } X_i\text{'s in the } j\text{th interval } [a_{j-1}, a_j)$$

for $j = 1, 2, \ldots, k$. (Note that $\sum_{j=1}^{k} N_j = n$.) Next, we compute the expected *proportion* p_j of the X_i's that would fall in the jth interval if we were sampling from the fitted distribution. In the continuous case,

$$p_j = \int_{a_{j-1}}^{a_j} \hat{f}(x)\, dx$$

where $\hat{f}$ is the density of the fitted distribution. For discrete data,

$$p_j = \sum_{a_{j-1} \le x_i < a_j} \hat{p}(x_i)$$

where $\hat{p}$ is the mass function of the fitted distribution. Finally, the test statistic is

$$\chi^2 = \sum_{j=1}^{k} \frac{(N_j - np_j)^2}{np_j}$$

Since np_j is the expected number of the n X_i's that would fall in the jth interval if H_0 were true (see Prob. 6.17), we would expect χ^2 to be small if the fit were good. Therefore, we reject H_0 if χ^2 is too large. The precise form of the test depends on whether we have estimated any of the parameters of the fitted distribution from our data.

First, suppose that all parameters of the fitted distribution are known; that is, we specified the fitted distribution without making use of the data in any way. [This all-parameters-known case might appear to be of little practical use, but there are at least two applications for it in simulation: (1) In the Poisson-process test (later in this section), we test to see whether times of arrival can be regarded as being IID U(0, T) random variables, where T is a constant independent of the data; and (2) in empirical testing of random-number generators (Sec. 7.4.1), we test for a U(0, 1) distribution.] Then if H_0 is true, χ^2 converges in distribution (as $n \to \infty$) to a chi-square distribution with $k - 1$ df, which is the same as the gamma[$(k - 1)/2$, 2] distribution. Thus, for large n, a test with *approximate* level α is obtained by rejecting H_0 if $\chi^2 > \chi^2_{k-1,1-\alpha}$ (see Fig. 6.45), where $\chi^2_{k-1,1-\alpha}$ is the upper $1 - \alpha$ critical point for a chi-square distribution with $k - 1$ df. (Values for $\chi^2_{k-1,1-\alpha}$ can be found in Table T.2 at the end of the book.) Note that the chi-square test is only *valid*, i.e., is of level α, asymptotically as $n \to \infty$.

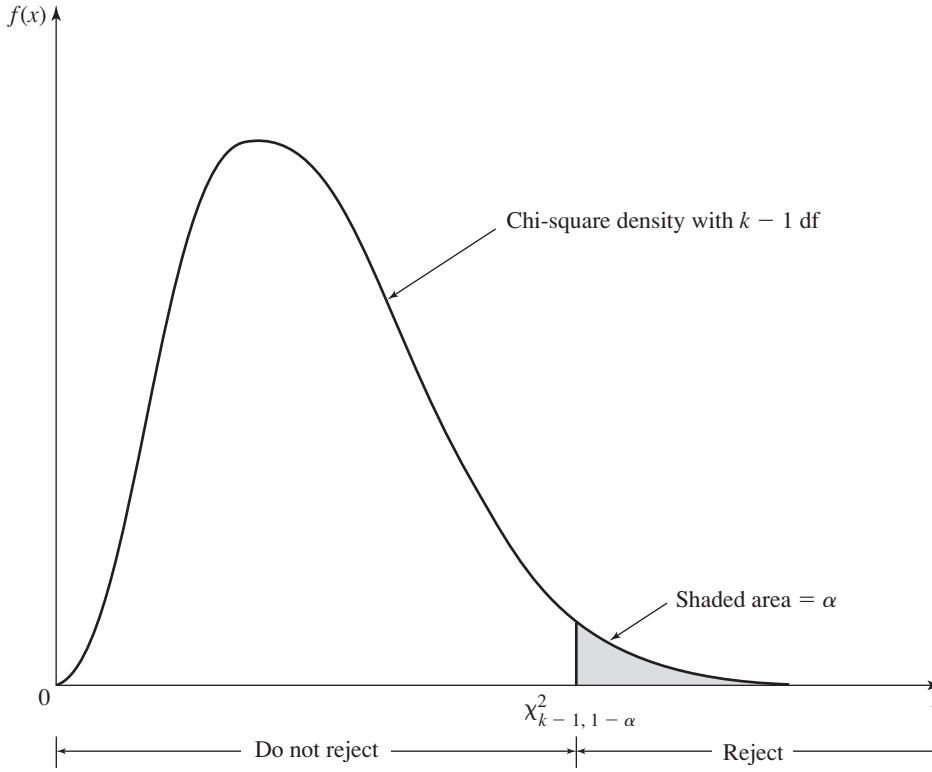


FIGURE 6.45
The chi-square test when all parameters are known.

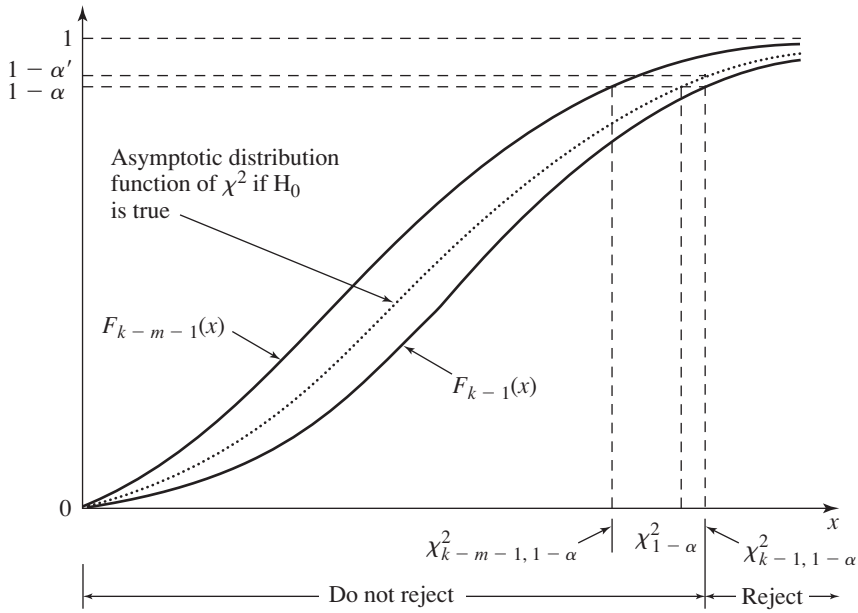


FIGURE 6.46
The chi-square test when m parameters are estimated by their MLEs.

Second, suppose that in order to specify the fitted distribution, we had to estimate m parameters ($m \geq 1$) from the data. When MLEs are used, Chernoff and Lehmann (1954) showed that if H_0 is true, then as $n \to \infty$ the distribution function of χ^2 converges to a distribution function that lies *between* the distribution functions of chi-square distributions with $k - 1$ and $k - m - 1$ df. (See Fig. 6.46, where F_{k-1} and F_{k-m-1} represent the distribution functions of chi-square distributions with $k - 1$ and $k - m - 1$ df, respectively, and the dotted distribution function is the one to which the distribution function of χ^2 converges as $n \to \infty$.) If we let $\chi^2_{1-\alpha}$ be the upper $1 - \alpha$ critical point of the asymptotic distribution of χ^2, then

$$\chi^2_{k-m-1,1-\alpha} \leq \chi^2_{1-\alpha} \leq \chi^2_{k-1,1-\alpha}$$

as shown in Fig. 6.46; unfortunately, the value of $\chi^2_{1-\alpha}$ will not be known in general. It is clear that we should reject H_0 if $\chi^2 > \chi^2_{k-1,1-\alpha}$ and we should not reject H_0 if $\chi^2 < \chi^2_{k-m-1,1-\alpha}$; an ambiguous situation occurs when

$$\chi^2_{k-m-1,1-\alpha} \leq \chi^2 \leq \chi^2_{k-1,1-\alpha}$$

It is often recommended that we reject H_0 only if $\chi^2 > \chi^2_{k-1,1-\alpha}$, since this is *conservative*; that is, the actual probability α' of committing a Type I error [rejecting H_0 when it is true (see Sec. 4.5)] is at *least* as small as the *stated* probability α (see Fig. 6.46). This choice, however, will entail loss of power (probability of rejecting a false H_0) of the test. Usually, m will be no more than 2, and if k is fairly large, the

difference between $\chi^2_{k-m-1,1-\alpha}$ and $\chi^2_{k-1,1-\alpha}$ will not be too great. Thus, we reject H_0 if (and only if) $\chi^2 > \chi^2_{k-1,1-\alpha}$, as in the all-parameters-known case. The rejection region for χ^2 is indicated in Fig. 6.46.

The most troublesome aspect of carrying out a chi-square test is choosing the number and size of the intervals. This is a difficult problem, and no definitive prescription can be given that is guaranteed to produce good results in terms of validity (actual level of the test close to the desired level α) and high power for all alternative distributions and all sample sizes. There are, however, a few guidelines that are often followed. First, some of the ambiguity in interval selection is eliminated if the intervals are chosen so that $p_1 = p_2 = \cdots = p_k$, which is called the *equiprobable approach*. In the continuous case, this might be inconvenient to do for some distributions since the distribution function of the fitted distribution must be inverted (see Example 6.15 below). Furthermore, for discrete distributions, we will generally be able to make the p_j's only approximately equal (see Example 6.16).

We now discuss how to choose the intervals to ensure "validity" of the test. Let $a = \min_{1 \le j \le k} np_j$, and let $y(5)$ be the number of np_j's less than 5. Based on extensive theoretical and empirical investigations (for the all-parameters-known case), Yarnold (1970) states that the chi-square test will be approximately valid if $k \ge 3$ and $a \ge 5y(5)/k$. For equiprobable intervals, these conditions will be satisfied if $k \ge 3$ and $np_j \ge 5$ for all j.

We now turn our attention to the power of the chi-square test. A test is said to be *unbiased* if it is more likely to reject H_0 when it is false than when it is true or, in other words, power is greater than the probability of a Type I error. A test without this property would certainly be undesirable. It can be shown that the chi-square test is always unbiased for the equiprobable approach [see Kendall and Stuart (1979, pp. 455–461)]. If the np_j's are not equal (and many are small), it is possible to obtain a valid test that is highly biased [see Haberman (1988)].

In general, there is no rule for choosing the intervals so that high power is obtained for all alternative distributions. For a particular null distribution, a fixed sample size n, and the equiprobable approach, Kallenberg, Oosterhoff, and Schriever (1985) showed empirically that power is an increasing function of the number of intervals k for some alternative distributions, and a decreasing function of k for other alternative distributions. Surprisingly, they also found in certain cases that the power was greater when the np_j's were smaller in the tails (see Prob. 6.18).

In the absence of a definitive guideline for choosing the intervals, we recommend the equiprobable approach and $np_j \ge 5$ for all j in the continuous case. This guarantees a valid and unbiased test. In the discrete case, we suggest making the np_j's approximately equal and all at least 5. The lack of a clear prescription for interval selection is the major drawback of the chi-square test. In some situations entirely different conclusions can be reached from the *same* data set depending on how the intervals are specified, as illustrated in Example 6.17. The chi-square test nevertheless remains in wide use, since it can be applied to any hypothesized distribution; as we shall see below, other goodness-of-fit tests do not enjoy such a wide range of applicability.

TABLE 6.12
A chi-square goodness-of-fit test for the interarrival-time data

j	Interval	N_j	np_j	$\frac{(N_j - np_j)^2}{np_j}$
1	[0, 0.020)	8	10.950	0.795
2	[0.020, 0.042)	11	10.950	0.000
3	[0.042, 0.065)	14	10.950	0.850
4	[0.065, 0.089)	14	10.950	0.850
5	[0.089, 0.115)	16	10.950	2.329
6	[0.115, 0.142)	10	10.950	0.082
7	[0.142, 0.172)	7	10.950	1.425
8	[0.172, 0.204)	5	10.950	3.233
9	[0.204, 0.239)	13	10.950	0.384
10	[0.239, 0.277)	12	10.950	0.101
11	[0.277, 0.319)	7	10.950	1.425
12	[0.319, 0.366)	7	10.950	1.425
13	[0.366, 0.419)	12	10.950	0.101
14	[0.419, 0.480)	10	10.950	0.082
15	[0.480, 0.553)	20	10.950	7.480
16	[0.553, 0.642)	9	10.950	0.347
17	[0.642, 0.757)	11	10.950	0.000
18	[0.757, 0.919)	9	10.950	0.347
19	[0.919, 1.195)	14	10.950	0.850
20	[1.195, ∞)	10	10.950	0.082
				$\chi^2 = 22.188$

EXAMPLE 6.15. We now use a chi-square test to compare the $n = 219$ interarrival times of Table 6.7 with the fitted exponential distribution having distribution function $\hat{F}(x) = 1 - e^{-x/0.399}$ for $x \geq 0$. If we form, say, $k = 20$ intervals with $p_j = 1/k = 0.05$ for $j = 1, 2, \ldots, 20$, then $np_j = (219)(0.05) = 10.950$, so that this satisfies the guidelines that the intervals be chosen with equal p_j's and $np_j \geq 5$. In this case, it is easy to find the a_j's, since $\hat{F}$ can be inverted. That is, we set $a_0 = 0$ and $a_{20} = \infty$, and for $j = 1, 2, \ldots, 19$ we want a_j to satisfy $\hat{F}(a_j) = j/20$; this is equivalent to setting $a_j = -0.399 \ln(1 - j/20)$ for $j = 1, 2, \ldots, 19$ since $a_j = \hat{F}^{-1}(j/20)$. (For continuous distributions such as the normal, gamma, and beta, the inverse of the distribution function does not have a simple closed form. In these cases, however, F^{-1} can be evaluated by numerical methods; consult the references given in Table 6.11.) The computations for the test are given in Table 6.12, and the value of the test statistic is $\chi^2 = 22.188$. Referring to Table T.2, we see that $\chi^2_{19,0.90} = 27.204$, which is not exceeded by χ^2, so we would not reject H_0 at the $\alpha = 0.10$ level. (Note that we would also not reject H_0 for certain larger values of α such as 0.25.) Thus, this test gives us no reason to conclude that our data are poorly fitted by the expo(0.399) distribution.

EXAMPLE 6.16. As an illustration of the chi-square test in the discrete case, we test how well the fitted geom(0.346) distribution agrees with the demand-size data of Table 6.9. As is usually the case for discrete distributions, we cannot make the p_j's exactly equal, but by grouping together adjacent points on which the mass function $\hat{p}(x)$ is defined (here, the nonnegative integers), we can define intervals that make the p_j's roughly the same. One way to do this is to note that the mode of the fitted distribution is 0; thus, $\hat{p}(0) = 0.346$ is the largest value of the mass function. The large value for the

TABLE 6.13
A chi-square goodness-of-fit test for the demand-size data

j	Interval	N_j	np_j	$\frac{(N_j - np_j)^2}{np_j}$
1	{0}	59	53.960	0.471
2	{1, 2}	50	58.382	1.203
3	{3, 4, . . .}	47	43.658	0.256
				$\chi^2 = 1.930$

mode limits our choice of intervals, and we end up with the three intervals given in Table 6.13, where the calculations for the chi-square test are also presented. In particular, $\chi^2 = 1.930$, which is less than the critical value $\chi^2_{2,0.90} = 4.605$. Thus, we would not reject H_0 at the $\alpha = 0.10$ level, and we have no reason to believe that the demand-size data are not fitted well by a geom(0.346) distribution.

EXAMPLE 6.17. If we fit the log-logistic distribution to the 856 ship-loading times of Fig. 6.3, then we obtain the MLEs $\hat{\alpha} = 8.841$ and $\hat{\beta} = 0.822$ for the shape and scale parameters, respectively. We now perform a chi-square test at level $\alpha = 0.1$ using $k = 10$, 20, and 40 equiprobable intervals, with the results given in Table 6.14. (Note that all three choices for k satisfy the recommendation that $np_j \geq 5$.) We see that the log-logistic distribution is rejected for 20 intervals, but is not rejected for 10 or 40 intervals.

Kolmogorov-Smirnov Tests

As we just saw, chi-square tests can be thought of as a more formal comparison of a histogram of the data with the density or mass function of the fitted distribution. We also identified a real difficulty in using a chi-square test in the continuous case, namely, that of deciding how to specify the intervals. *Kolmogorov-Smirnov* (K-S) *tests* for goodness of fit, on the other hand, compare an empirical *distribution* function with the *distribution* function $\hat{F}$ of the hypothesized distribution. As we shall see, K-S tests do not require us to group the data in any way, so no information is lost; this also eliminates the troublesome problem of interval specification. Another advantage of K-S tests is that they are valid (exactly) for any sample size n (in the all-parameters-known case), whereas chi-square tests are valid only in an asymptotic sense. Finally, K-S tests tend to be more powerful than chi-square tests against many alternative distributions; see, for example, Stephens (1974).

TABLE 6.14
Chi-square goodness-of-fit tests for the ship-loading data

k	Statistic	Critical value	Result of test
10	11.383	14.684	Do not reject
20	27.645	27.204	Reject
40	50.542	50.660	Do not reject

Nevertheless, K-S tests do have some drawbacks, at least at present. Most seriously, their range of applicability is more limited than that for chi-square tests. First, for discrete data, the required critical values are not readily available and must be computed using a complicated set of formulas [see Conover (1999, pp. 435–437), Gleser (1985), and Pettitt and Stephens (1977)]. Second, the *original* form of the K-S test is valid only if *all* the parameters of the hypothesized distribution are *known* and the distribution is continuous; i.e., the parameters cannot have been estimated from the data. However, the K-S test has been extended to allow for estimation of the parameters in the cases of normal (lognormal), exponential, Weibull, and log-logistic distributions. Although the K-S test in its original (all-parameters-known) form has often been applied directly for any continuous distribution with estimated parameters and for discrete distributions, this will, in fact, produce a conservative test [see Conover (1999, pp. 432, 442)]. That is, the probability of a Type I error will be smaller than specified, with a corresponding loss of power.

To define the K-S statistic, we will use the empirical distribution function $F_n(x)$ defined by Eq. (6.5), which is a (right-continuous) step function such that $F_n(X_{(i)}) = i/n$ for $i = 1, 2, \ldots, n$ (see Prob. 6.19). If $\hat{F}(x)$ is the fitted distribution function, a natural assessment of goodness of fit is some kind of measure of the closeness between the functions F_n and $\hat{F}$. The K-S test statistic D_n is simply the *largest* (vertical) distance between $F_n(x)$ and $\hat{F}(x)$ *for all values of* x and is defined formally by

$$D_n = \sup_x \{|F_n(x) - \hat{F}(x)|\}$$

[The "sup" of a set of numbers A is the smallest value that is greater than or equal to all members of A. The "sup" is used here instead of the more familiar "max" since, in some cases, the maximum may not be well defined. For example, if $A = (0, 1)$, there is no maximum but the "sup" is 1.] The statistic D_n can be computed by calculating

$$D_n^+ = \max_{1 \le i \le n} \left\{ \frac{i}{n} - \hat{F}(X_{(i)}) \right\}, \qquad D_n^- = \max_{1 \le i \le n} \left\{ \hat{F}(X_{(i)}) - \frac{i-1}{n} \right\}$$

and finally letting

$$D_n = \max \{D_n^+, D_n^-\}$$

An example is given in Fig. 6.47 for $n = 4$, where $D_n = D_n^+$. [*Beware!* Incorrect computational formulas are often given for D_n. In particular, one sometimes sees

$$D'_n = \max_{1 \le i \le n} \left\{ \left| \frac{i}{n} - \hat{F}(X_{(i)}) \right| \right\}$$

as a "formula" for D_n. For the situation of Fig. 6.47, it *is* true that $D'_n = D_n$. Consider, however, Fig. 6.48, where $D'_n = \hat{F}(X_{(2)}) - \frac{2}{4}$ but the correct value for D_n is $\hat{F}(X_{(2)}) - \frac{1}{4}$, which occurs *just before* $x = X_{(2)}$. Clearly, $D'_n \neq D_n$ in this case.] Direct computation of D_n^+ and D_n^- requires sorting the data to obtain the $X_{(i)}$'s. However, Gonzalez, Sahni, and Franta (1977) provide an algorithm for computing D_n^+ and D_n^- without sorting.

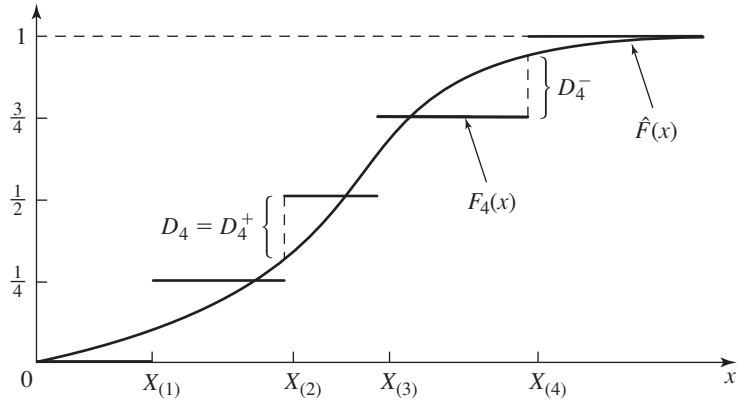


FIGURE 6.47
Geometric meaning of the K-S test statistic D_n for $n = 4$.

Clearly, a large value of D_n indicates a poor fit, so that the form of the test is to reject the null hypothesis H_0 if D_n exceeds some constant $d_{n,1-\alpha}$, where α is the specified level of the test. The numerical value of the critical point $d_{n,1-\alpha}$ depends on how the hypothesized distribution was specified, and we must distinguish several cases.

Case 1

If *all parameters of $\hat{F}$ are known*, i.e., none of the parameters of $\hat{F}$ is estimated in any way from the data, the distribution of D_n does not depend on $\hat{F}$, assuming (of course) that $\hat{F}$ is continuous. This rather remarkable fact means that a single table of values for $d_{n,1-\alpha}$ will suffice for all continuous distribution forms; these tables are widely available [see, for example, Owen (1962)]. Stephens (1974) devised an

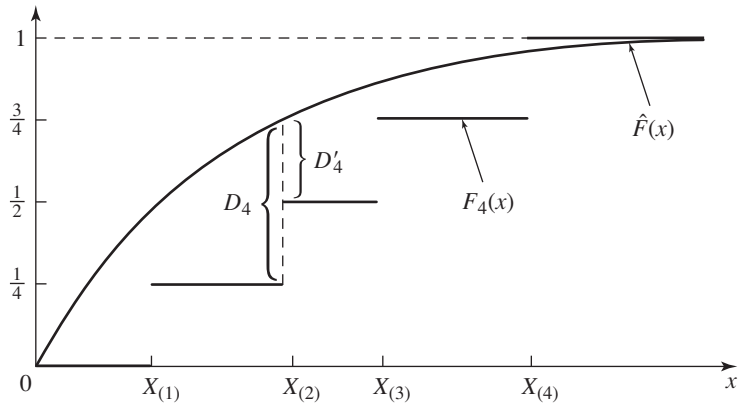


FIGURE 6.48
An example in which the K-S test statistic D_n is not equal to D_n'.

TABLE 6.15
Modified critical values $c_{1-\alpha}$, $c'_{1-\alpha}$, and $c''_{1-\alpha}$ for adjusted K-S test statistics

		$1-\alpha$				
Case	**Adjusted test statistic**	**0.850**	**0.900**	**0.950**	**0.975**	**0.990**
All parameters known	$\left(\sqrt{n} + 0.12 + \frac{0.11}{\sqrt{n}}\right) D_n$	1.138	1.224	1.358	1.480	1.628
N($\overline{X}(n)$, $S^2(n)$)	$\left(\sqrt{n} - 0.01 + \frac{0.85}{\sqrt{n}}\right) D_n$	0.775	0.819	0.895	0.955	1.035
expo($\overline{X}(n)$)	$\left(D_n - \frac{0.2}{n}\right)\left(\sqrt{n} + 0.26 + \frac{0.5}{\sqrt{n}}\right)$	0.926	0.990	1.094	1.190	1.308

accurate approximation that eliminates the need for all but a tiny table; instead of testing for $D_n > d_{n,1-\alpha}$, we reject H_0 if

$$\left(\sqrt{n} + 0.12 + \frac{0.11}{\sqrt{n}}\right) D_n > c_{1-\alpha}$$

where values for $c_{1-\alpha}$ (which do not depend on n) are given in the all-parameters-known row of Table 6.15. This all-parameters-known case is the original form of the K-S test.

Case 2

Suppose that the hypothesized distribution is N(μ, σ^2) with both μ and σ^2 unknown. We can estimate μ and σ^2 by $\overline{X}(n)$ and $S^2(n)$, respectively, and define the distribution function $\hat{F}$ to be that of the N($\overline{X}(n)$, $S^2(n)$) distribution; i.e., let $\hat{F}(x) = \Phi\{[x - \overline{X}(n)]/\sqrt{S^2(n)}\}$, where Φ is the distribution function of the standard normal distribution. Using this $\hat{F}$ (which *has* estimated parameters), D_n is computed in the same way, but different critical points must be used. Lilliefors (1967) estimated (via Monte Carlo simulation) the critical points of D_n as a function of n and $1 - \alpha$. Stephens (1974) performed further Monte Carlo simulations and provided an accurate approximation that obviates the need for large tables; namely, we reject H_0 if

$$\left(\sqrt{n} - 0.01 + \frac{0.85}{\sqrt{n}}\right) D_n > c'_{1-\alpha}$$

where values for $c'_{1-\alpha}$ are given in the N($\overline{X}(n)$, $S^2(n)$) row of Table 6.15. (This case includes a K-S test for the lognormal distribution if the X_i's are the *logarithms* of the basic data points we have hypothesized to have a lognormal distribution; see Sec. 6.2.2.)

Case 3

Suppose the hypothesized distribution is expo(β) with β unknown. Now β is estimated by its MLE $\overline{X}(n)$, and we define $\hat{F}$ to be the expo($\overline{X}(n)$) distribution function; that is, $\hat{F}(x) = 1 - e^{-x/\overline{X}(n)}$ for $x \geq 0$. In this case, critical points for D_n were originally estimated by Lilliefors (1969) in a Monte Carlo study, and exact tables

TABLE 6.16
Modified critical values $c^*_{1-\alpha}$ for the K-S test for the Weibull distribution

	$1-\alpha$			
n	**0.900**	**0.950**	**0.975**	**0.990**
10	0.760	0.819	0.880	0.944
20	0.779	0.843	0.907	0.973
50	0.790	0.856	0.922	0.988
∞	0.803	0.874	0.939	1.007

were later obtained by Durbin (1975) [see also Margolin and Maurer (1976)]. Stephens's (1974) approximation in this case is to reject H_0 if

$$\left(D_n - \frac{0.2}{n}\right)\left(\sqrt{n} + 0.26 + \frac{0.5}{\sqrt{n}}\right) > c''_{1-\alpha}$$

where $c''_{1-\alpha}$ can be found in the expo($\overline{X}(n)$) row of Table 6.15.

Case 4

Suppose the hypothesized distribution is Weibull with both shape parameter α and scale parameter β unknown; we estimate these parameters by their respective MLEs $\hat{\alpha}$ and $\hat{\beta}$. (See the discussion of the Weibull MLEs in Sec. 6.2.2.) And $\hat{F}$ is taken to be the Weibull($\hat{\alpha}$, $\hat{\beta}$) distribution function $\hat{F}(x) = 1 - \exp[-(x/\hat{\beta})^{\hat{\alpha}}]$ for $x \geq 0$, and D_n is computed in the usual fashion. Then H_0 is rejected if the adjusted K-S statistic $\sqrt{n}D_n$ is greater than the modified critical value $c^*_{1-\alpha}$ [see Chandra, Singpurwalla, and Stephens (1981)] given in Table 6.16. Note that critical values are available only for certain sample sizes n, and that the critical values for $n = 50$ and ∞ (an extremely large sample size) are, fortunately, very similar. [Critical values for other n less than 50 are given by Littell, McClave, and Offen (1979).]

Case 5

Suppose that the hypothesized distribution is log-logistic with both shape parameter α and scale parameter β unknown. Let the X_i's here be the *logarithms* of the basic data points. Estimate the parameters by their respective MLEs $\hat{\alpha}$ and $\hat{\beta}$ based on the X_i's (see Sec. 6.2.2). Also $\hat{F}(x)$ is taken to be the logistic distribution function

$$\hat{F}(x) = \left(1 + e^{[-(x - \ln \hat{\beta})]\hat{\alpha}}\right)^{-1} \qquad \text{for } -\infty < x < \infty$$

and D_n is computed in the usual fashion. Then H_0 is rejected if the adjusted K-S statistic $\sqrt{n}D_n$ is greater than the modified critical value $c^{\dagger}_{1-\alpha}$ [see Stephens (1979)] given in Table 6.17. Note that the critical values are available only for certain sample sizes n, and that the critical values for $n = 50$ and ∞ are, fortunately, very similar.

EXAMPLE 6.18. In Example 6.15 we used a chi-square test to check the goodness of fit of the fitted expo(0.399) distribution for the interarrival-time data of Table 6.7. We can also apply a K-S test with $\hat{F}(x) = 1 - e^{-x/0.399}$ for $x \geq 0$, by using Case 3 above.

TABLE 6.17
Modified critical values $c^{\dagger}_{1-\alpha}$ for the K-S test for the log-logistic distribution

	$1-\alpha$			
n	**0.900**	**0.950**	**0.975**	**0.990**
10	0.679	0.730	0.774	0.823
20	0.698	0.755	0.800	0.854
50	0.708	0.770	0.817	0.873
∞	0.715	0.780	0.827	0.886

Using the formulas for D^{+}_{219} and D^{-}_{219}, we found that $D_{219} = 0.047$, so that the adjusted test statistic is

$$\left(D_{219} - \frac{0.2}{219}\right)\left(\sqrt{219} + 0.26 + \frac{0.5}{\sqrt{219}}\right) = 0.696$$

Since 0.696 is less than $0.990 = c''_{0.90}$ (from the last row of Table 6.15), we do not reject H_0 at the $\alpha = 0.10$ level.

Note that critical values for the K-S test in the case of the gamma distribution with estimated parameters have been published by Tadikamalla (1990), but, unfortunately, the largest sample size considered is 40.

Anderson-Darling Tests*

One possible drawback of K-S tests is that they give the same weight to the difference $|F_n(x) - \hat{F}(x)|$ for every value of x, whereas many distributions of interest differ primarily in their tails. The *Anderson-Darling* (A-D) *test* [see Anderson and Darling (1954)], on the other hand, is designed to detect discrepancies in the tails and has higher power than the K-S test against many alternative distributions [see Stephens (1974)]. The A-D statistic A^2_n is defined by

$$A^2_n = n\int_{-\infty}^{\infty} [F_n(x) - \hat{F}(x)]^2\, \psi(x)\hat{f}(x)\, dx$$

where the *weight function* $\psi(x) = 1/\{\hat{F}(x)[1 - \hat{F}(x)]\}$. Thus, A^2_n is just the weighted average of the squared differences $[F_n(x) - \hat{F}(x)]^2$, and the weights are the largest for $\hat{F}(x)$ close to 1 (right tail) and $\hat{F}(x)$ close to 0 (left tail). If we let $Z_i = \hat{F}(X_{(i)})$ for $i = 1, 2, \ldots, n$, then it can be shown that

$$A^2_n = \left(-\left\{\sum_{i=1}^{n} (2i - 1)[\ln Z_i + \ln (1 - Z_{n+1-i})]\right\}/n\right) - n$$

which is the form of the statistic used for actual computations. Since A^2_n is a "weighted distance," the form of the test is to reject the null hypothesis H_0 if A^2_n exceeds some critical value $a_{n,1-\alpha}$, where α is the level of the test.

*Skip this section on the first reading.

TABLE 6.18
Modified critical values for adjusted A-D test statistics

		$1-\alpha$			
Case	**Adjusted test statistic**	**0.900**	**0.950**	**0.975**	**0.990**
All parameters known	A_n^2 for $n \geq 5$	1.933	2.492	3.070	3.857
N($\overline{X}(n)$, $S^2(n)$)	$\left(1 + \frac{4}{n} - \frac{25}{n^2}\right)A_n^2$	0.632	0.751	0.870	1.029
Expo($\overline{X}(n)$)	$\left(1 + \frac{0.6}{n}\right)A_n^2$	1.062	1.321	1.591	1.959
Weibull($\hat{\alpha}$, $\hat{\beta}$)	$\left(1 + \frac{0.2}{\sqrt{n}}\right)A_n^2$	0.637	0.757	0.877	1.038
Log-logistic($\hat{\alpha}$, $\hat{\beta}$)	$\left(1 + \frac{0.25}{\sqrt{n}}\right)A_n^2$	0.563	0.660	0.769	0.906

Critical values $a_{n,1-\alpha}$ are available for the A-D test for the same five continuous distributions [see Stephens (1974, 1976, 1977, 1979) and D'Agostino and Stephens (1986, p. 134)] as for the K-S test. [See Gleser (1985) for a discussion of the discrete case.] Furthermore, $\hat{F}(x)$ is computed in the same manner as before; see Example 6.19 below. Performance of the A-D test is facilitated by the use of adjusted test statistics (except for the all-parameters-known case) and modified critical values, which are given in Table 6.18. If the adjusted test statistic is greater than the modified critical value, then H_0 is rejected.

D'Agostino and Stephens (1986, pp. 151–156) give a procedure for performing an A-D test for the gamma distribution, where the critical values are obtained by interpolating in a table. An A-D test can also be performed for the Pearson type V distribution by using the fact that if X has a Pearson type V distribution, then $1/X$ has a gamma distribution (see Sec. 6.2.2). Cheng and Currie (2009) show how bootstrapping can be used to estimate critical values for the A-D test in the case of distributions with no published tables.

EXAMPLE 6.19. We can use Case 3 of the A-D test to see whether the fitted exponential distribution $\hat{F}(x) = 1 - e^{-x/0.399}$ provides a good model for the interarrival-time data at level $\alpha = 0.10$. We found that $A_{219}^2 = 0.558$, so that the adjusted test statistic is

$$\left(1 + \frac{0.6}{219}\right)A_{219}^2 = 0.560$$

Since 0.560 is less than the modified critical value 1.062 (from the third row of Table 6.18), we do not reject H_0 at level 0.10.

Poisson-Process Tests*

Suppose that we observe a Poisson process (see Sec. 6.12.1) for a *fixed* interval of time $[0, T]$, where T is a constant that is decided upon before we start our

*Skip this section on the first reading.

observation. Let n be the number of events we observe in the interval $[0, T]$, and let t_i be the time of the ith event for $i = 1, 2, \ldots, n$. {Thus, $0 \le t_1 \le t_2 \le \cdots \le t_n \le T$. If $t_n < T$, then no events occurred in the interval $(t_n, T]$.} Then the joint distribution of $t_1, t_2, \ldots, t_n$ is related to the U(0, T) distribution in the following way. Assume that $Y_1, Y_2, \ldots, Y_n$ (the same n as above) are IID random variables with the U(0, T) distribution, and let $Y_{(1)}, Y_{(2)}, \ldots, Y_{(n)}$ be their corresponding order statistics (see Sec. 6.2.4). Then a property of the Poisson process is that $t_1, t_2, \ldots, t_n$ have the same joint distribution as $Y_{(1)}, Y_{(2)}, \ldots, Y_{(n)}$. [See Ross (2003, p. 303) for a proof.]

One way of interpreting this property is that if someone simply showed us the *values* of $t_1, t_2, \ldots, t_n$ without telling us that t_i was obtained as the time of the ith event in some sequence of events, it would appear (in a statistical sense) that these n numbers had been obtained by taking a sample of n IID random values from the U(0, T) distribution and then sorting them into increasing order. Alternatively, one could think of this property as saying that if we consider $t_1, t_2, \ldots, t_n$ as unordered random variables, they are IID with the U(0, T) distribution. This is why we sometimes see a Poisson process described as one in which events occur "at random," since the instants at which events occur are uniformly distributed over time.

In any case, this property provides us with a different way of testing the null hypothesis that an observed sequence of events was generated by a Poisson process. (We have already seen one way this hypothesis can be tested, namely, testing whether the interevent times appear to be IID exponential random variables; see Sec. 6.12.1 and Examples 6.15, 6.18, and 6.19.) We simply test whether the event times $t_1, t_2, \ldots, t_n$ appear to be IID U(0, T) random variables using any applicable testing procedure.

EXAMPLE 6.20. The interarrival-time data of Table 6.7 were collected over a fixed 90-minute period, and $n = 220$ arrivals were recorded during this interval. (It was decided beforehand to start observing the process at exactly 5 P.M., rather than at the first time after 5:00 when an arrival happened to take place. Also, data collection terminated promptly at 6:30 P.M., regardless of any arrivals that occurred later. It is important for the validity of this test that the data collection be designed in this way, i.e., independent of the actual event times.) The times of arrivals were $t_1 = 1.53$, $t_2 = 1.98, \ldots, t_{220} = 88.91$ (in minutes after 5 P.M.). To test whether these numbers can be regarded as being independent with the U(0,90) distribution, we used the *all-parameters-known* cases of the chi-square and K-S tests. [The density and distribution functions of the fitted distribution are, respectively, $\hat{f}(x) = 1/90$ and $\hat{F}(x) = x/90$, for $0 \le x \le 90$. Note also that our "data" points are already sorted, conveniently.] We carried out a chi-square test with the $k = 17$ equal-size intervals $[0, 5.294), [5.294, 10.588), \ldots, [84.706, 90]$, so that $np_j = 220/17 = 12.941$ for $j = 1, 2, \ldots, 17$. The resulting value of χ^2 was 13.827, and since $\chi^2_{16,0.90} = 23.542$, we cannot reject the null hypothesis that the arrivals occurred in accordance with a Poisson process at level 0.10. The K-S test resulted in $D_{220} = 0.045$, and the value of the adjusted test statistic from the all-parameters-known row of Table 6.15 is thus 0.673. Since this is well below $c_{0.90} = 1.224$, once again we cannot reject the null hypothesis at level 0.10.

6.7 THE ExpertFit SOFTWARE AND AN EXTENDED EXAMPLE

Performing the statistical procedures discussed in this chapter can, in some cases, be difficult, time-consuming, and prone to error. For example, the chi-square test with equiprobable intervals requires the availability of the inverse of the distribution function, which is not available in closed form for some distributions (e.g., normal and gamma). Thus, in these cases a numerical approximation to the inverse of the distribution function would have to be obtained and programmed. Also, the K-S test is often misstated or misapplied in textbooks and software packages. These considerations led to the development of the ExpertFit distribution-fitting software.

The commercial versions of ExpertFit [see Averill M. Law & Associates (2013)] will automatically and accurately determine which of 40 probability distributions best represents a data set. The selected distribution is then put into the proper format for direct input to a large number of different simulation packages. ExpertFit contains the following four modules that are used sequentially to determine the best distribution:

- "Data" reads or imports data into ExpertFit, displays summary statistics, and makes histograms; makes correlation plots and scatter diagrams; and performs the Kruskal-Wallis test for homogeneity of data sets (see Sec. 6.13).
- "Models" fits distributions to the data by using the method of maximum likelihood, ranks the distributions in terms of quality of fit, and determines whether the "best" distribution is actually good enough to use in a simulation model. (Otherwise, it recommends the use of an empirical distribution.)
- "Comparisons" compares the best distribution(s) to the data to further determine the quality of fit, using density-histogram plots, distribution-function-differences plots, probability plots, goodness-of-fit tests, etc.
- "Applications" displays and computes characteristics of a fitted distribution such as the density function, moments, probabilities, and quantiles; it also puts the selected distribution into the proper format for a chosen simulation package.

ExpertFit has the following documentation:

- Context-sensitive help for all menus and all results tables and graphs
- Online feature index and tutorials (see the "Help" pull-down menu in the Menu Bar at the top of the screen)
- User's Guide with eight complete examples

There is a Student Version of ExpertFit on the book's website, www.mhhe.com/law, that can be used to analyze most data sets corresponding to the examples and problems in this chapter. It cannot, however, be used to analyze your own data sets.

We now use ExpertFit to perform a comprehensive analysis of the $n = 200$ service times of Example 6.1. However, for expository convenience we limit our analysis to the exponential, gamma, Weibull, lognormal, log-logistic, Pearson type V, Pearson type VI, and normal distributions. In Table 6.19 we present the "Data Summary" (i.e., summary statistics) for the service times, and Fig. 6.49 is a corresponding histogram based on $k = 12$ intervals of width $\Delta b = 0.2$. (The interval width was

TABLE 6.19
Data summary for the service-time data

Data characteristic	Value
Source file	EXAMPLE61
Observation type	Real valued
Number of observations	200
Minimum observation	0.054
Maximum observation	2.131
Mean	0.888
Median	0.849
Variance	0.210
Coefficient of variation	0.515
Skewness	0.513

determined by trial and error.) The shape of the histogram strongly suggests that the underlying distribution is skewed to the right, which tends to rule out the normal distribution. This is supported by noting that $\overline{X}(200) = 0.888 > 0.849 = \hat{x}_{0.5}(200)$ and $\hat{\nu}(200) = 0.513 > 0$. Furthermore, $\widehat{cv}(200) = 0.515$ makes it fairly unlikely that the true distribution could be exponential, which has a coefficient of variation of 1; this conclusion is supported by the shape of the histogram.

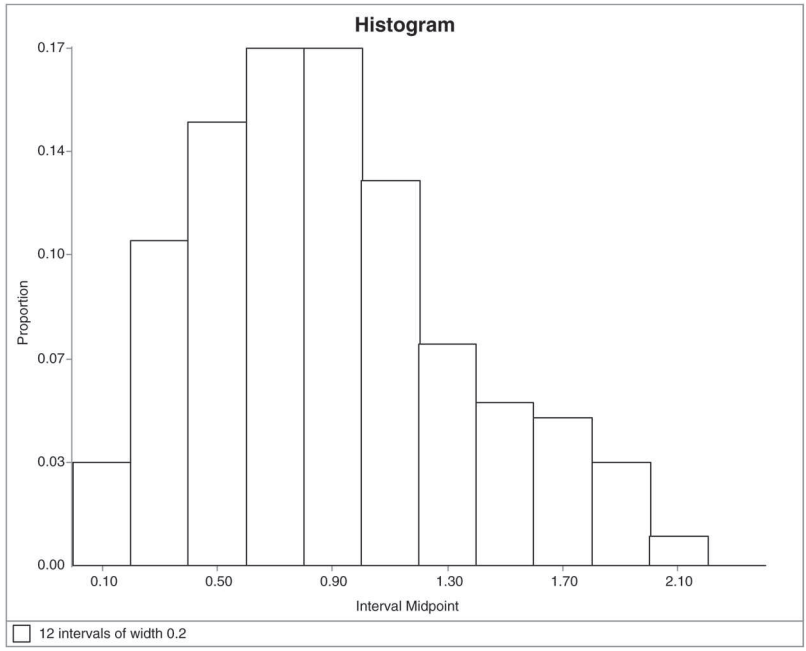
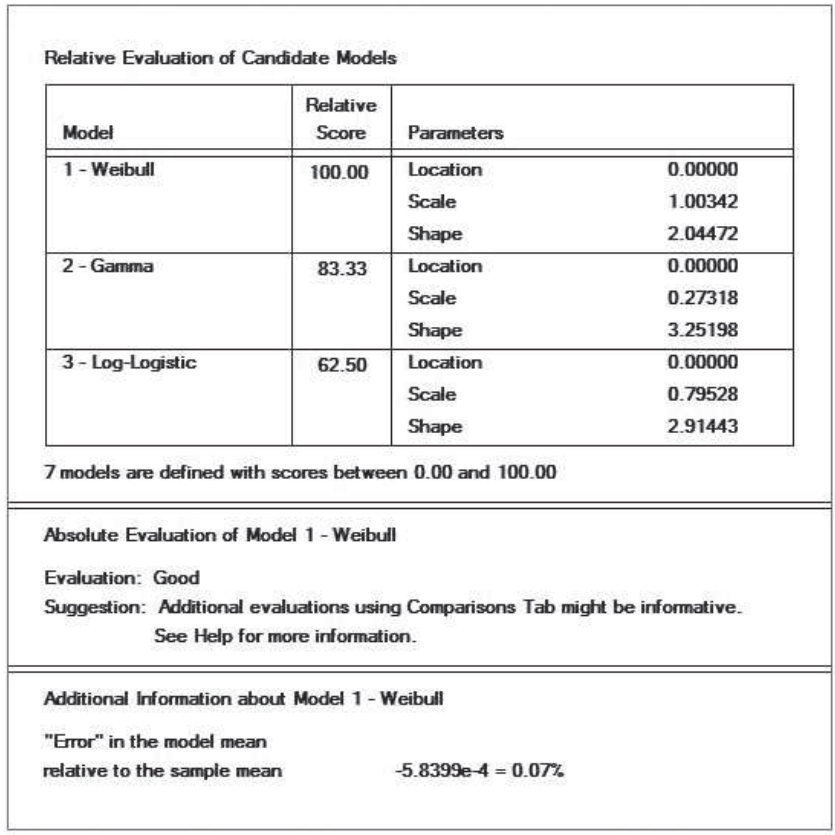


FIGURE 6.49
Histogram of 200 service times with $\Delta b = 0.2$.



Relative Evaluation of Candidate Models

Model	Relative Score	Parameters	
1 - Weibull	100.00	Location	0.00000
		Scale	1.00342
		Shape	2.04472
2 - Gamma	83.33	Location	0.00000
		Scale	0.27318
		Shape	3.25198
3 - Log-Logistic	62.50	Location	0.00000
		Scale	0.79528
		Shape	2.91443

7 models are defined with scores between 0.00 and 100.00

Absolute Evaluation of Model 1 - Weibull

Evaluation: Good
Suggestion: Additional evaluations using Comparisons Tab might be informative.
See Help for more information.

Additional Information about Model 1 - Weibull

"Error" in the model mean
relative to the sample mean -5.8399e-4 = 0.07%

FIGURE 6.50
ExpertFit results screen for the service-time data.

The "Automated Fitting" option, which is in the Models module, was then used to fit, rank, and evaluate the specified distributions other than the normal distribution automatically. The normal distribution was not automatically fit to the data, since it can take on negative values that are inconsistent with the range of service times. (If desired, the normal distribution could be fit to this data set manually by using "Fit Individual Models" in the Models module.) The resulting ExpertFit results screen is shown in Fig. 6.50. From the "Relative Evaluation of Candidate Models," it can be seen that the Weibull distribution is ranked first and received a "Relative Score" (see below) of 100.00 followed by the gamma distribution and the log-logistic distribution with Relative Scores of 83.33 and 62.50, respectively. The maximum-likelihood estimates for the best-fitting Weibull distribution are $\hat{\alpha} = 2.045$ and $\hat{\beta} = 1.003$.

Even if a distribution is ranked first, this does not necessarily mean that it is good enough to use in a simulation. However, since the "Absolute Evaluation" is "Good," there is no current evidence for not using the Weibull distribution. On the other hand, it is prudent to obtain further confirmation using the Comparisons module. If the highest-ranked distribution receives an Absolute Evaluation of "Bad,"

then it is not suitable for use in a simulation model and ExpertFit will recommend the use of an empirical distribution (see Sec. 6.2.4).

The ExpertFit ranking and evaluation algorithm was developed as follows. We had 15 heuristics that were thought to have some ability to discriminate between a good-fitting and bad-fitting distribution. (The chi-square statistic was not considered because it depends on an arbitrary choice of intervals.) To determine which of these heuristics was actually the best, a random sample of size n was generated from a known "parent" distribution, and each of the 15 heuristics was applied to see if it could, in fact, choose the correct distribution. This was repeated for 200 independent samples, giving an estimated probability that each heuristic would pick the parent distribution for the specified sample size. This whole process was repeated for 175 parent distribution/sample-size pairs, resulting in several heuristics that appeared to be superior. These heuristics were combined to give the overall algorithm for ranking the fitted distributions and for computing the relative scores. The 35,000 generated data sets were also used to develop the rules for determining the Absolute Evaluations.

The ranking screen also shows that the error in the mean of the Weibull distribution relative to the sample mean is only 0.07 percent.

As suggested by the Absolute Evaluation, we will now try to obtain additional confirmation for the Weibull distribution by using graphical plots and goodness-of-fit tests. In Fig. 6.51 we give density-histogram plots for the Weibull and gamma

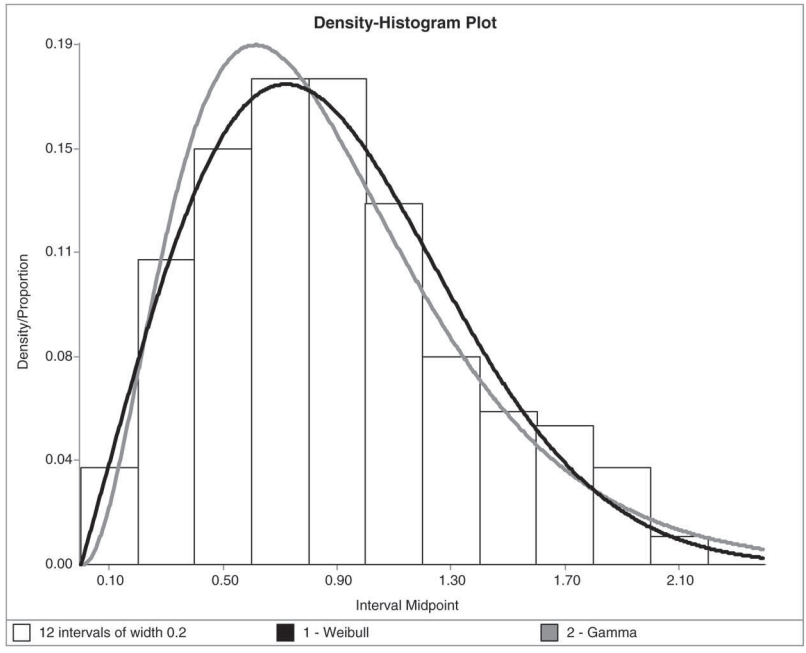


FIGURE 6.51
Density-histogram plots for the service-time data and the Weibull and gamma distributions.

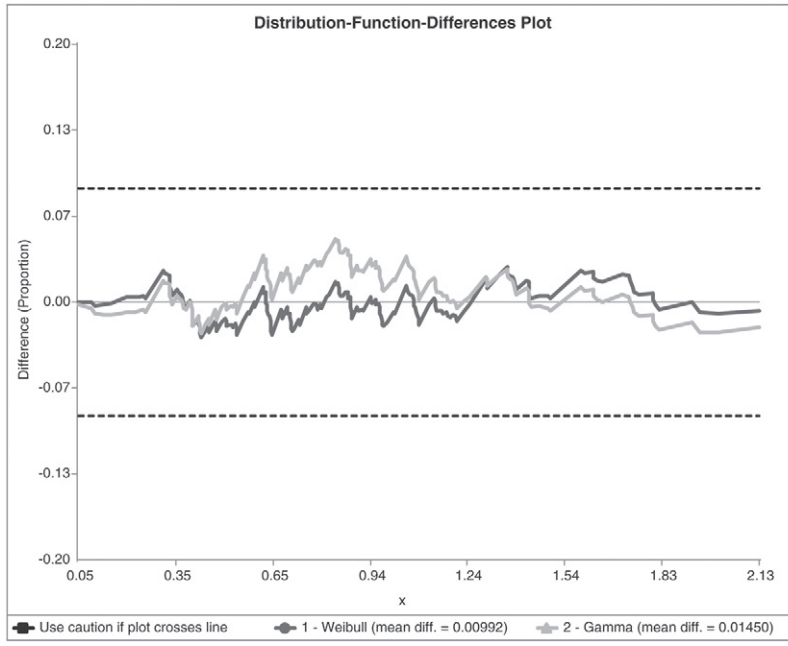


FIGURE 6.52
Distribution-function-differences plots for the service-time data and the Weibull and gamma distributions.

distributions (see also the Color Plates). It can be seen that the Weibull distribution matches the histogram well and that the gamma distribution is clearly inferior. Figure 6.52 gives distribution-function-differences plots for the two distributions and, once again, the superiority of the Weibull distribution can be seen (see the Color Plates). The *P–P* plots in Fig. 6.53 also show that the Weibull distribution is preferable.

We next performed an equiprobable chi-square test for the Weibull distribution at level $\alpha = 0.05$ using $k = 20$ intervals. The chi-square statistic was 15.6, which is less than the critical value 30.144; therefore, this particular chi-square test gives us no reason to reject the Weibull distribution. The adjusted test statistic for the K-S test was 0.428, which is less than the $\alpha = 0.05$ modified critical value 0.874. Once again we have no reason to reject the Weibull distribution. Finally, the test statistic for the A-D test was 0.264, which is less than the $\alpha = 0.05$ critical value 0.746, giving us no reason to reject the Weibull distribution.

Thus, based on the Absolute Evaluation, the graphical plots, and the goodness-of-fit tests, there is no reason to think that the Weibull distribution is not a good representation for the service-time data.

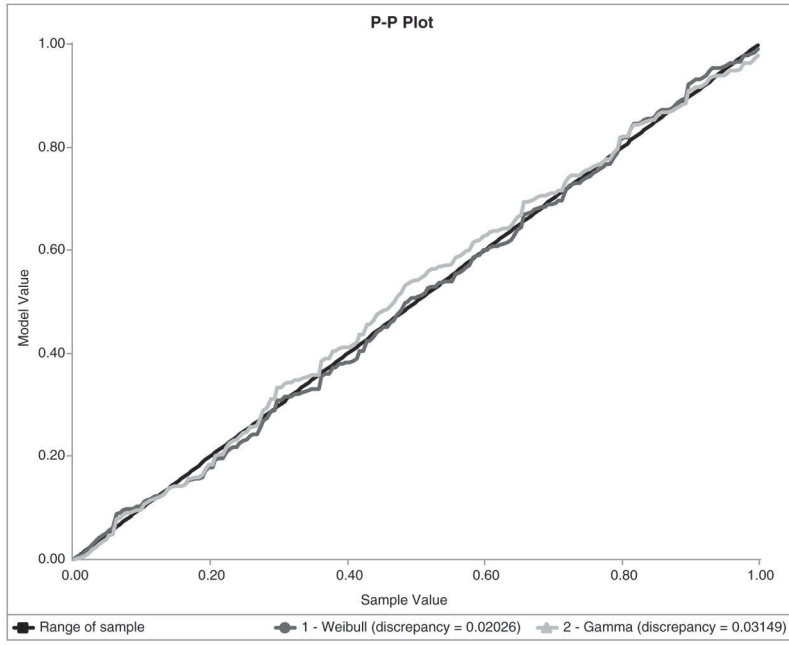


FIGURE 6.53
P–P plots for the service-time data and the Weibull and gamma distributions.

6.8 SHIFTED AND TRUNCATED DISTRIBUTIONS

The exponential, gamma, Weibull, lognormal, Pearson type V, Pearson type VI, and log-logistic distributions, discussed in Sec. 6.2.2, have range $[0, \infty)$. Thus, if the random variable X has any of these distributions, it can take on arbitrarily small positive values. However, frequently in practice if X represents the time to complete some task (such as customer service), it is simply impossible for X to be less than some fixed positive number. For example, in a bank it is probably not possible to serve anyone in less than, say, 30 seconds; this will be reflected in the service-time data we might collect on the bank's operation. Thus, in reality $P(X < 30 \text{ seconds}) = 0$; however, for a fitted gamma distribution, for instance, there is a *positive* probability of generating a random value that is less than 30 seconds. Thus, it would appear that a modification of these distribution forms would provide a more realistic model and might result in a better fit *in some cases*.

This change can be effected by *shifting* the distribution some distance to the right. What this really amounts to is generalizing the density function of the distribution in question to include a location parameter (see Sec. 6.2.1).

For example, the gamma distribution shifted to the right by an amount $\gamma > 0$ has density

$$f(x) = \begin{cases} \dfrac{\beta^{-\alpha}\,(x-\gamma)^{\alpha-1}\,e^{-(x-\gamma)/\beta}}{\Gamma(\alpha)} & \text{if } x > \gamma \\ 0 & \text{otherwise} \end{cases}$$

which has the same shape and scale parameters as the gamma (α, β) distribution but is shifted γ units to the right. (This is often called the *three-parameter gamma distribution*.) Shifted versions of the other distributions discussed above are defined similarly, by replacing x by $x - \gamma$ in the density functions and their domains of definition. The range of these shifted distributions is $[\gamma, \infty)$.

With these shifted distributions, we then have to estimate γ as well as the other parameters. In theory, this can be done by finding the MLE for γ in addition to the MLEs for the original parameters. For the shifted exponential, $\hat{\gamma}$ and $\hat{\beta}$ are relatively easy to find (see Prob. 6.12). However, finding MLEs for the three-parameter distributions is considerably more problematic. For example, in the case of the gamma, Weibull, and lognormal distributions, it is known that (global) MLEs are not well defined [see Cheng and Amin (1983), Cohen and Whitten (1980), and Zanakis (1979a)]. That is, the likelihood function L can be made infinite by choosing $\hat{\gamma} = X_{(1)}$ (the smallest observation in the sample), which results in inadmissible values for the other parameters. A simple approach to the three-parameter estimation problem is first to estimate the location parameter γ by

$$\tilde{\gamma} = \frac{X_{(1)}X_{(n)} - X_{(k)}^2}{X_{(1)} + X_{(n)} - 2X_{(k)}}$$

where k is the smallest integer in $\{2, 3, \ldots, n-1\}$ such that $X_{(k)} > X_{(1)}$ [see Dubey (1967)]. It can be shown that $\tilde{\gamma} < X_{(1)}$ if and only if $X_{(k)} < [X_{(1)} + X_{(n)}]/2$, which is very likely to occur; see Prob. 6.23. [Zanakis (1979b) has shown empirically the accuracy of $\tilde{\gamma}$ for the Weibull distribution.] Given the value $\tilde{\gamma}$, we next define X_i' as follows:

$$X_i' = X_i - \tilde{\gamma} \geq 0 \qquad \text{for } i = 1, 2, \ldots, n$$

Finally, MLE estimators of the scale and shape parameters are obtained by applying the usual two-parameter MLE procedures to the observations $X_1', X_2', \ldots, X_n'$.

EXAMPLE 6.21. In Fig. 6.54, we give a histogram (with $\Delta b = 0.2$) of the times (in hours) to unload $n = 808$ coal trains, each consisting of approximately 110 cars. (Figure 6.54 is actually a density-histogram plot.) The shape of the histogram suggests that a fitted distribution would require a positive location parameter. Since $X_{(1)} = 3.37$, $X_{(2)} = 3.68$, and $X_{(808)} = 6.32$, we get $\hat{\gamma} = 3.329$. The values $X_i' = X_i - 3.329$ for $i = 1, 2, \ldots, 808$ were then used to obtain the MLEs $\hat{\alpha} = 7.451$ and $\hat{\beta} = 1.271$ for the log-logistic distribution, whose density function is also given in Fig. 6.54. In general, the agreement between the shifted log-logistic density function and the histogram seems quite good.

In some situations a fitted distribution might provide a good model for observed data generally, but other information says that, for instance, no value can be larger

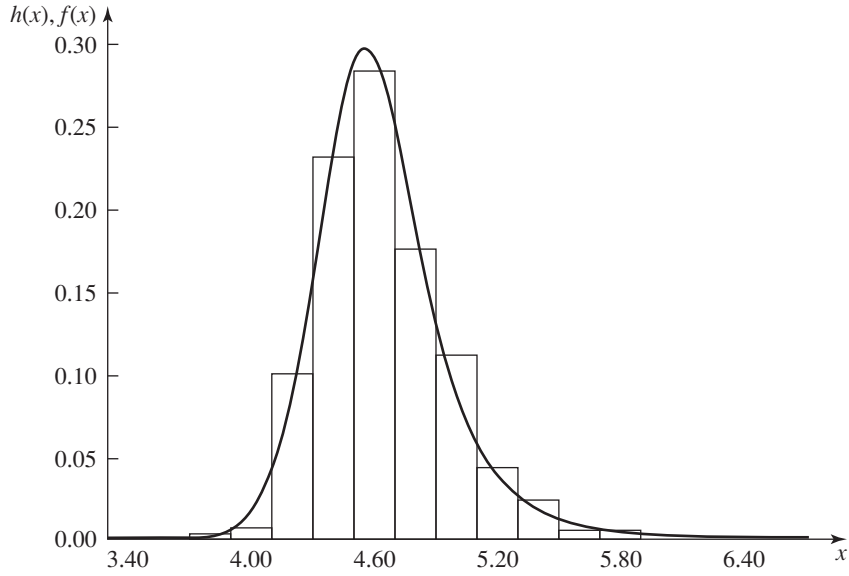


FIGURE 6.54
Density-histogram plot for the train-unloading data and the fitted log-logistic distribution.

than some finite constant $b > 0$. If the range of the fitted density f is $[0, \infty)$, this is incompatible with the upper limit b, so we might instead use a truncated density

$$f^*(x) = f(x)/F(b) \text{ for } 0 \le x \le b \text{ (and 0 otherwise), where } F(b) = \int_0^b f(x)\,dx < 1.$$

A method to generate random values from f^* is given in Sec. 8.2.1.

EXAMPLE 6.22. If a gamma distribution is found to provide a good model for service times in a bank, the density function might be truncated above $b = 15$ minutes if larger values than this are extremely unlikely.

6.9 BÉZIER DISTRIBUTIONS

There is a fourth approach [Wagner and Wilson (1996a, 1996b)] for specifying a probability distribution that models a set of observed data $X_1, X_2, \ldots, X_n$ (see Sec. 6.1 for the other three approaches). If X is a continuous random variable with finite range $[a, b]$ and a distribution function $F(x)$ having any shape, then $F(x)$ can be approximated arbitrarily closely by a *Bézier distribution* function with sufficiently high degree m. Let $\{\mathbf{p}_0, \mathbf{p}_1, \ldots, \mathbf{p}_m\}$ be a set of *control points*, where

$\mathbf{p}_i = (y_i, z_i)$ $(i = 1, 2, \ldots, m - 1)$, $\mathbf{p}_0 = (a, 0)$, and $\mathbf{p}_m = (b, 1)$. The Bézier distribution function $\mathbf{P}(t)$ is given parametrically by

$$\mathbf{P}(t) = \sum_{i=0}^{m} B_{m,i}(t)\mathbf{p}_i \qquad \text{for } t \in [0, 1] \tag{6.6}$$

where

$$B_{m,i}(t) = \frac{m!}{i!(m - i)!} t^i (1 - t)^{m-i}$$

Let $\mathbf{y}$ be the vector of y_i's, and let $\mathbf{z}$ be the vector of z_i's. Furthermore, let $F_n(x)$ be the empirical distribution function defined by Eq. (6.5), and let $F(x; m, \mathbf{y}, \mathbf{z})$ be the Bézier distribution function given by Eq. (6.6). For fixed m, $F(x; m, \mathbf{y}, \mathbf{z})$ is fit to the X_i's by using a suitable optimization technique (e.g., least-squares estimation) to find the minimum distance between $F_n(x)$ and $F(x; m, \mathbf{y}, \mathbf{z})$ over all possible $\mathbf{y}$ and $\mathbf{z}$, and subject to certain constraints. (The optimization determines $\mathbf{p}_i$ for $i = 1, 2, \ldots, m - 1$.)

A Bézier distribution is an alternative to an empirical distribution for modeling a data set that is not represented well by a standard theoretical distribution. Furthermore, a software package for fitting Bézier distributions has been developed by Wagner and Wilson. There is, however, a difficulty in using Bézier distributions, at least at present. Bézier distributions are not implemented in most simulation packages, and doing so on one's own could be difficult in some software.

6.10 SPECIFYING MULTIVARIATE DISTRIBUTIONS, CORRELATIONS, AND STOCHASTIC PROCESSES

So far in this chapter we have considered only the specification and estimation of the distribution of a single, univariate random variable at a time. If the simulation model needs input of only such scalar random variables, and if they are all independent of each other across the model, then repeated application to each input of the methods we've discussed up to now in this chapter will suffice. Indeed, this is the standard mode of operation in most simulation projects, and it is the one supported by most simulation packages.

There are systems, however, in which the input random variables are statistically related to each other in some way:

- Some of the input random variables together form a random *vector* with some multivariate (or joint) probability distribution (see Sec. 4.2) to be specified by the modeler.
- In other cases we may not want (or be able) to go quite so far as to specify the complete multivariate distribution, but nonetheless suspect that there could be *correlation* between different input random variables having their own individual, or *marginal,* distributions, without knowledge or specification of the complete multivariate distribution. We would like our inputs to reflect this correlation even if we can't specify the entire multivariate distribution.

- In yet other cases, we may want to specify an entire input stochastic process (see Sec. 4.3) in which the marginal distribution of the individual random variables composing the process is to be specified, as well as the autocorrelations between them out through some desired lag. This could be regarded as an infinite-dimensional input random vector.

It is easy to think of physical situations where such input might occur:

- Consider a maintenance shop that can be modeled as a tandem queue with two service stations. At the first station, incoming parts are inspected, and any defects are marked for repair at the second station. Since a badly damaged part would probably require above-average times for both inspection and repair, we might expect the two service times for a given part to be positively correlated. Mitchell et al. (1977) found that ignoring this correlation in modeling a system can lead to serious inaccuracies in simulation results.
- In a model of a communications system, the sizes of (and perhaps interarrival times between) arriving messages could form a stochastic process with some stationary marginal univariate distribution of message size, as well as some kind of correlation out through several lags. For instance, it could be that large messages tend to come in groups, as do small messages, resulting in positive autocorrelation within the message-size input process. Livny et al. (1993) show that autocorrelation in either the service-time or interarrival-time input process of a simple *M*/*M*/1 queue can have a major effect on the output performance measures.
- In a model of an inventory or production system, the stream of incoming orders could display negative lag-one autocorrelation if a large order in one period tends to be followed by a small order in the next period, and vice versa.

Thus, if the modeler has evidence of some kind of statistical relationship between a simulation's various scalar input random variables, or if an input stream is a process over time that exhibits autocorrelation within itself, then consideration might be given to modeling these relationships and generating them during the simulation to avoid possible problems with model validity.

In the remainder of this section we briefly discuss some of these issues with regard to specification and estimation [see the book chapter by Biller and Ghosh (2006) for a comprehensive discussion], and in Sec. 8.5 we discuss generating the corresponding observations for input to the simulation as it runs. In Sec. 6.12 we take up the related issue of modeling arrival processes; in Sec. 8.6 we discuss generation methods for such processes. Many of these issues are discussed in Leemis (2004) and in Sec. 5 of Nelson and Yamnitsky (1998).

6.10.1 Specifying Multivariate Distributions

Let $\mathbf{X} = (X_1, X_2, \ldots, X_d)^T$ be an input random vector of dimension d (A^T denotes the transpose of a vector or matrix A, so that $\mathbf{X}$ is a $d \times 1$ column vector). For instance, in the maintenance-shop example above, $d = 2$ (in which case $\mathbf{X}$ is called *bivariate*), X_1 is the inspection time of a part, and X_2 is the subsequent

repair time of that *same* part. Let $\mathbf{X}_k = (X_{1k}, X_{2k}, \ldots, X_{dk})^T$ be the kth of n IID observations on this d-dimensional random vector; in the maintenance shop we would have n pairs

$$\begin{pmatrix} X_{11} \\ X_{21} \end{pmatrix}, \begin{pmatrix} X_{12} \\ X_{22} \end{pmatrix}, \ldots, \begin{pmatrix} X_{1n} \\ X_{2n} \end{pmatrix}$$

of observed data corresponding to the inspection and repair times of n different parts, and we would further want to be able to generate such a sequence of 2-vectors as input to the simulation.

Note that while we allow correlation *within* the components of a specific $\mathbf{X}_k$, we are assuming here that the component random variables *across* different $\mathbf{X}_k$'s are independent, i.e., that the random vectors $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n$ are independent of each other. In the maintenance-shop situation this means that while the inspection and repair times of a given part may be related, there is no relation between either of these times across different parts—the parts are assumed to behave independently of each other. [In Sec. 6.10.3, we allow autocorrelation in the $\{\mathbf{X}_1, \mathbf{X}_2, \ldots\}$ sequence, but mostly in the univariate (scalar) case $d = 1$.]

The *multivariate* (or *joint*) *distribution function* of the random vector $\mathbf{X}$ is defined as

$$F(\mathbf{x}) = P(\mathbf{X} \le \mathbf{x}) = P(X_1 \le x_1, X_2 \le x_2, \ldots, X_d \le x_d)$$

for any fixed d-vector $\mathbf{x} = (x_1, x_2, \ldots, x_d)^T$. This includes all the cases of continuous, discrete, or mixed individual marginal distributions. In addition to implying the marginal distributions, all the information about relationships between the individual component random variables is embodied in the multivariate distribution function, including their correlations.

As with univariate scalar random variables, there are a variety of multivariate distributions that have been developed and parameterized in various ways and into various families; see Chap. XI of Devroye (1986), Johnson (1987), Johnson et al. (1997), and Kotz et al. (2000). However, it may be difficult or impractical to estimate the entire multivariate distribution of a random vector from observed data, particularly if the sample size n is not large. Thus, we restrict ourselves in the remainder of this subsection to certain special cases of multivariate-distribution estimation that have been found useful in simulation, and we discuss in Sec. 6.10.2 what can be done if our estimation goals are more modest than that of specifying the entire multivariate distribution. And since our interest is ultimately in simulation input, we must also pay attention to how realizations of such random vectors can be generated, as discussed in Sec. 8.5.

Multivariate Normal

This is probably the best-known special case of a multivariate distribution. While this distribution might have somewhat limited direct utility as a simulation-input model since all its marginal distributions are symmetric and have infinite tails in both directions, it does serve as a springboard to other more useful input-modeling distributions.

The multivariate normal density function (see Sec. 4.2) is defined as

$$f(\mathbf{x}) = (2\pi)^{-n/2} |\Sigma|^{-1/2} \exp\left[-\frac{(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu})}{2} \right]$$

for any vector $\mathbf{x}$ in d-dimensional real space. Here, $\boldsymbol{\mu} = (\mu_1, \mu_2, \ldots, \mu_d)^T$ is the mean vector, Σ is the covariance matrix with (i, j)th entry $\sigma_{ij} = \sigma_{ji} = \text{Cov}(X_i, X_j)$ (so Σ is symmetric and positive definite), $|\Sigma|$ is the determinant of Σ, and Σ^{-1} is the matrix inverse of Σ. The marginal distribution of X_i is $\text{N}(\mu_i, \sigma_{ii})$. We denote the multivariate normal distribution as $\text{N}_d(\boldsymbol{\mu}, \Sigma)$.

The *correlation coefficient* between X_i and X_j is $\rho_{ij} = \sigma_{ij}/\sqrt{\sigma_{ii}\sigma_{jj}} = \rho_{ji}$, which is always between -1 and $+1$. Thus, since $\sigma_{ij} = \sigma_{ji} = \rho_{ij}\sqrt{\sigma_{ii}\sigma_{jj}}$, an alternative parameterization of the multivariate normal distribution would be to replace Σ by the parameters σ_{ii} and ρ_{ij} for $i = 1, 2, \ldots, d$ and $j = i + 1, i + 2, \ldots, d$.

Note that in the multivariate-normal case, the entire joint distribution is uniquely determined by the marginal distributions and the correlation coefficients. This is generally not true for other, non-normal multivariate distributions; i.e., there could be several different joint distributions that result in the same set of marginal distributions and correlations.

To fit a multivariate normal distribution to observed d-dimensional data $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n$, the mean vector $\boldsymbol{\mu}$ is estimated by the MLE

$$\hat{\boldsymbol{\mu}} = \overline{\mathbf{X}} = (\overline{X}_1, \overline{X}_2, \ldots, \overline{X}_d)^T \tag{6.7}$$

where $\overline{X}_i = \sum_{k=1}^{n} X_{ik}/n$, and the covariance matrix Σ is estimated by the $d \times d$ matrix $\hat{\Sigma}$ whose (i, j)th entry is

$$\hat{\sigma}_{ij} = \frac{\sum_{k=1}^{n} (X_{ik} - \overline{X}_i)(X_{jk} - \overline{X}_j)}{n} \tag{6.8}$$

The correlation coefficient ρ_{ij} is estimated by the MLE

$$\hat{\rho}_{ij} = \frac{\hat{\sigma}_{ij}}{\sqrt{\hat{\sigma}_{ii}\hat{\sigma}_{jj}}} = \hat{\rho}_{ji} \tag{6.9}$$

With a multivariate normal distribution so estimated, generation from it is possible by methods given in Sec. 8.5.2.

Multivariate Lognormal

This multivariate distribution affords the modeler positively skewed marginal distributions on $[0, \infty)$ with the possibility of correlation between them.

Rather than giving the explicit definition of its full joint density function, it is more useful for simulation purposes to describe the multivariate lognormal in terms of its transformational relation to the multivariate normal. We say that $\mathbf{X} = (X_1, X_2, \ldots, X_d)^T$ has a multivariate lognormal distribution if and only if

$$\mathbf{Y} = (Y_1, Y_2, \ldots, Y_d)^T = (\ln X_1, \ln X_2, \ldots, \ln X_d)^T$$

has a multivariate normal distribution $N_d(\boldsymbol{\mu}, \Sigma)$; see Jones and Miller (1966) and Johnson and Ramberg (1978). Put another way, the multivariate lognormal random vector $\mathbf{X}$ can be represented as

$$\mathbf{X} = \left(e^{Y_1}, e^{Y_2}, \ldots, e^{Y_d}\right)^T$$

where $\mathbf{Y}$ is multivariate normal $N_d(\boldsymbol{\mu}, \Sigma)$. The marginal distribution of X_i is univariate lognormal $LN(\mu_i, \sigma_{ii})$ where μ_i is the ith element of $\boldsymbol{\mu}$ and σ_{ii} is the ith diagonal entry in Σ.

Since the multivariate normal random vector $\mathbf{Y}$ in the above logarithmic transformation of $\mathbf{X}$ has mean vector $\boldsymbol{\mu} = (\mu_1, \mu_2, \ldots, \mu_d)^T$ and covariance matrix Σ with (i, j)th entry σ_{ij} (so the correlation coefficients are $\rho_{ij} = \sigma_{ij}/\sqrt{\sigma_{ii}\sigma_{jj}}$), it turns out that

$$E(X_i) = e^{\mu_i + \sigma_{ii}/2} \tag{6.10}$$

$$\text{Var}(X_i) = e^{2\mu_i + \sigma_{ii}}(e^{\sigma_{ii}} - 1) \tag{6.11}$$

and

$$\text{Cov}(X_i, X_j) = (e^{\sigma_{ij}} - 1)\exp\left(\mu_i + \mu_j + \frac{\sigma_{ii} + \sigma_{jj}}{2}\right) \tag{6.12}$$

This implies that the correlation coefficient between X_i and X_j is

$$\text{Cor}(X_i, X_j) = \frac{e^{\sigma_{ij}} - 1}{\sqrt{(e^{\sigma_{ii}} - 1)(e^{\sigma_{jj}} - 1)}} \tag{6.13}$$

Note that $\boldsymbol{\mu}$ and Σ are *not* the mean vector and covariance matrix of the multivariate lognormal random vector $\mathbf{X}$, but rather are the mean and covariance of the corresponding multivariate normal random vector $\mathbf{Y}$. The mean vector and covariance matrix (and correlations) of $\mathbf{X}$ are given by Eqs. (6.10) through (6.13) above.

To fit a multivariate lognormal distribution to a sample $\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n$ of d-dimensional vectors, take the natural logarithm of each component scalar observation in each observed data vector to get the data vectors $\mathbf{Y}_1, \mathbf{Y}_2, \ldots, \mathbf{Y}_n$; treat these $\mathbf{Y}_k$'s as multivariate normal with unknown mean vector $\boldsymbol{\mu}$ and covariance matrix Σ; and estimate $\boldsymbol{\mu}$ and Σ, respectively, by Eqs. (6.7) and (6.8) above.

Generation from a fitted multivariate lognormal distribution is discussed in Sec. 8.5.2.

Multivariate Johnson Translation System

The univariate Johnson translation system, which includes the normal, lognormal, Johnson S_B (see Sec. 6.2.2), and Johnson S_U distributions, affords considerable flexibility in terms of range and shape of fitted distributions. This family of distributions has been extended to the multivariate case; see Chap. 5 of Johnson (1987), Stanfield et al. (1996), and Wilson (1997).

As in the univariate case, the multivariate Johnson translation system permits great flexibility in terms of range and shape to obtain good fits to a wide variety of observed multivariate data vectors, certainly far more flexibility than given by the multivariate normal or lognormal distributions discussed above. In particular, in the method developed by Stanfield et al. (1996), the first four moments of the marginal distributions from the fitted multivariate distribution match those of the observed

data, and the correlation structure in the fitted distribution matches the empirical correlations as well. Fitting such a distribution to observed data involves several steps and uses methods to fit univariate Johnson distributions; for details on this fitting procedure, as well as generating random vectors from the fitted multivariate Johnson distribution, see Stanfield et al. (1996).

Bivariate Bézier

Univariate Bézier distributions, as described in Sec. 6.9, have been extended to the bivariate case ($d = 2$ dimensions) by Wagner and Wilson (1995). Software is also described there that allows graphical interactive adjustment of the fitted distribution; in addition, generating random vectors is discussed there. Further results and methods concerning bivariate Bézier distributions can be found in Wagner and Wilson (1996a).

Extension of Bézier distributions to three or more dimensions is described by Wagner and Wilson (1995) as "feasible but cumbersome."

6.10.2 Specifying Arbitrary Marginal Distributions and Correlations

In Sec. 6.10.1 we discussed several cases where a complete multivariate distribution might be specified to model the joint behavior of d possibly related input random variables that together compose an input random vector. In each of these cases, the fitted member of the multivariate distribution family involved (normal, lognormal, Johnson, or Bézier) determined the correlation between pairs of the component random variables in the vector, as well as their marginal distributions; it also imposed a more general and complete description of the joint variation of the component random variables as embodied in the joint density function itself.

Sometimes we need greater flexibility than that. We may want to allow for possible correlation between various pairs of input random variables to our simulation model, yet not impose an overall multivariate distribution forcing the fitted marginal distributions all to be members of the same family. In other words, we would like to be free to specify arbitrary univariate distributions to model the input random variables separately, as described in Secs. 6.1 through 6.9, yet also estimate correlations between them quite apart from their marginal distributions. In fact, we might even want some of the component input random variables to be continuous, others to be discrete, and still others to be mixed continuous-discrete, yet still allowing for correlations between them.

A very simple, and fairly obvious, procedure for doing this is just to fit distributions to each of the univariate random variables involved, one at a time and in isolation from the others, and then to estimate suspected correlations between pairs of input random variables by Eq. (6.9) above. Gather these random variables together into an input random vector, which then by construction has the desired univariate marginal distributions and desired correlation structure. However, it is important to note that this procedure does not specify, or "control," the resulting joint distribution of the random vector as a whole—in fact, we generally won't even know what this joint distribution is. Thus, the suggested procedure allows for greater flexibility on the marginal distributions and correlations, but exerts less overall control. Another

caution is that the form and parameters of the marginal distributions can impose restrictions on what correlations are possible; see Whitt (1976).

While specification of such a situation seems, at least in principle, relatively straightforward, we need to make sure that whatever we specify here can be generated from during the simulation. We discuss this in Sec. 8.5.5, based on work in Hill and Reilly (1994) and Cario et al. (2002).

6.10.3 Specifying Stochastic Processes

As mentioned earlier, there are situations in which a sequence of input random variables on the same phenomenon are appropriately modeled as being draws from the same (marginal) distribution, yet might exhibit some autocorrelation between themselves within the sequence. For instance, if $\{X_1, X_2, \ldots\}$ denote the sizes of successive messages arriving to a communications node, the X_i's might be from the same (stationary) distribution, but $\text{Cor}(X_i, X_{i+l})$ could be nonzero for lags $l = 1, 2, \ldots, p$, where the longest autocorrelation lag p would be specified as part of the modeling activity. In this case, the X_i's are identically distributed, but they are not independent and so form a stationary stochastic process with possible autocorrelation out through lag p. As mentioned earlier, such autocorrelation in an input stream can have a major impact on a simulation's results, as demonstrated by Livny et al. (1993).

In this subsection we briefly describe some models for this situation, and in Sec. 8.5.6 we discuss how realizations from such models can be generated as input to the simulation. Except for VARTA processes, we consider only the case where the points X_i in the process are univariate (scalar) random variables, rather than being themselves multivariate random vectors.

AR and ARMA Processes

Standard *autoregressive* (AR) or *autoregressive moving-average* (ARMA) models, developed in Box et al. (2008) for time-series data analysis, might first come to mind for modeling an input time series. While there are many different parameterizations of these processes in the literature, one version of a stationary AR(p) model with mean μ is

$$X_i = \mu + \phi_1(X_{i-1} - \mu) + \phi_2(X_{i-2} - \mu) + \cdots + \phi_p(X_{i-p} - \mu) + \varepsilon_i \quad (6.14)$$

where the ε_i's are IID normal random variables with mean 0 and variance chosen to control $\text{Var}(X_i)$, and the ϕ_i's are constants that must obey a condition for the X_i's to have a stationary marginal distribution. The definition of ARMA models adds weighted proportions of past values of the ε_i's to the above recursion [see Box et al. (2008) for complete details].

To fit such models to observed data, a linear-regression approach is taken to estimate the unknown parameters of the process. The marginal distribution of the X_i's is generally restricted to being normal, however, making this model of limited direct use in simulation-input modeling since the range is infinite in both directions. AR processes do serve, however, as "base" processes for ARTA models discussed below, which are more flexible and useful as simulation input process models.

Gamma Processes

These processes, developed by Lewis et al. (1989), yield marginal distributions having a gamma distribution, as well as autocorrelation between points within the process. They are constructed by a kind of autoregressive operation, similar in spirit to the normal AR processes described in Eq. (6.14) above. This includes the case of exponential marginals, known as *exponential autoregressive* (EAR) processes.

ARTA Processes

Cario and Nelson (1996) developed *autoregressive-to-anything* (ARTA) processes, which seek to model any stationary marginal distribution and any autocorrelation structure. ARTA processes can exactly match the desired autocorrelation structure out to a specified lag p, as well as the desired stationary marginal distribution; in addition, they are specified by an automated procedure requiring no subjective interactive manipulation.

To define an ARTA process, start by specifying a standard stationary AR process $\{Z_i\}$ with N(0, 1) marginal distribution ($\{Z_i\}$ is called the *base process*). Then define the final input process to the simulation as

$$X_i = F^{-1}[\Phi(Z_i)] \tag{6.15}$$

where F^{-1} is the inverse of the desired stationary marginal distribution F and Φ denotes the N(0, 1) distribution function. Since $\Phi(Z_i)$ has a U(0, 1) distribution by a basic result known as the *probability integral transform* [see, e.g., Mood, Graybill, and Boes (1974, pp. 202–203)], applying F^{-1} to this U(0, 1) random variable results in one that has distribution function F. Thus, it is clear that the marginal distribution of X_i will be the desired F.

The principal work in specifying the desired ARTA process, however, is to specify the autocorrelation structure of the base process $\{Z_i\}$ so that the resulting final input process $\{X_i\}$ will exhibit the autocorrelation structure desired. Cario and Nelson (1998) developed numerical methods to do so, as well as a software package to carry out the calculations. The software assumes that the marginal distribution and the autocorrelations for the $\{X_i\}$ process are *given,* although it will compute sample autocorrelations from a set of observed time-series data if desired.

Biller and Nelson (2005, 2008) present a statistical methodology and software for fitting an ARTA process with marginal distributions from the Johnson translation system (see Sec. 6.10.1) to a set of observed univariate time-series data.

VARTA Processes

Biller and Nelson (2003) provided a methodology for modeling and generating stationary multivariate stochastic processes $\{\mathbf{X}_1, \mathbf{X}_2, \ldots\}$, which they call *vector-autoregressive-to-anything* (VARTA) processes. Let $\mathbf{X}_i = (X_{1i}, X_{2i}, \ldots, X_{d,i})^T$ be the input random vector of dimension d at time i, for $i = 1, 2, \ldots$. Let F_j be the distribution function of $X_{j,i}$ for $j = 1, 2, \ldots, d$ and $i = 1, 2, \ldots$. Also, let $\rho_{j,k,l}(\mathbf{X}) = \text{Cor}(X_{j,i}, X_{k,i+l})$ for $j,k = 1, 2, \ldots, d$ and lag $l = 0, 1, \ldots, p$, where $\rho_{j,j,0}(\mathbf{X}) = 1$. Their methodology assumes that F_j is a *given* member of the Johnson translation system and that the correlations $\rho_{j,k,l}(\mathbf{X})$ are *specified.* (In general, F_j will be different for each value of j.)

The principal work in specifying the desired VARTA process is to specify the autocorrelation structure of a Gaussian vector-autoregressive base process $\{\mathbf{Z}_i\}$ so that the resulting final input process $\{\mathbf{X}_i\}$ will exhibit the autocorrelation structure desired.

Biller (2009) generalizes VARTA processes using copula theory to allow representing dependence structures that arise in situations where extreme component realizations occur together.

6.11 SELECTING A DISTRIBUTION IN THE ABSENCE OF DATA

In some simulation studies it may not be possible to collect data on the random variables of interest, so the techniques of Secs. 6.4 through 6.6 are not applicable to the problem of selecting corresponding probability distributions. For example, if the system being studied does not currently exist in some form, then collecting data from the system is obviously not possible. This difficulty can also arise for existing systems, if the number of required probability distributions is large and the time available for the simulation study prohibits the necessary data collection and analysis. Also, sometimes data are collected by an automated data-collection system, which doesn't provide the data in a suitable format. In this section we discuss four heuristic procedures for choosing a distribution in the absence of data.

Let us assume that the random quantity of interest is a continuous random variable X. It will also be useful to think of this random variable as being the time to perform some task, e.g., the time required to repair a piece of equipment when it fails. The first step in using the triangular-distribution or beta-distribution approaches is to identify an interval $[a, b]$ (where a and b are real numbers such that $a < b$) in which it is felt that X will lie with probability close to 1; that is, $P(a \leq X \leq b) \approx 1$. To obtain *subjective* estimates of a and b, subject-matter experts (SMEs) are asked for their most optimistic and pessimistic estimates, respectively, of the time to perform the task. Once an interval $[a, b]$ has been identified subjectively, the next step is to place a probability density function on $[a, b]$ that is thought to be representative of X.

In the triangular-distribution approach, the SMEs are also asked for their subjective estimate of the most-likely time to perform the task. This most-likely value m is the mode of the distribution of X. Given a, b, and m, the random variable X is then considered to have a triangular distribution (see Sec. 6.2.2) on the interval $[a, b]$ with mode m. A graph of a triangular density function is given in Fig. 6.17. Furthermore, an algorithm for generating a triangular random variate is given in Sec. 8.3.15.

One difficulty with the triangular approach is that it requires subjective estimates of the absolute minimum and maximum possible values a and b, which can be problematic. For example, is the value b the maximum over the next 3 months or the maximum over a lifetime? A second major problem with the triangular distribution is that it cannot have a long right tail, as is often the case with density functions for the time to perform some task. [Alternative triangular distributions are discussed in Keefer and Bodily (1983).]

A second approach to placing a density function on $[a, b]$ is to assume that the random variable X has a beta distribution (see Sec. 6.2.2) on this interval with shape

parameters α_1 and α_2. This approach offers greater modeling flexibility because of the variety of shapes that the beta density function can assume (see Fig. 6.11). On the other hand, it is not clear how to choose the parameters α_1 and α_2 so as to specify the distribution completely. We can suggest several possible ideas. If one is willing to assume that X is equally likely to take on any value between a and b, choose $\alpha_1 = \alpha_2 = 1$, which results in the U(a, b) distribution (see Fig. 6.11). (This model might be used if very little is known about the random variable X other than its range [a, b].) An alternative idea, which we feel is generally more realistic, is to assume that the density function of X is skewed to the right. (Our experience with real-world data indicates that density functions corresponding to a task time usually have this shape.) This density shape corresponds to $\alpha_2 > \alpha_1 > 1$ in the beta distribution (see Fig. 6.11). Furthermore, such a beta distribution has a mean μ and a mode m, given by

$$\mu = a + \frac{\alpha_1(b - a)}{\alpha_1 + \alpha_2} \qquad \text{and} \qquad m = a + \frac{(\alpha_1 - 1)(b - a)}{\alpha_1 + \alpha_2 - 2}$$

Given subjective estimates of μ and m, these equations can be solved to obtain the following estimates of α_1 and α_2:

$$\tilde{\alpha}_1 = \frac{(\mu - a)(2m - a - b)}{(m - \mu)(b - a)} \qquad \text{and} \qquad \tilde{\alpha}_2 = \frac{(b - \mu)\tilde{\alpha}_1}{\mu - a}$$

Note, however, that μ must be greater than m for the density to be skewed to the right; if $\mu < m$, it will be skewed to the left. Algorithms for generating a beta random variate are given in Sec. 8.3.8.

A difficulty with the second idea for specifying a beta distribution is that some SMEs will have trouble differentiating between the mean and the mode of a distribution. Keefer and Bodily (1983) suggest alternative ways of specifying the parameters of a beta distribution.

People sometimes use the triangular or beta distribution to model a source of randomness even when it is feasible to collect and analyze the necessary data. This might be done just because the analyst doesn't want to be bothered collecting data, or because the analyst doesn't understand the importance of choosing an appropriate distribution. Example 6.23 shows that the cavalier use of the triangular (or beta) distribution can sometimes result in very erroneous results.

EXAMPLE 6.23. Consider a single-server queueing system with exponential interarrival times with mean 1 and lognormal service times with mean 0.9 and variance 1.39 ($\mu = -0.605$ and $\sigma^2 = 1$), as shown in Fig. 6.55. However, the service-time distribution is actually *unknown* to the analyst, and he first tries to approximate this distribution by a triangular distribution with $a = 0$, $m = 0.2$, and $b = 1.97$ (the 0.9-quantile for the lognormal distribution). Note that a and m have been guessed correctly. Using the formula for the steady-state average delay in queue d for an $M/G/1$ queue in App. 1B, it can be shown that $d = 11.02$ for the lognormal distribution, but $d = 1.30$ for this triangular distribution. Thus, approximating the lognormal distribution by this triangular distribution results in an 88.2 percent error (see Table 6.20 and Fig. 6.55).

Alternatively, suppose the analyst tries to approximate the unknown lognormal distribution by a triangular distribution with $a = 0$, $m = 0.2$, and a mean of 0.9 (correct),

TABLE 6.20
Approximating a lognormal distribution by triangular or beta distributions

Service-time distribution	Steady-state average delay in queue, d	Percent error
Lognormal(−0.605, 1)	11.02	0
Triangular(0, 1.97, 0.2)	1.30	88.2
Triangular(0, 2.5, 0.2)	5.66	48.7
2.5 Beta(1.08, 1.92)	5.85	46.9

which results in $b = 2.5$. [The mean of a triangular distribution is $(a + b + m)/3$.] In this case, $d = 5.66$, which is still an error of 48.7 percent.

Finally, suppose that the analyst tries to approximate the lognormal distribution by a beta distribution with $a = 0$, $b = 2.5$, $\mu = 0.9$, and $m = 0.2$ (the same as for the second triangular distribution), resulting in $\tilde{\alpha}_1 = 1.08$ and $\tilde{\alpha}_2 = 1.92$. In this case, $d = 5.85$, which is a 46.7 percent error.

In summary, we have seen that approximating an unknown distribution by a triangular or beta distribution can result in a large error in the simulation output.

*Because of the shortcomings of the triangular and beta approaches, we now develop two new models for representing a task time in the absence of data that are

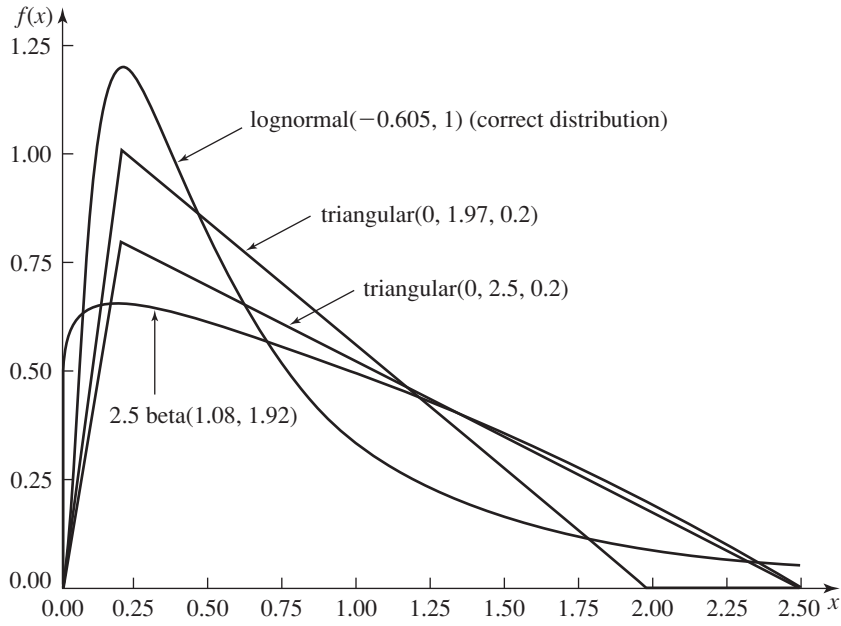


FIGURE 6.55
Lognormal distribution and approximating triangular and beta distributions.

*Skip the remainder of this section on the first reading.

based on the lognormal and Weibull distributions. These models require subjective estimates of the location parameter γ, the most-likely task time m, and the q-quantile ($100q$th percentile) x_q, of the task-time distribution. The location parameter γ plays a role similar to that played by the minimum value a above, but now X must be greater than γ. We also assume that $0 \leq \gamma < m < x_q < \infty$.

We begin with the lognormal distribution. If Y has normal distribution with mean μ and variance σ^2, then $V = e^Y$ has a (two-parameter) lognormal distribution with scale parameter e^μ ($e^\mu > 0$) (see Sec. 6.2.2) and shape parameter σ ($\sigma > 0$). If $X = V + \gamma$, then X has a three-parameter lognormal distribution (see Sec. 6.8) with location parameter γ, scale parameter e^μ, and shape parameter σ, denoted by LN(γ, μ, σ^2). It follows from the discussion of the lognormal distribution in Sec. 6.2.2 that the mode of X is given by

$$m = \gamma + e^{\mu - \sigma^2} \tag{6.16}$$

Furthermore, it can be shown that (see Prob. 6.28)

$$x_q = \gamma + e^{\mu + z_q \sigma} \tag{6.17}$$

where z_q is the q-quantile of a N(0,1) random variable.

If we substitute e^μ from Eq. (6.16) into Eq. (6.17), then we get the following quadratic equation in σ:

$$\sigma^2 + z_q \sigma + c = 0$$

where $c = \ln[(m - \gamma)/(x_q - \gamma)] < 0$. Solving this equation for σ gives the following expression for σ:

$$\sigma = \frac{-z_q \pm \sqrt{z_q^2 - 4c}}{2}$$

Since σ must be positive, we take the "+" root and get the following estimate $\tilde{\sigma}$ for the shape parameter σ:

$$\tilde{\sigma} = \frac{-z_q + \sqrt{z_q^2 - 4c}}{2} \tag{6.18}$$

Substituting $\tilde{\sigma}$ into (6.16), we get the following estimate $\tilde{\mu}$ for μ:

$$\tilde{\mu} = \ln(m - \gamma) + (\tilde{\sigma})^2 \tag{6.19}$$

EXAMPLE 6.24. Suppose that we want a lognormal distribution with a location parameter of $\gamma = 1$, a most-likely value of $m = 4$, and a 0.9-quantile (90th percentile) of $x_{0.9} = 10$. From Eqs. (6.18) and (6.19), we get $\tilde{\sigma} = 0.588$ and $\tilde{\mu} = 1.444$, and the resulting lognormal density function is shown in Fig. 6.56.

We now consider the Weibull distribution. Suppose that the random variable Y has a (two-parameter) Weibull distribution with shape parameter α ($\alpha > 0$) and scale parameter β ($\beta > 0$). We will further assume that $\alpha > 1$, so that the mode is greater than zero. If $X = Y + \gamma$, then X has a three-parameter Weibull distribution (see Sec. 6.8) with location parameter γ, shape parameter α, and scale parameter β, denoted by Weibull(γ, α, β). The mode of X is given by (see Sec. 6.2.2)

$$m = \gamma + \beta\left(\frac{\alpha - 1}{\alpha}\right)^{1/\alpha}$$

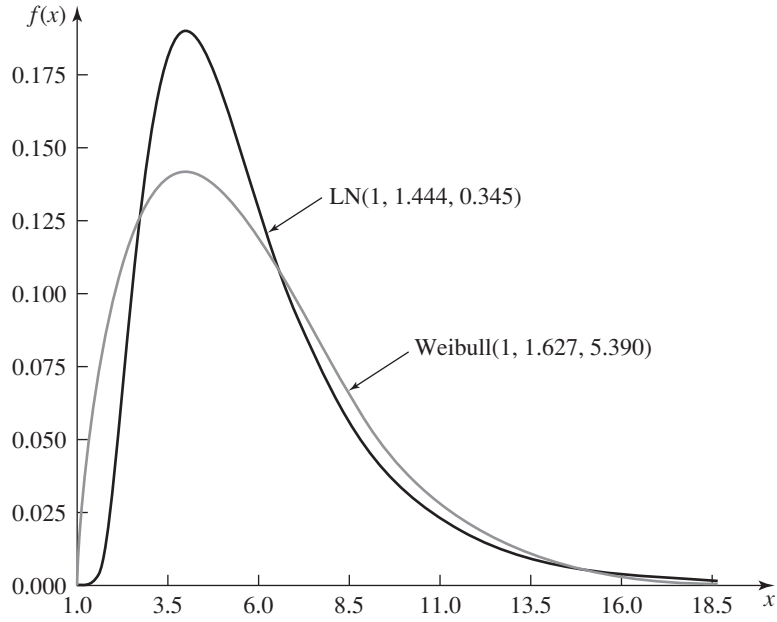


FIGURE 6.56
Specified lognormal and Weibull distributions.

which can be rewritten as

$$\beta = \frac{m - \gamma}{[(\alpha - 1)/\alpha]^{1/\alpha}} \tag{6.20}$$

Furthermore, the distribution function of X evaluated at x_q, $F_X(x_q)$, is given by (see Sec. 6.8)

$$F_X(x_q) = 1 - e^{-[(x_q - \gamma)/\beta]^{\alpha}} = q$$

which can be rewritten as

$$\beta = \frac{x_q - \gamma}{\{\ln[1/(1 - q)]\}^{1/\alpha}} \tag{6.21}$$

Equating Eqs. (6.20) and (6.21) gives the following expression in α:

$$\frac{m - \gamma}{x_q - \gamma} = \left\{\frac{\alpha - 1}{\alpha \ln[1/(1 - q)]}\right\}^{1/\alpha} \tag{6.22}$$

This equation cannot be solved in closed form, but can be solved iteratively by using Newton's method [see, e.g., Press et al. (2007)] to obtain an estimate $\tilde{\alpha}$ of the shape parameter α (see Prob. 6.29). Then $\tilde{\alpha}$ can be substituted into Eq. (6.20) to get an estimate $\tilde{\beta}$ of the scale parameter β:

$$\tilde{\beta} = \frac{m - \gamma}{[(\tilde{\alpha} - 1)/\tilde{\alpha}]^{1/\tilde{\alpha}}} \tag{6.23}$$

EXAMPLE 6.25. Suppose that we want a Weibull distribution with a location parameter of $\gamma = 1$, a most-likely value of $m = 4$, and a 0.9-quantile of $x_{0.9} = 10$. From Eqs. (6.22) and (6.23), we get $\tilde{\alpha} = 1.627$ and $\tilde{\beta} = 5.390$, and the resulting Weibull density function is also shown in Fig. 6.56. Note that the calculation of the estimates $\tilde{\alpha}$ and $\tilde{\beta}$ was done using ExpertFit (see Sec. 6.7).

The lognormal and Weibull distributions can take on arbitrary large values, albeit with a very small probability. Thus, if it is known that the corresponding random variable can never take on values larger than $b\,(b > x_q)$, then it might be desirable to truncate the distribution at b (see Sec. 6.8).

Note that it is also possible to specify a triangular distribution based on subjective estimates of a (the minimum value), m, and x_q (see Prob. 6.30).

6.12 MODELS OF ARRIVAL PROCESSES

In many simulations we need to generate a sequence of random points in time $0 = t_0 \le t_1 \le t_2 \le \cdots$, such that the ith event of some kind occurs at time t_i ($i = 1, 2, \ldots$) and the distribution of the event times $\{t_i\}$ follows some specified form. Let $N(t) = \max\{i: t_i \le t\}$ be the number of events to occur at or before time t for $t \ge 0$. We call the stochastic process $\{N(t), t \ge 0\}$ an *arrival process* since, for our purposes, the events of interest are usually arrivals of customers to a service facility of some kind. In what follows, we call $A_i = t_i - t_{i-1}$ (where $i = 1, 2, \ldots$) the *interarrival time* between the $(i - 1)$st and ith customers.

In Sec. 6.12.1 we discuss the Poisson process, which is an arrival process for which the A_i's are IID exponential random variables. The Poisson process is probably the most commonly used model for the arrival process of customers to a queueing system. Section 6.12.2 discusses the nonstationary Poisson process, which is often used as a model of the arrival process to a system when the arrival rate varies with time. Finally, in Sec. 6.12.3 we describe an approach to modeling arrival processes where each event is really the arrival of a "batch" of customers.

A general reference for this section is Çinlar (1975, Chap. 4).

6.12.1 Poisson Processes

In this section we define a Poisson process, state some of its important properties, and in the course of doing so explain why the interarrival times for many real-world systems closely resemble IID exponential random variables.

The stochastic process $\{N(t), t \ge 0\}$ is said to be a *Poisson process* if:

1. Customers arrive one at a time.
2. $N(t + s) - N(t)$ (the number of arrivals in the time interval $(t, t + s]$) is independent of $\{N(u), 0 \le u \le t\}$.
3. The distribution of $N(t + s) - N(t)$ is independent of t for all $t, s \ge 0$.

Properties 1 and 2 are characteristic of many actual arrival processes. Property 1 would not hold if customers arrived in batches; see Sec. 6.12.3. Property 2 says that

the number of arrivals in the interval $(t, t + s]$ is independent of the number of arrivals in the earlier time interval $[0, t]$ and also of the times at which these arrivals occur. This property could be violated if, for example, a large number of arrivals in $[0, t]$ caused some customers arriving in $(t, t + s]$ to balk, i.e., to go away immediately without being served, because they find the system highly congested. Property 3, on the other hand, will be violated by many real-life arrival processes since it implies that the arrival rate of customers does not depend on the time of day, etc. If, however, the time period of interest for the system is relatively short, say, a 1- or 2-hour period of peak demand, we have found that for many systems (but certainly not all) the arrival rate is reasonably constant over this interval and the Poisson process is a good model for the process during this interval. (See Theorem 6.2 below and then Example 6.4.)

The following theorem, proved in Çinlar (1975, pp. 74–76), explains where the Poisson process gets its name.

THEOREM 6.1. If $\{N(t), t \geq 0\}$ is a Poisson process, then the number of arrivals in any time interval of length s is a Poisson random variable with parameter λs (where λ is a positive real number). That is,

$$P[N(t + s) - N(t) = k] = \frac{e^{-\lambda s}(\lambda s)^k}{k!} \qquad \text{for } k = 0, 1, 2, \ldots \text{ and } t, s \geq 0$$

Therefore, $E[N(s)] = \lambda s$ (see Sec. 6.2.3) and, in particular, $E[N(1)] = \lambda$. Thus, λ is the expected number of arrivals in any interval of length 1. We call λ the *rate* of the process.

We now see that the interarrival times for a Poisson process are IID exponential random variables; see Çinlar (1975, pp. 79–80).

THEOREM 6.2. If $\{N(t), t \geq 0\}$ is a Poisson process with rate λ, then its corresponding interarrival times $A_1, A_2, \ldots$ are IID exponential random variables with mean $1/\lambda$.

This result together with our above discussion explains why we have found that interarrival times during a restricted time period are often approximately IID exponential random variables. For example, recall that the interarrival times of cars for the drive-up bank of Example 6.4 were found to be approximately exponential during a 90-minute period.

The converse of Theorem 6.2 is also true. Namely, if the interarrival times $A_1, A_2, \ldots$ for an arrival process $\{N(t), t \geq 0\}$ are IID exponential random variables with mean $1/\lambda$, then $\{N(t), t \geq 0\}$ is a Poisson process with rate λ [Çinlar (1975, p. 80)].

6.12.2 Nonstationary Poisson Processes

Let $\lambda(t)$ be the arrival rate of customers to some system at time t. [See below for some insight into the meaning of $\lambda(t)$.] If customers arrive at the system in accordance with a Poisson process with constant rate λ, then $\lambda(t) = \lambda$ for all $t \geq 0$. However, for many real-world systems, $\lambda(t)$ is actually a function of t. For example, the arrival rate of customers to a fast-food restaurant will be larger during the noon rush hour than in the middle of the afternoon. Also, traffic on a freeway will

be heavier during the morning and evening rush hours. If the arrival rate $\lambda(t)$ does in fact change with time, then the interarrival times $A_1, A_2, \ldots$ are *not* identically distributed; thus, it is not appropriate to fit a single probability distribution to the A_i's by using the techniques discussed in Secs. 6.4 through 6.6. In this section we discuss a commonly used model for arrival processes with time-varying arrival rates.

The stochastic process $\{N(t),\ t \geq 0\}$ is said to be a *nonstationary Poisson process* if:

1. Customers arrive one at a time.
2. $N(t + s) - N(t)$ is independent of $\{N(u),\ 0 \leq u \leq t\}$.

Thus, for a nonstationary Poisson process, customers must still arrive one at a time, and the numbers of arrivals in disjoint intervals are independent, but now the arrival rate $\lambda(t)$ is allowed to be a function of time.

Let $\Lambda(t) = E[N(t)]$ for all $t \geq 0$. If $\Lambda(t)$ is differentiable for a particular value of t, we formally define $\lambda(t)$ as

$$\lambda(t) = \frac{d}{dt}\Lambda(t)$$

Intuitively, $\lambda(t)$ will be large in intervals for which the expected number of arrivals is large. We call $\Lambda(t)$ and $\lambda(t)$ the *expectation function* and the *rate function*, respectively, for the nonstationary Poisson process.

The following theorem shows that the number of arrivals in the interval $(t, t + s]$ for a nonstationary Poisson process is a Poisson random variable whose parameter depends on *both* t and s.

THEOREM 6.3. If $\{N(t),\ t \geq 0\}$ is a nonstationary Poisson process with continuous expectation function $\Lambda(t)$, then

$$P[N(t + s) - N(t) = k] = \frac{e^{-b(t,s)}[b(t, s)]^k}{k!} \qquad \text{for } k = 0, 1, 2, \ldots \text{ and } t, s \geq 0$$

where $b(t, s) = \Lambda(t + s) - \Lambda(t) = \int_t^{t+s} \lambda(y)dy$, the last equality holding if $d\Lambda(t)/dt$ is bounded on $[t, t + s]$ and if $d\Lambda(t)/dt$ exists and is continuous for all but finitely many points in $[t, t + s]$ (see Prob. 6.25).

We have not yet addressed the question of how to estimate $\lambda(t)$ [or $\Lambda(t)$] from a set of observations on an arrival process of interest. The following example gives a heuristic but practical approach, and other approaches are briefly discussed after the example.

EXAMPLE 6.26. A simulation model was developed for a xerographic copy shop, and data were collected on the times of arrival of customers between 11 A.M. and 1 P.M. for eight different days. From observing the characteristics of the arriving customers, it was felt that properties 1 and 2 for the nonstationary Poisson process were applicable and, in addition, that $\lambda(t)$ varied over the 2-hour interval. To obtain an estimate of $\lambda(t)$, the 2-hour interval was divided into the following 12 subintervals:

[11:00, 11:10), [11:10, 11:20), . . . , [12:40, 12:50), [12:50, 1:00)

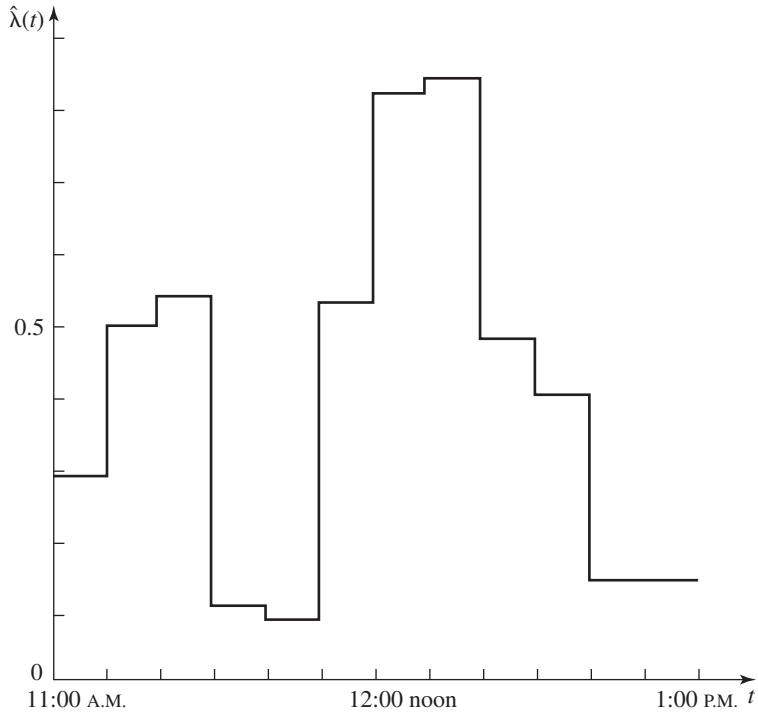


FIGURE 6.57
Plot of the estimated rate function $\hat{\lambda}(t)$ in customers per minute for the arrival process to a copy shop between 11 A.M. and 1 P.M.

For each day, the number of arrivals in each of these subintervals was determined. Then, for each subinterval, the average number of arrivals in that subinterval over the 8 days was computed. These 12 averages are estimates of the expected numbers of arrivals in the corresponding subintervals. Finally, for each subinterval, the average number of arrivals in that subinterval was divided by the subinterval length, 10 minutes, to obtain an estimate of the arrival rate for that subinterval. The estimated arrival rate $\hat{\lambda}(t)$ (in customers per minute) is plotted in Fig. 6.57. Note that the estimated arrival rate varies substantially over the 2-hour period.

One might legitimately ask how we decided on these subintervals of length 10 minutes. Actually, we computed estimates of $\lambda(t)$ in the above manner for subintervals of length 5, 10, and 15 minutes. The estimate of $\lambda(t)$ based on subintervals of length 5 minutes was rejected because it was felt that the corresponding plot of $\hat{\lambda}(t)$ was too ragged; i.e., a subinterval length of 5 minutes was too small. On the other hand, the estimate of $\lambda(t)$ based on subintervals of length 15 minutes was not chosen because the corresponding plot of $\hat{\lambda}(t)$ seemed too "smooth," meaning that information on the true nature of $\lambda(t)$ was being lost. In general, the problem of choosing a subinterval length here is similar to that of choosing the interval width for a histogram (see Sec. 6.4.2).

While the piecewise-constant method of specifying $\lambda(t)$ in Example 6.26 is certainly quite simple and fairly flexible, it does require somewhat arbitrary judgment about the boundaries and widths of the constant-rate time intervals. Other methods

for specifying and estimating $\lambda(t)$ [or, alternatively, $\Lambda(t)$] have been developed, which we briefly mention here:

- Several authors have proposed procedures where the estimated rate function $\hat{\lambda}(t)$ is specified to be a generalization of what was done in Example 6.26, in terms of allowing for piecewise-linear or piecewise-polynomial forms. These include Kao and Chang (1988), Lewis and Shedler (1976), and Klein and Roberts (1984).
- Leemis (1991) developed an intuitive and simple nonparametric method to specify a piecewise-linear estimate of the expectation function $\Lambda(t)$, where the breakpoints are determined by the observed arrival times in a superposition of several realizations of the process. He shows that the estimator converges, with probability 1, to the true underlying expectation function as the number of observed realizations increases, and also derives a confidence band around the true $\Lambda(t)$ that could be useful in input-specification sensitivity analysis. Since the estimated $\Lambda(t)$ is piecewise linear, generating observations (see Sec. 8.6.2) is simple and efficient. A generalization of this method is given in Arkin and Leemis (2000).
- A different approach to specifying and estimating the rate function $\lambda(t)$ is to assume that it has some specific parametric (functional) form that is general enough in structure and has a sufficient number of parameters to allow it to fit observed data well. The parametric form should allow for trends and cycles, and should admit rigorous statistical methods for parameter estimation, such as maximum-likelihood or least-squares methods. Such functions, together with software for estimation and generation, are developed in Lee et al. (1991), Johnson et al. (1994a, 1994b), Kuhl, Damerji, and Wilson (1997), Kuhl, Wilson, and Johnson (1997), and Kuhl and Wilson (2000).
- Kuhl and Wilson (2001) and Kuhl et al. (2006) give a combined nonparametric/parametric approach and software for estimating an expectation function $\Lambda(t)$ of a nonstationary Poisson process with a long-run trend or cyclic effects that may exhibit nontrigonometric characteristics.
- Gerhardt and Nelson (2009) provide methodologies for modeling nonstationary arrival processes that are more or less variable than a nonstationary Poisson process.

6.12.3 Batch Arrivals

For some real-world systems, customers arrive in *batches*, or groups, so that property 1 of the Poisson process and of the nonstationary Poisson process is violated. For example, people arriving at a sporting event or at a cafeteria often come in batches. We now consider how one might model such an arrival process.

Let $N(t)$ now be the number of *batches* of individual customers that have arrived by time t. By applying the techniques discussed previously in this chapter to the times of arrivals of the successive batches, we can develop a model for the process $\{N(t), t \geq 0\}$. For example, if the interarrival times of batches appear to be approximately IID exponential random variables, $\{N(t), t \geq 0\}$ can be modeled as a Poisson process. Next, we fit a discrete distribution to the sizes of the successive

batches; the batch sizes will be positive integers. Thus, for the original arrival process, it is assumed that batches of customers arrive in accordance with the arrival process $\{N(t), t \geq 0\}$ and that the number of customers in each batch is a random variable with the fitted discrete distribution.

The above informal discussion can be made more precise. If $X(t)$ is the total number of individual customers to arrive by time t, and if B_i is the number of customers in the ith batch, then $X(t)$ is given by

$$X(t) = \sum_{i=1}^{N(t)} B_i \qquad \text{for } t \geq 0$$

If the B_i's are assumed to be IID random variables that are also independent of $\{N(t), t \geq 0\}$, and if $\{N(t), t \geq 0\}$ is a Poisson process, then the stochastic process $\{X(t), t \geq 0\}$ is said to be a *compound Poisson process*.

6.13 ASSESSING THE HOMOGENEITY OF DIFFERENT DATA SETS

Sometimes an analyst collects k sets of observations on a random phenomenon independently and would like to know whether these data sets are homogeneous and thus can be merged. For example, it might be of interest to know whether service times of customers in a bank collected on different days are homogeneous. If they are, then the service times from the different days can be merged and the combined sample used to find *the* service-time distribution. Otherwise, more than one service-time distribution is needed. In this section, we discuss the Kruskal-Wallis hypothesis test for homogeneity. It is a nonparametric test since no assumptions are made about the distributions of the data.

Suppose that we have k independent samples of possibly different sizes, and that the samples themselves are independent. Denote the ith sample of size n_i by X_{i1}, $X_{i2}, \ldots, X_{in_i}$ for $i = 1, 2, \ldots, k$; and let n denote the total number of observations

$$n = \sum_{i=1}^{k} n_i$$

Then we would like to test the null hypothesis

H_0: All the population distribution functions are identical

against the alternative hypothesis

H_1: At least one of the populations tends to yield larger observations than at least one of the other populations

To construct the Kruskal-Wallis (K-W) statistic, assign rank 1 to the smallest of the n observations, rank 2 to the second smallest, and so on to the largest of the

n observations, which receives rank n. Let $R(X_{ij})$ represent the rank assigned to X_{ij}, and let R_i be the sum of the ranks assigned to the ith sample, that is,

$$R_i = \sum_{j=1}^{n_i} R(X_{ij}) \qquad \text{for } i = 1, 2, \ldots, k$$

Then the K-W test statistic T is defined as

$$T = \frac{12}{n(n+1)} \sum_{i=1}^{k} \frac{R_i^2}{n_i} - 3(n+1)$$

We reject the null hypothesis H_0 at level α if $T > \chi^2_{k-1,1-\alpha}$, where $\chi^2_{k-1,1-\alpha}$ is the upper $1-\alpha$ critical value for a chi-square distribution with $k-1$ degrees of freedom. The above expression for T assumes that no two observations are equal. If this is not the case, then a different expression for T must be used [see Conover (1999, pp. 288–290)].

EXAMPLE 6.27. A simulation model was developed for the hub operations of an overnight air-delivery service for the purpose of determining the amount of unloading equipment required. The model included provision for the fact that planes may arrive before or after their scheduled arrival times. Data were available on the actual times of arrival for two different incoming flight numbers (each corresponding to a different origin city) for 57 different days. (Each flight number arrives once a day, 5 days a week.) Let X_{ij} be the scheduled time of arrival minus the actual time of arrival (in minutes) for day j and flight number i, for $j = 1, 2, \ldots, 57$ and $i = 1, 2$. If $X_{ij} < 0$, then flight number i was late on day j. We want to perform a K-W test to determine whether the X_{1j}'s and X_{2j}'s are homogeneous, i.e., whether the arrival patterns for the two flight numbers are similar. We computed the K-W statistic and obtained $T = 4.317$, which is greater than the critical value $2.706 = \chi^2_{1,0.90}$. Therefore, we rejected the null hypothesis at level $\alpha = 0.10$, and the arrival patterns for the two flight numbers had to be modeled separately. The observed differences were to a large extent due to different weather conditions in the two origin cities.

APPENDIX 6A TABLES OF MLEs FOR THE GAMMA AND BETA DISTRIBUTIONS

TABLE 6.21
$\hat{\alpha}$ as a function of T, gamma distribution

T	$\hat{\alpha}$	T	$\hat{\alpha}$	T	$\hat{\alpha}$	T	$\hat{\alpha}$
0.01	0.010	1.40	0.827	5.00	2.655	13.00	6.662
0.02	0.019	1.50	0.879	5.20	2.755	13.50	6.912
0.03	0.027	1.60	0.931	5.40	2.856	14.00	7.163
0.04	0.036	1.70	0.983	5.60	2.956	14.50	7.413
0.05	0.044	1.80	1.035	5.80	3.057	15.00	7.663
0.06	0.052	1.90	1.086	6.00	3.157	15.50	7.913
0.07	0.060	2.00	1.138	6.20	3.257	16.00	8.163

(*continued*)

TABLE 6.21 (*continued*)

T	$\hat{\alpha}$	T	$\hat{\alpha}$	T	$\hat{\alpha}$	T	$\hat{\alpha}$
0.08	0.068	2.10	1.189	6.40	3.357	16.50	8.413
0.09	0.076	2.20	1.240	6.60	3.458	17.00	8.663
0.10	0.083	2.30	1.291	6.80	3.558	17.50	8.913
0.11	0.090	2.40	1.342	7.00	3.658	18.00	9.163
0.12	0.098	2.50	1.393	7.20	3.759	18.50	9.414
0.13	0.105	2.60	1.444	7.40	3.859	19.00	9.664
0.14	0.112	2.70	1.495	7.60	3.959	19.50	9.914
0.15	0.119	2.80	1.546	7.80	4.059	20.00	10.164
0.16	0.126	2.90	1.596	8.00	4.159	20.50	10.414
0.17	0.133	3.00	1.647	8.20	4.260	21.00	10.664
0.18	0.140	3.10	1.698	8.40	4.360	21.50	10.914
0.19	0.147	3.20	1.748	8.60	4.460	22.00	11.164
0.20	0.153	3.30	1.799	8.80	4.560	22.50	11.414
0.30	0.218	3.40	1.849	9.00	4.660	23.00	11.664
0.40	0.279	3.50	1.900	9.20	4.760	23.50	11.914
0.50	0.338	3.60	1.950	9.40	4.860	24.00	12.164
0.60	0.396	3.70	2.001	9.60	4.961	24.50	12.414
0.70	0.452	3.80	2.051	9.80	5.061	25.00	12.664
0.80	0.507	3.90	2.101	10.00	5.161	30.00	15.165
0.90	0.562	4.00	2.152	10.50	5.411	35.00	17.665
1.00	0.616	4.20	2.253	11.00	5.661	40.00	20.165
1.10	0.669	4.40	2.353	11.50	5.912	45.00	22.665
1.20	0.722	4.60	2.454	12.00	6.162	50.00	25.166
1.30	0.775	4.80	2.554	12.50	6.412		

TABLE 6.22

$\hat{\alpha}_1$ and $\hat{\alpha}_2$ as functions of G_1 and G_2, beta distribution

If $G_1 \leq G_2$, use the first line of labels; if $G_2 \leq G_1$, use the second line of labels

G_1 G_2	G_2 G_1	$\hat{\alpha}_1$ $\hat{\alpha}_2$	$\hat{\alpha}_2$ $\hat{\alpha}_1$	G_1 G_2	G_2 G_1	$\hat{\alpha}_1$ $\hat{\alpha}_2$	$\hat{\alpha}_2$ $\hat{\alpha}_1$
0.01	0.01	0.112	0.112	0.15	0.35	0.405	0.563
0.01	0.05	0.126	0.157	0.15	0.40	0.432	0.653
0.01	0.10	0.135	0.192	0.15	0.45	0.464	0.762
0.01	0.15	0.141	0.223	0.15	0.50	0.502	0.903
0.01	0.20	0.147	0.254	0.15	0.55	0.550	1.090
0.01	0.25	0.152	0.285	0.15	0.60	0.612	1.353
0.01	0.30	0.157	0.318	0.15	0.65	0.701	1.752
0.01	0.35	0.163	0.354	0.15	0.70	0.842	2.429
0.01	0.40	0.168	0.395	0.15	0.75	1.111	3.810
0.01	0.45	0.173	0.441	0.15	0.80	1.884	8.026
0.01	0.50	0.179	0.495	0.15	0.84	7.908	42.014

(*continued*)

TABLE 6.22 (*continued*)

G_1 G_2	G_2 G_1	$\hat{\alpha}_1$ $\hat{\alpha}_2$	$\hat{\alpha}_2$ $\hat{\alpha}_1$	G_1 G_2	G_2 G_1	$\hat{\alpha}_1$ $\hat{\alpha}_2$	$\hat{\alpha}_2$ $\hat{\alpha}_1$
0.01	0.55	0.185	0.559	0.20	0.20	0.395	0.395
0.01	0.60	0.192	0.639	0.20	0.25	0.424	0.461
0.01	0.65	0.200	0.741	0.20	0.30	0.456	0.537
0.01	0.70	0.210	0.877	0.20	0.35	0.491	0.626
0.01	0.75	0.221	1.072	0.20	0.40	0.531	0.735
0.01	0.80	0.237	1.376	0.20	0.45	0.579	0.873
0.01	0.85	0.259	1.920	0.20	0.50	0.640	1.057
0.01	0.90	0.299	3.162	0.20	0.55	0.720	1.314
0.01	0.95	0.407	8.232	0.20	0.60	0.834	1.701
0.01	0.98	0.850	42.126	0.20	0.65	1.016	2.352
0.05	0.05	0.180	0.180	0.20	0.70	1.367	3.669
0.05	0.10	0.195	0.223	0.20	0.75	2.388	7.654
0.05	0.15	0.207	0.263	0.20	0.79	10.407	39.649
0.05	0.20	0.217	0.302	0.25	0.25	0.500	0.500
0.05	0.25	0.228	0.343	0.25	0.30	0.543	0.588
0.05	0.30	0.238	0.387	0.25	0.35	0.592	0.695
0.05	0.35	0.248	0.437	0.25	0.40	0.651	0.830
0.05	0.40	0.259	0.494	0.25	0.45	0.724	1.007
0.05	0.45	0.271	0.560	0.25	0.50	0.822	1.254
0.05	0.50	0.284	0.640	0.25	0.55	0.962	1.624
0.05	0.55	0.299	0.739	0.25	0.60	1.186	2.243
0.05	0.60	0.317	0.867	0.25	0.65	1.620	3.486
0.05	0.65	0.338	1.037	0.25	0.70	2.889	7.230
0.05	0.70	0.366	1.280	0.25	0.74	12.905	37.229
0.05	0.75	0.403	1.655	0.30	0.30	0.647	0.647
0.05	0.80	0.461	2.305	0.30	0.35	0.717	0.777
0.05	0.85	0.566	3.682	0.30	0.40	0.804	0.947
0.05	0.90	0.849	8.130	0.30	0.45	0.920	1.182
0.05	0.94	2.898	45.901	0.30	0.50	1.086	1.532
0.10	0.10	0.245	0.245	0.30	0.55	1.352	2.115
0.10	0.15	0.262	0.291	0.30	0.60	1.869	3.280
0.10	0.20	0.278	0.337	0.30	0.65	3.387	6.779
0.10	0.25	0.294	0.386	0.30	0.69	15.402	34.780
0.10	0.30	0.310	0.441	0.35	0.35	0.879	0.879
0.10	0.35	0.327	0.503	0.35	0.40	1.013	1.101
0.10	0.40	0.345	0.576	0.35	0.45	1.205	1.430
0.10	0.45	0.365	0.663	0.35	0.50	1.514	1.975
0.10	0.50	0.389	0.770	0.35	0.55	2.115	3.060
0.10	0.55	0.417	0.909	0.35	0.60	3.883	6.313
0.10	0.60	0.451	1.093	0.35	0.64	17.897	32.315
0.10	0.65	0.497	1.356	0.40	0.40	1.320	1.320
0.10	0.70	0.560	1.756	0.40	0.45	1.673	1.827
0.10	0.75	0.660	2.443	0.40	0.50	2.358	2.832
0.10	0.80	0.846	3.864	0.40	0.55	4.376	5.837
0.10	0.85	1.374	8.277	0.40	0.59	20.391	29.841
0.10	0.89	5.406	44.239	0.45	0.45	2.597	2.597
0.15	0.15	0.314	0.314	0.45	0.50	4.867	5.354
0.15	0.20	0.335	0.367	0.45	0.54	22.882	27.359
0.15	0.25	0.357	0.424	0.49	0.49	12.620	12.620
0.15	0.30	0.380	0.489	0.49	0.50	24.873	25.371

PROBLEMS

6.1. Suppose that a man's job is to install 98 rivets in the right wing of an airplane under construction. If the random variable T is the total time required for one airplane, then what is the approximate distribution of T?

6.2. Prove comment 2 for the Weibull distribution in Table 6.3.

6.3. Prove comment 2 for the Pearson type VI distribution in Table 6.3.

6.4. Consider a four-parameter Pearson type VI distribution with shape parameters α_1 and α_2, scale parameter β, and location parameter γ. If $\alpha_1 = 1$, $\gamma = \beta = c > 0$, then the resulting density is

$$f(x) = \alpha_2 x^{-(\alpha_2 + 1)} c^{\alpha_2} \qquad \text{for } x > c$$

which is the density function of a *Pareto distribution* with parameters c and α_2, denoted Pareto(c, α_2). Show that $X \sim$ Pareto(c, α_2) if and only if $Y = \ln X \sim$ expo($\ln c$, $1/\alpha_2$), an exponential distribution with location parameter $\ln c$ and scale parameter $1/\alpha_2$.

6.5. For the empirical distribution given by $F(x)$ in Sec. 6.2.4, discuss the merit of defining $F(X_{(i)}) = i/n$ for $i = 1, 2, \ldots, n$, which seems like an intuitive definition. In this case, how would you define $F(x)$ for $0 \le x < X_{(1)}$?

6.6. Compute the expectation of the empirical distribution given by $F(x)$ in Sec. 6.2.4.

6.7. For discrete distributions, prove that the histogram (Sec. 6.4.2) is an unbiased estimator of the (unknown) mass function; i.e., show that $E(h_j) = p(x_j)$ for all j. *Hint:* For j fixed, define

$$Y_i = \begin{cases} 1 & \text{if } X_i = x_j \\ 0 & \text{otherwise} \end{cases} \qquad \text{for } i = 1, 2, \ldots, n$$

6.8. Suppose that the histogram of your observed data has several local modes (see Fig. 6.31), but that it is not possible to break the data into natural groups with a different probability distribution fitting each group. Describe an alternative approach for modeling your data.

6.9. For a geometric distribution with parameter p, explain why the MLE $\hat{p} = 1/[\overline{X}(n) + 1]$ is intuitive.

6.10. For each of the following distributions, derive formulas for the MLEs of the indicated parameters. Assume that we have IID data $X_1, X_2, \ldots, X_n$ from the distribution in question.

(a) U(0, b), MLE for b
(b) U(a, 0), MLE for a
(c) U(a, b), joint MLEs for a and b
(d) N(μ, σ^2), joint MLEs for μ and σ
(e) LN(μ, σ^2), joint MLEs for μ and σ
(f) Bernoulli(p), MLE for p
(g) DU(i, j), joint MLEs for i and j

(*h*) bin(t, p), MLE for p assuming that t is known
(*i*) negbin(s, p), MLE for p assuming that s is known
(*j*) U($\theta - 0.5$, $\theta + 0.5$), MLE for θ

6.11. For a Poisson distribution with parameter λ, derive an approximate $100(1 - \alpha)$ percent confidence interval for λ given the data $X_1, X_2, \ldots, X_n$. Use the asymptotic normality of the MLE $\hat{\lambda}$.

6.12. Consider the shifted (two-parameter) exponential distribution, which has density function

$$f(x) = \begin{cases} \dfrac{1}{\beta} e^{-(x-\gamma)/\beta} & \text{if } x \geq \gamma \\ 0 & \text{otherwise} \end{cases}$$

for $\beta > 0$ and any real number γ. Given a sample $X_1, X_2, \ldots, X_n$ of IID random values from this distribution, find formulas for the joint MLEs $\hat{\gamma}$ and $\hat{\beta}$. *Hint:* Remember that γ cannot exceed any X_i.

6.13. For a frequency comparison, show that r_j as given by Eq. (6.4) is actually the expected proportion of the n observations that would fall in the jth interval if the fitted distribution were in fact the true one.

6.14. For Q–Q plots, why it is inconvenient to have an empirical distribution function $\tilde{F}_n(x)$ such that $\tilde{F}_n(X_{(n)}) = 1$?

6.15. What difficulty arises when you try to define a Q–Q plot for a discrete distribution?

6.16. Suppose that the true distribution function $F(x)$ and the fitted distribution function $\hat{F}(x)$ are the same. For what distribution $F(x)$ will the Q–Q and P–P plots be essentially the same if the sample size n is large?

6.17. Suppose that the random variable M_j is the number of the n X_i's that would fall in the jth interval $[a_{j-1}, a_j)$ for a chi-square test if the fitted distribution were in fact the true one. What is the distribution of M_j, and what is its mean?

6.18. For the chi-square test, explain intuitively why Kallenberg, Oosterhoff, and Schriever (1985) found in certain cases that power was greater when the np_j's were smaller in the tails rather than all being equal.

6.19. Let $F_n(x)$ be the empirical distribution function used for the K-S test. Show that $F_n(x) \to F(x)$ as $n \to \infty$ (w.p. 1) for all x, where $F(x)$ is the true underlying distribution function.

6.20. Assume for the following analyses that each data set consists of IID observations from a continuous distribution. Use the Student Version of ExpertFit (see www.mhhe.com/law) to analyze these data (included in ExpertFit), following the steps in Sec. 6.7 (i.e., data summary, histogram, fitting and ranking distributions, density-histogram plot, distribution-function-differences plot, P–P plot, chi-square test, K-S test, and A-D test). Use a level of $\alpha = 0.05$ for the tests.
(*a*) 91 laboratory-processing times
(*b*) 1000 paper-roll yardages (use second-best model so tests are applicable)

(*c*) 218 post-office service times (use second-best model so tests are applicable)
(*d*) 1592 times at an automated-teller machine (ATM)
(*e*) 694 machine-repair times
(*f*) 3035 post-anesthesia recovery times

6.21. Assume for the following analyses that each data set consists of IID observations from a discrete distribution. Use the Student Version of ExpertFit to analyze these data, performing the following steps: data summary, histogram, fitting and ranking distributions, frequency-comparison plot, distribution-function-differences plot, *P–P* plot, and chi-square test. (K-S and A-D tests are not applicable to discrete data.) Use a level of $\alpha = 0.05$ for the test.
(*a*) 369 university test scores
(*b*) 200 deaths per corps per year by horse kick in the Prussian army in the late 1800s

6.22. There are two sets of IID machine-repair times named PROB622a and PROB622b. (The machines are from the same vendor.) Use the Student Version of ExpertFit (with "Advanced Mode") to perform the Kruskal-Wallis test (see Sec. 6.13) at level $\alpha = 0.05$ to determine whether the data sets are homogeneous.

6.23. For the location parameter estimator $\tilde{\gamma}$ in Sec. 6.8, show that $\tilde{\gamma} < X_{(1)}$ if and only if $X_{(k)} < [X_{(1)} + X_{(n)}]/2$.

6.24. Let LN(γ, μ, σ^2) denote the shifted (three-parameter) lognormal distribution, which has density

$$f(x) = \begin{cases} \dfrac{1}{(x-\gamma)\sqrt{2\pi\sigma^2}} \exp \dfrac{-[\ln(x-\gamma) - \mu]^2}{2\sigma^2} & \text{if } x > \gamma \\ 0 & \text{otherwise} \end{cases}$$

for $\sigma > 0$ and any real numbers γ and μ. [Thus, LN(0, μ, σ^2) is the original LN(μ, σ^2) distribution.]
(*a*) Verify that $X \sim \text{LN}(\gamma, \mu, \sigma^2)$ if and only if $X - \gamma \sim \text{LN}(\mu, \sigma^2)$.
(*b*) Show that for a fixed, known value of γ, the MLEs of μ and σ in the LN(γ, μ, σ^2) distribution are

$$\hat{\mu} = \frac{\sum_{i=1}^{n} \ln(X_i - \gamma)}{n} \qquad \text{and} \qquad \hat{\sigma} = \left\{ \frac{\sum_{i=1}^{n} [\ln(X_i - \gamma) - \hat{\mu}]^2}{n} \right\}^{1/2}$$

i.e., we simply shift the data by an amount $-\gamma$ and then treat them as being (unshifted) lognormal data.

6.25. For Theorem 6.3 in Sec. 6.12.2, explain intuitively why the expected number of arrivals in the interval $(t, t + s]$, $b(t, s)$, should be equal to $\int_t^{t+s} \lambda(y)\, dy$.

6.26. Provide an intuitive motivation for the definition of MLEs in the continuous case (see Sec. 6.5) by going through steps (*a*) through (*c*) below. As before, the observed data are $X_1, X_2, \ldots, X_n$, and are IID realizations of a random variable X with density f_θ. Bear in mind that the X_i's have already been observed, so are to be regarded as fixed numbers rather than variables.
(*a*) Let ε be a small (but strictly positive) real number, and define the phrase "getting a value of X near X_i" to be the event $\{X_i - \varepsilon < X < X_i + \varepsilon\}$. Use the mean-value

theorem from calculus to argue that $P(\text{getting a value of } X \text{ near } X_i) \approx 2\varepsilon f_\theta(X_i)$, for any $i = 1, 2, \ldots, n$.

(*b*) Define the phrase "getting a sample of n IID values of X near the observed data" to be the event (getting a value of X near X_1, getting a value of X near X_2, . . . , getting a value of X near X_n). Show that $P(\text{getting a sample of } n \text{ IID values of } X \text{ near the observed data}) \approx (2\varepsilon)^n f_\theta(X_1) f_\theta(X_2) \cdots f_\theta(X_n)$, and note that this is *proportional* to the likelihood function $L(\theta)$.

(*c*) Argue that the MLE $\hat{\theta}$ is the value of θ that maximizes the approximate probability of getting a sample of n IID values of X near the observed data, and in this sense "best explains" the data that were actually observed.

6.27. Why is the average delay in queue approximately equal to the corresponding average number in queue in Table 6.2?

6.28. Show that Eq. (6.17) in Sec. 6.11 is correct.

6.29. Develop the general recursive formula for Newton's method to estimate the shape parameter α for the Weibull distribution in Sec. 6.11 [see Eq. (6.22)]. The formula should be of the following form:

$$\tilde{\alpha}_{k+1} = \tilde{\alpha}_k - \frac{f(\tilde{\alpha}_k)}{f'(\tilde{\alpha}_k)}$$

where f' denotes the derivative of f.

6.30. In the absence of data (Sec. 6.11), show how to specify a triangular distribution based on subjective estimates of a, m, and x_q.

6.31. In the absence of data (Sec. 6.11), is it possible, in general, to specify a triangular distribution using a, b, and μ, rather than a, b, and m?